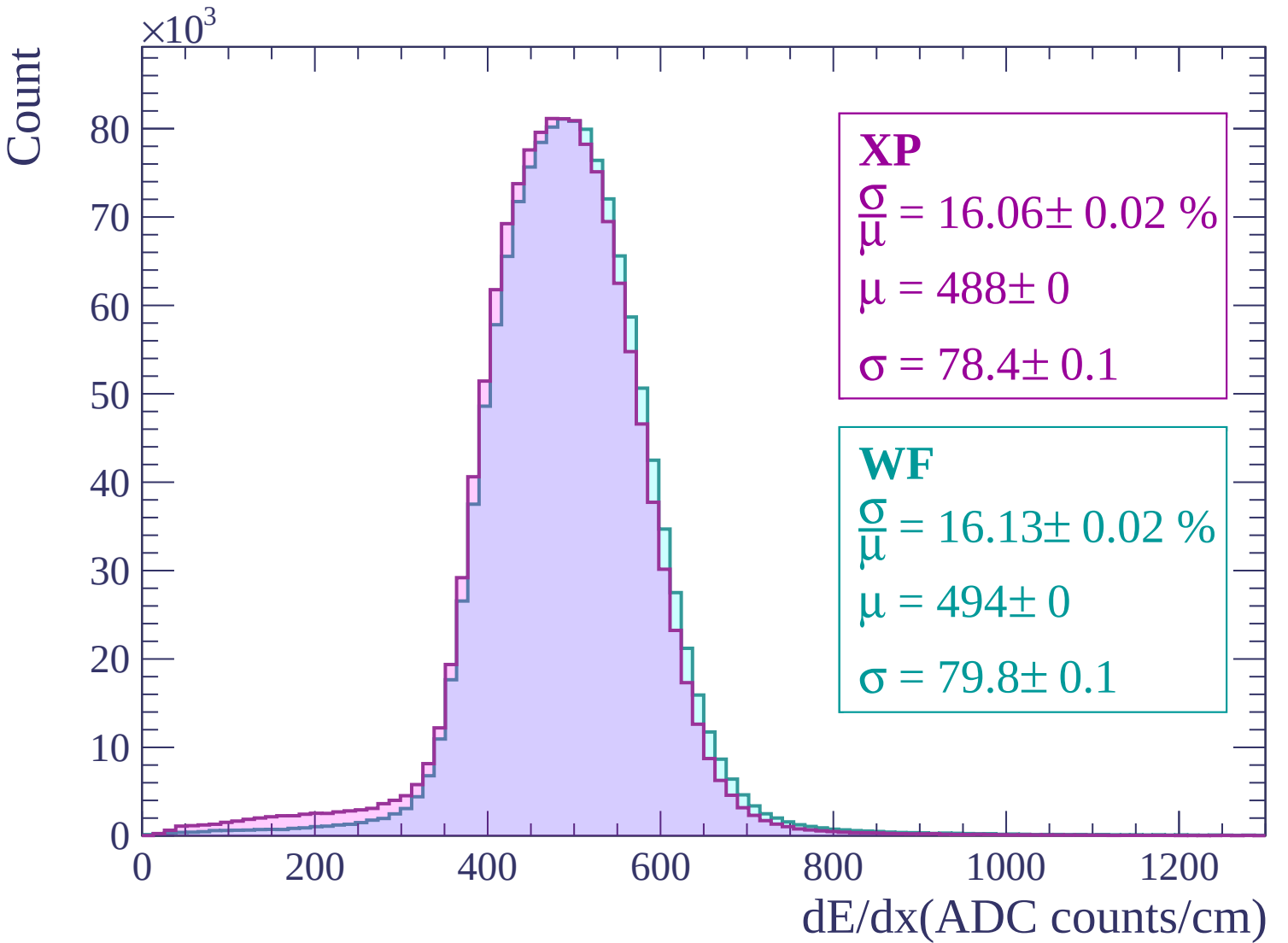
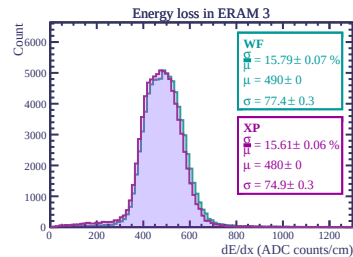
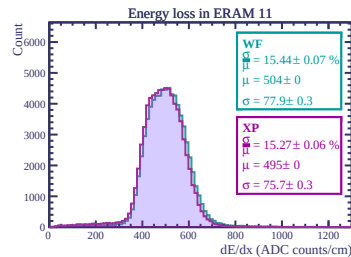
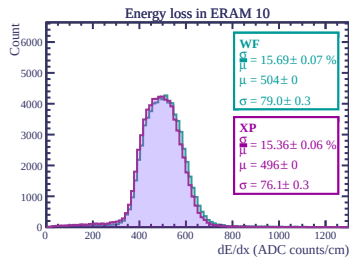
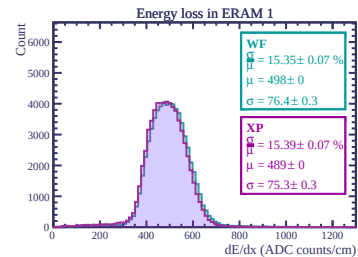
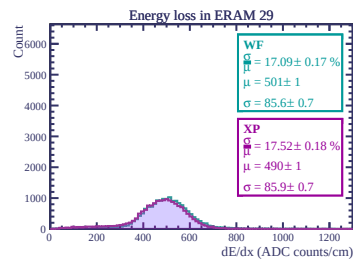
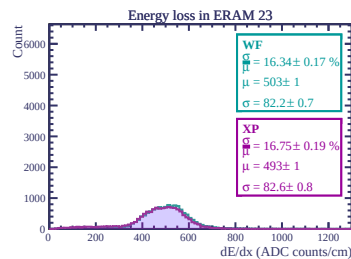
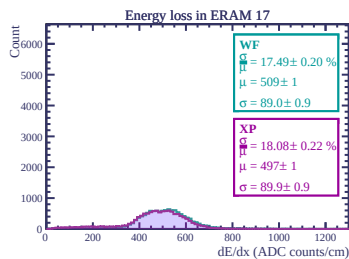
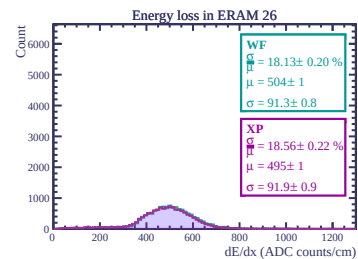
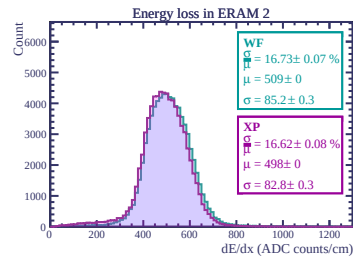
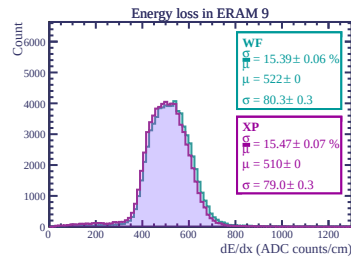
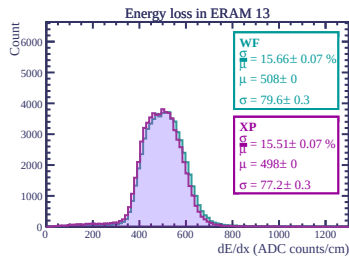
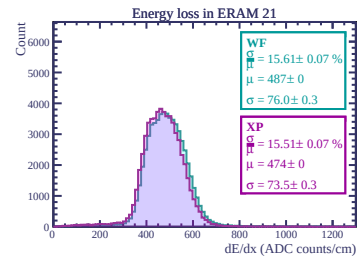
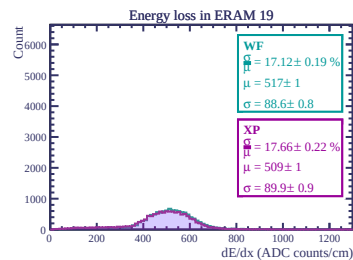
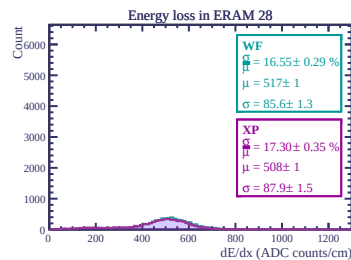
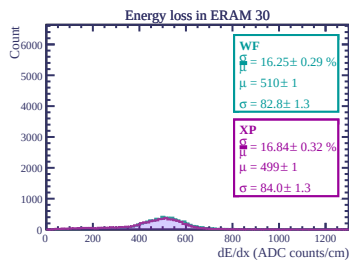
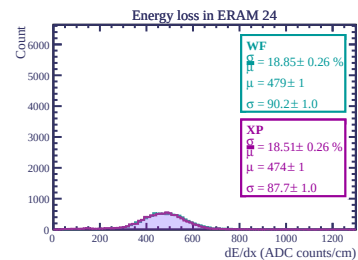
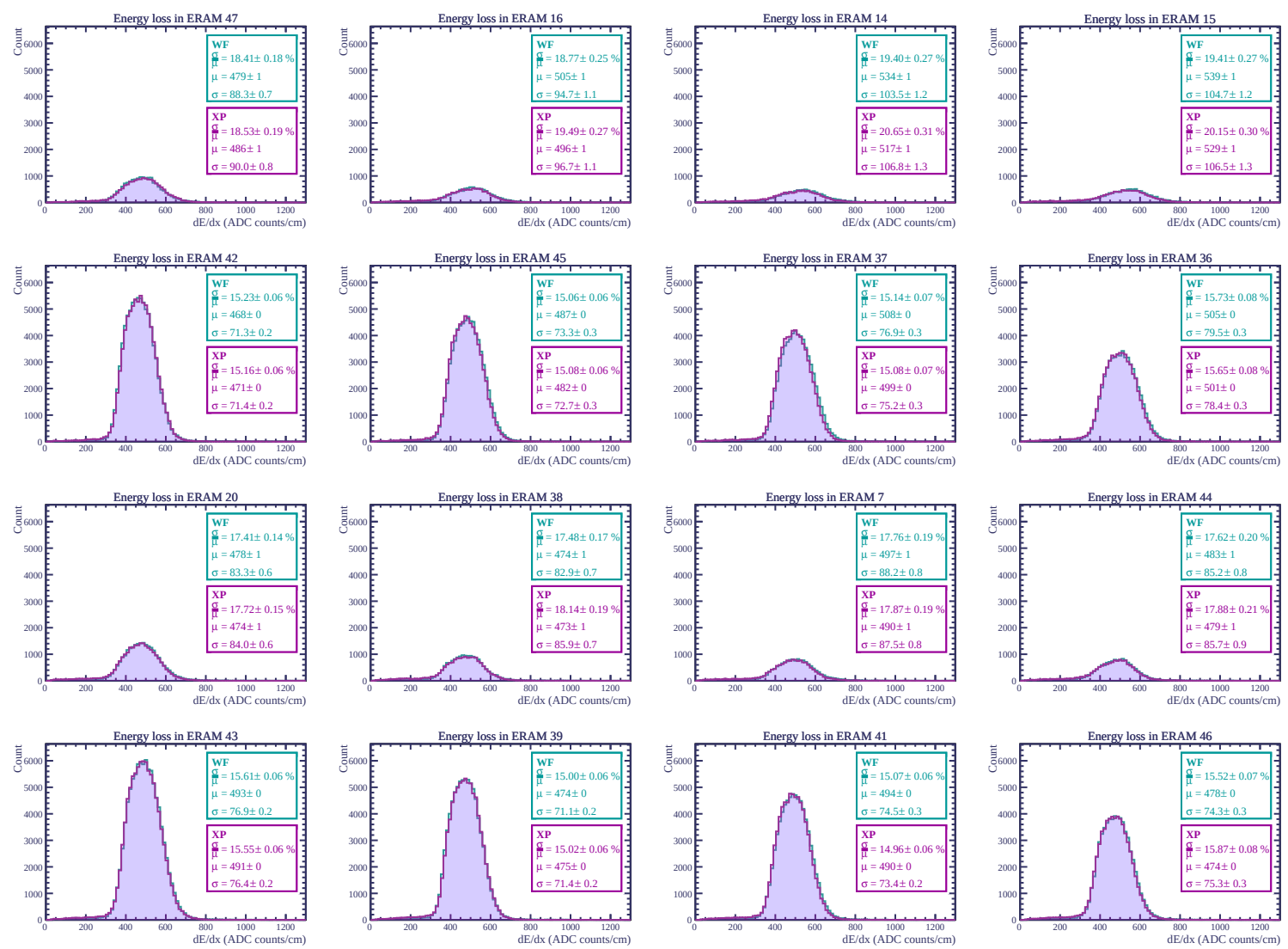
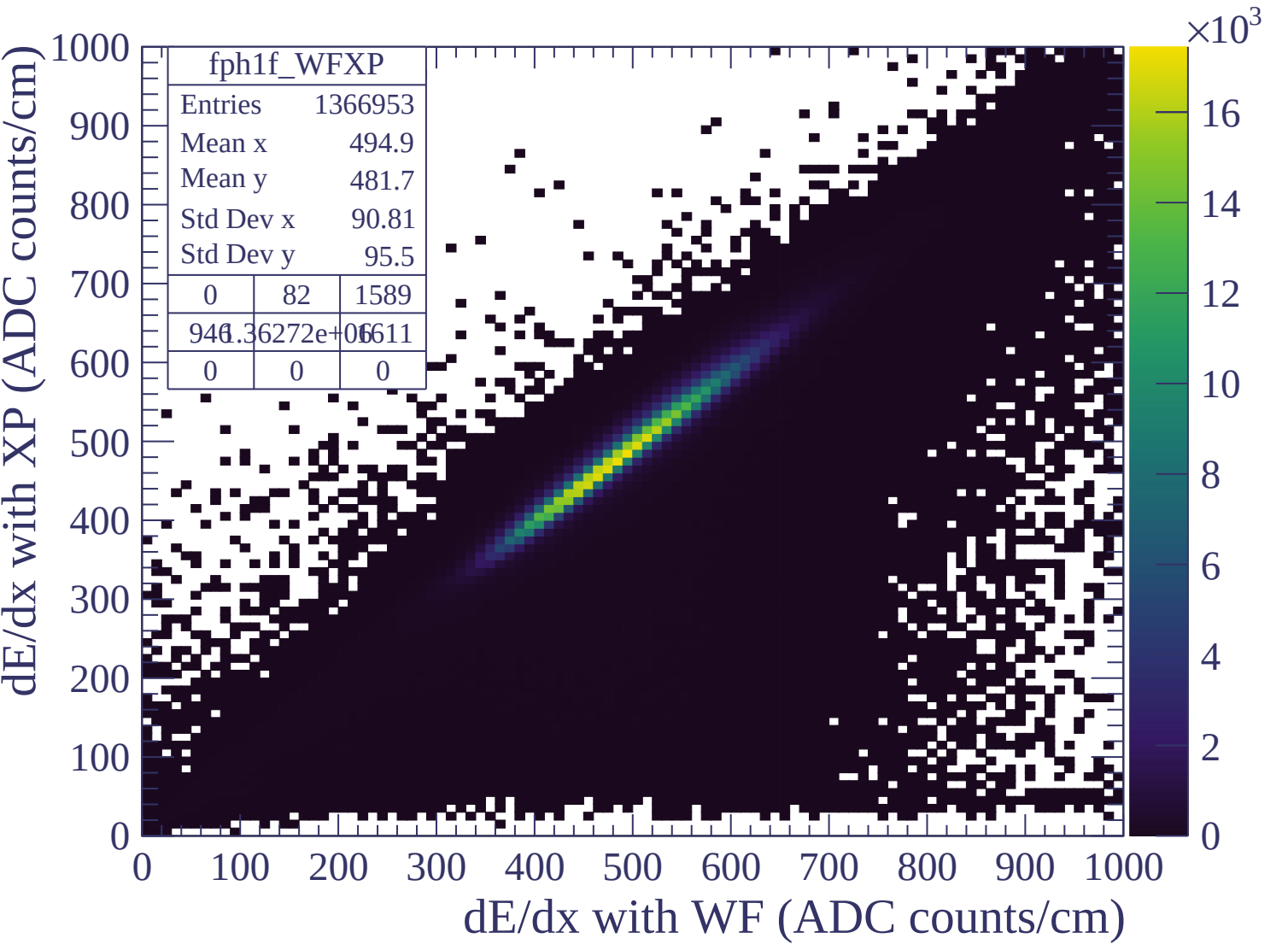
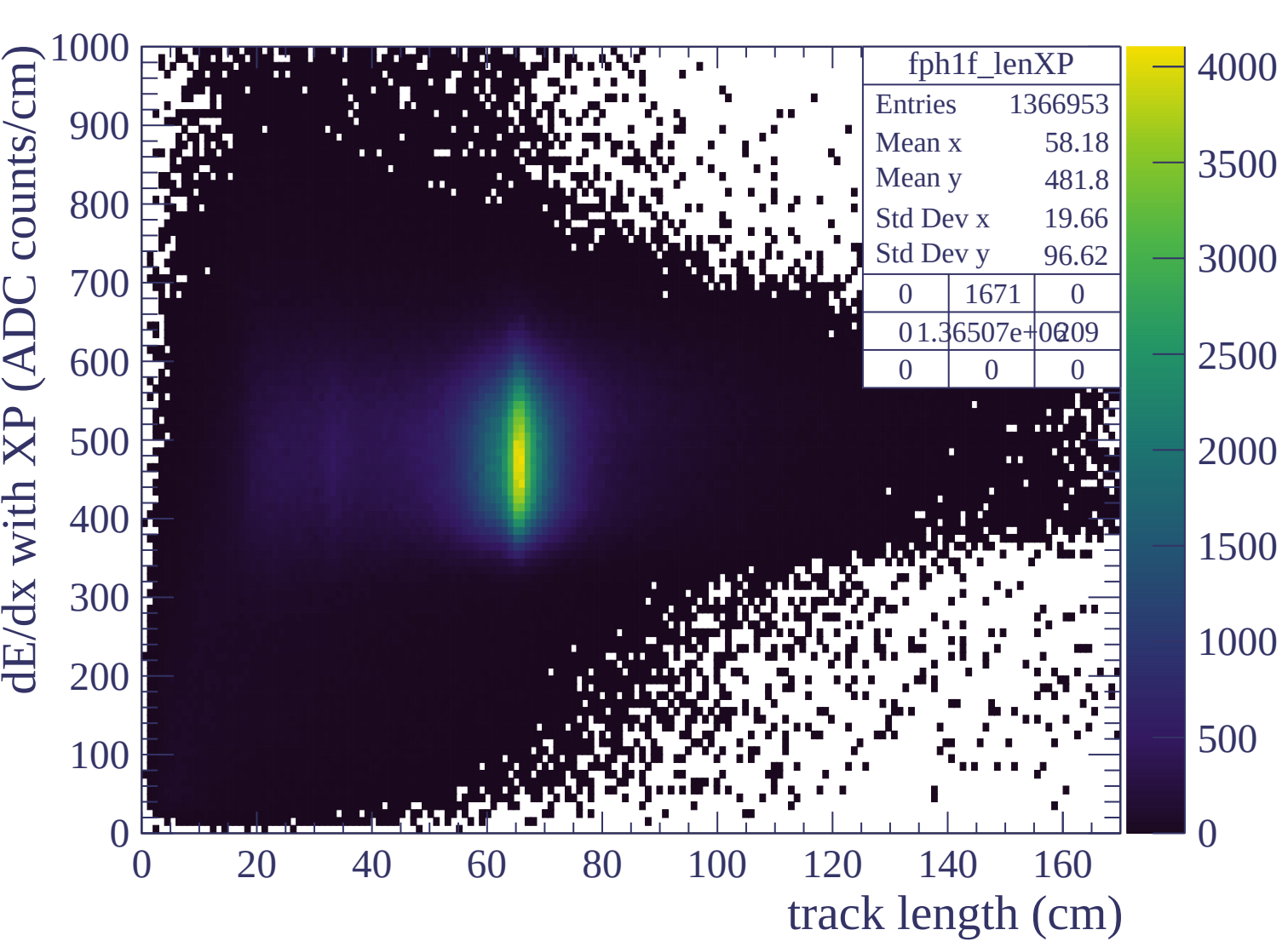


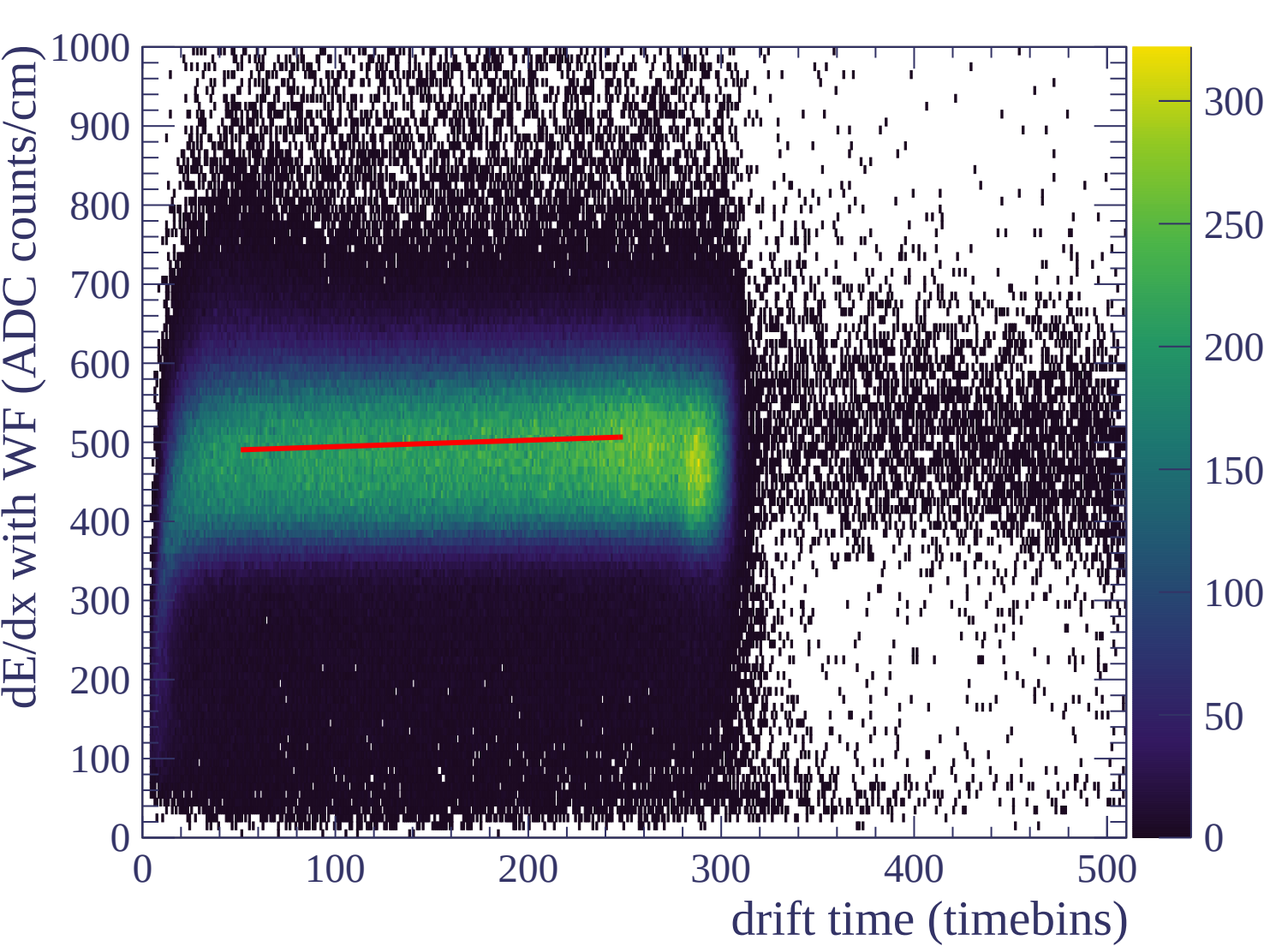


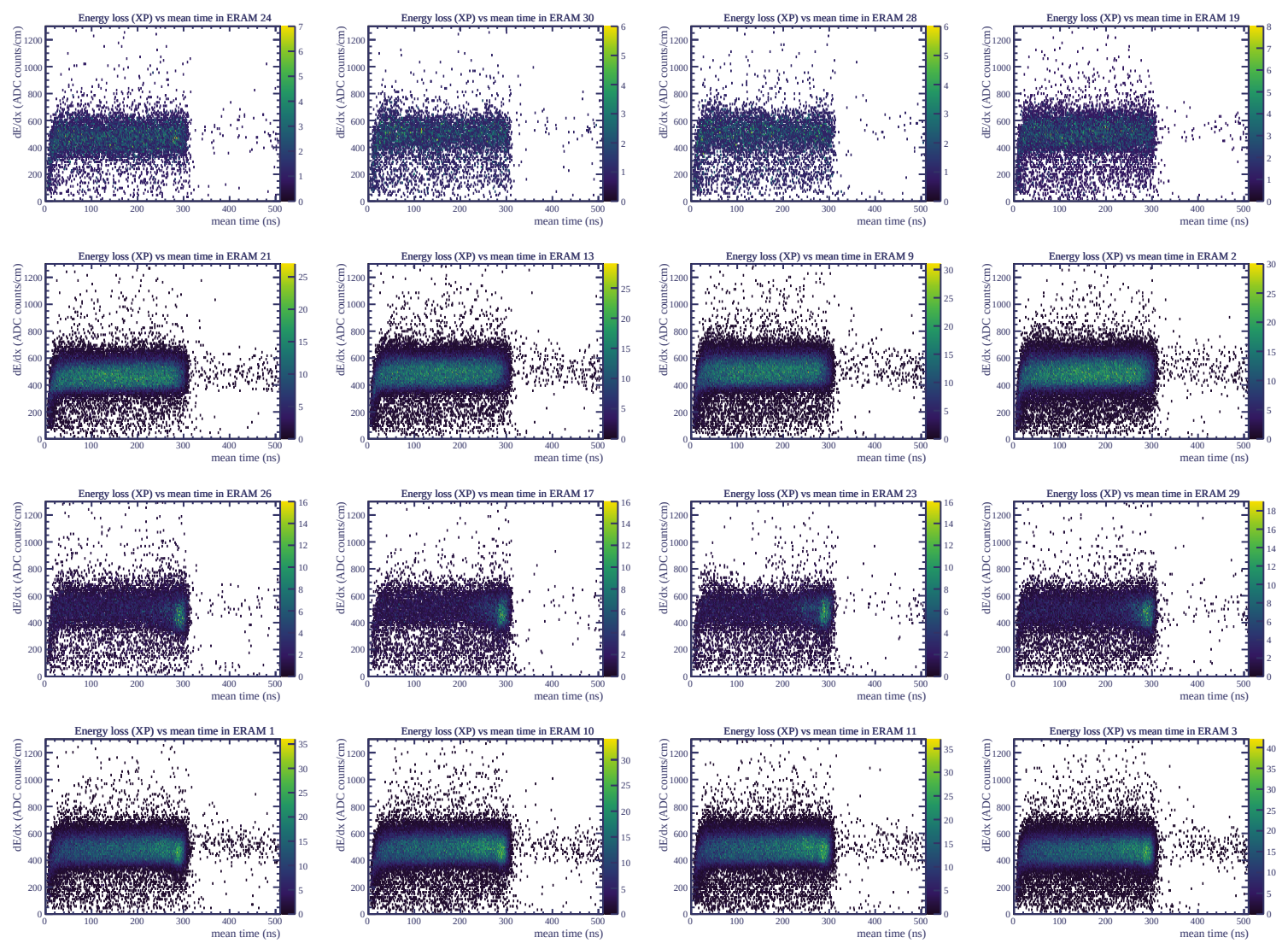


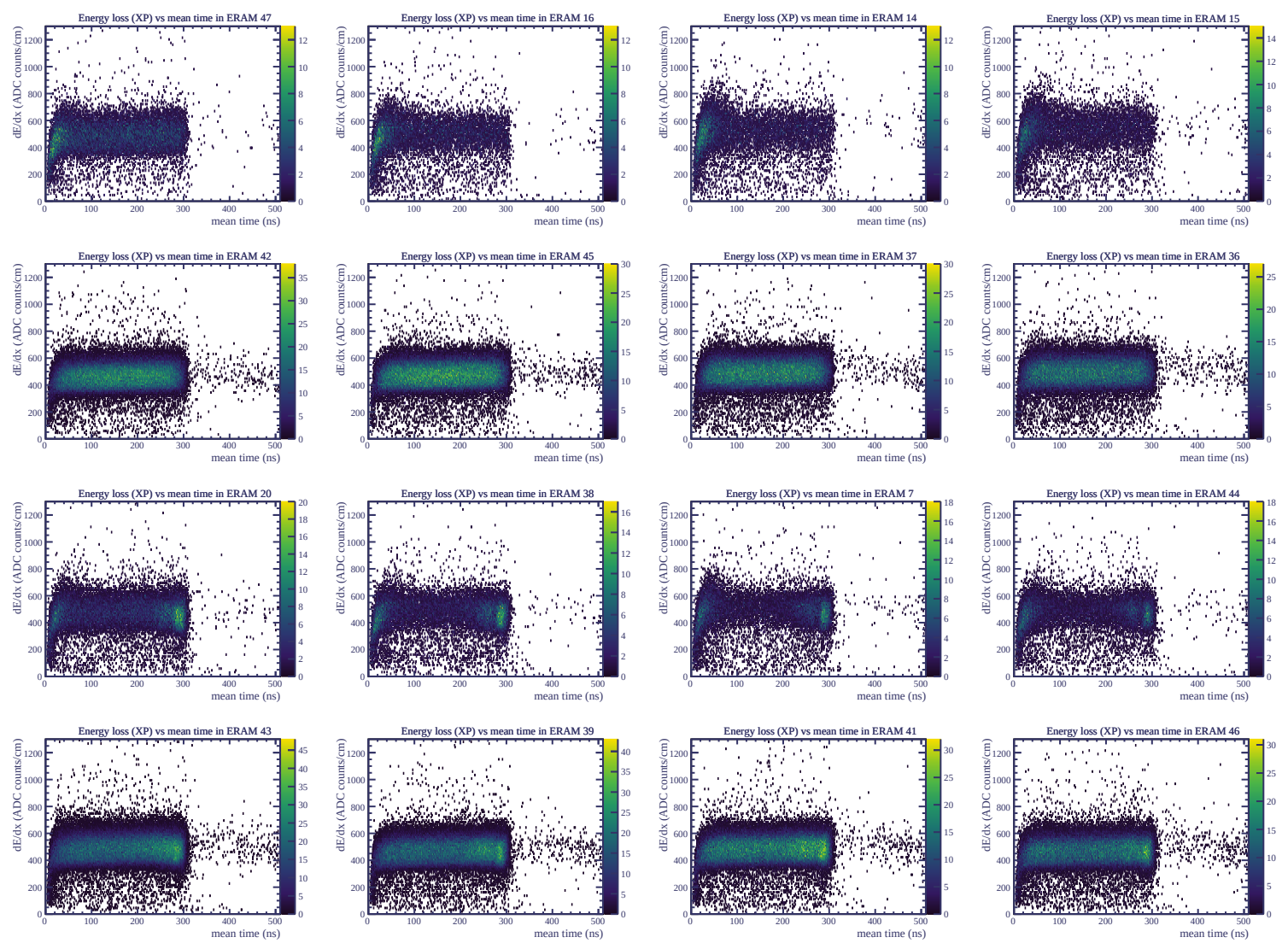




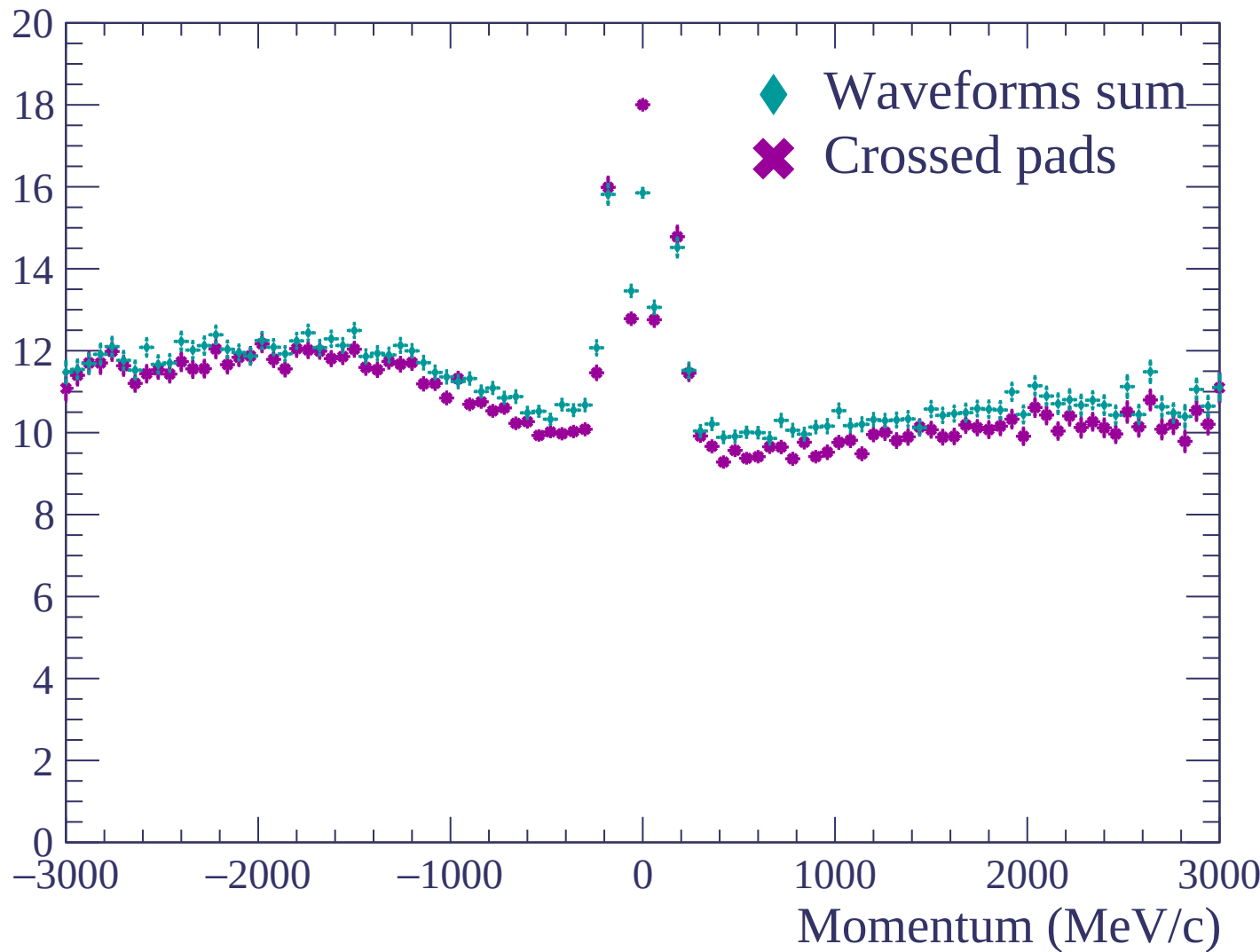


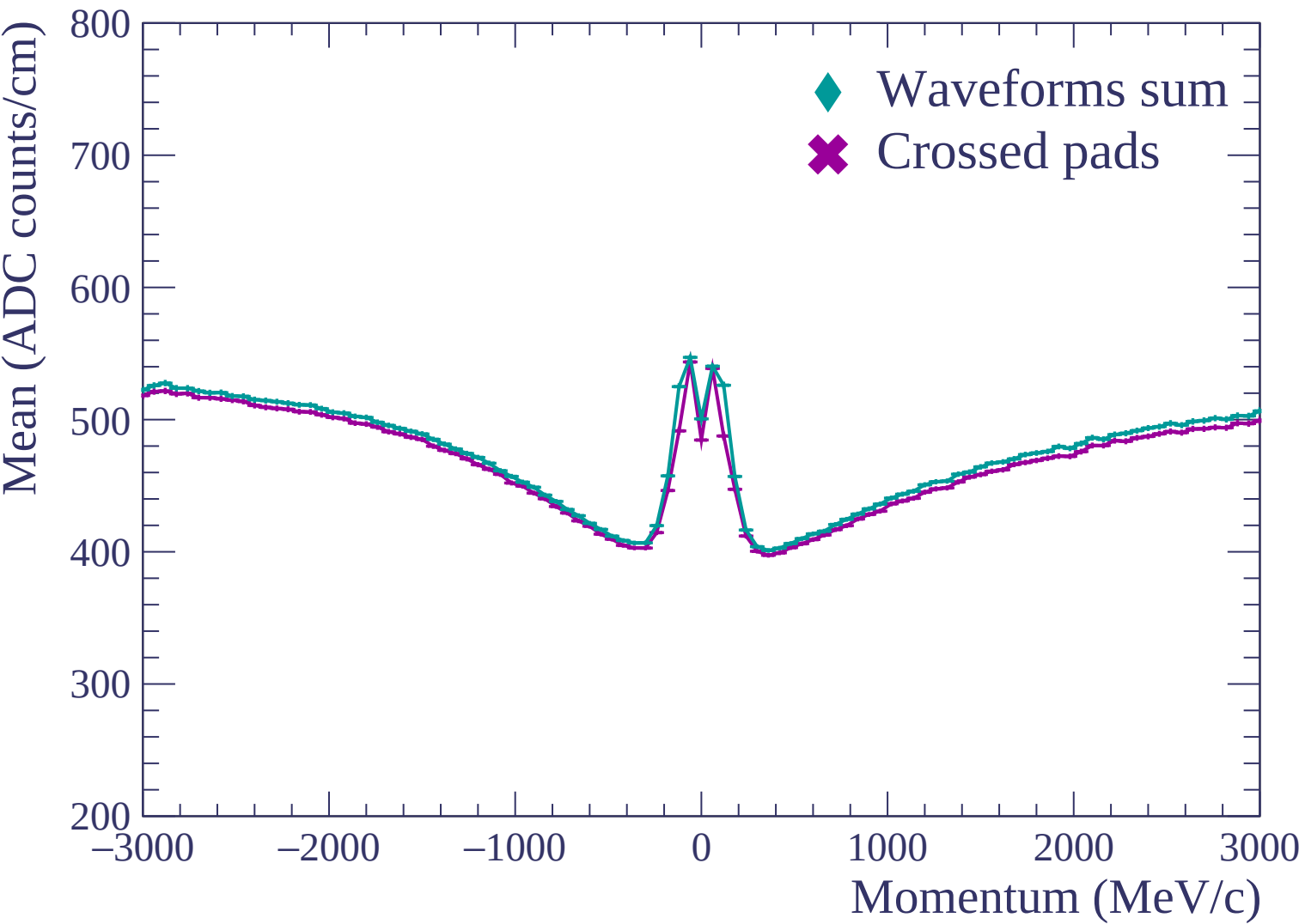


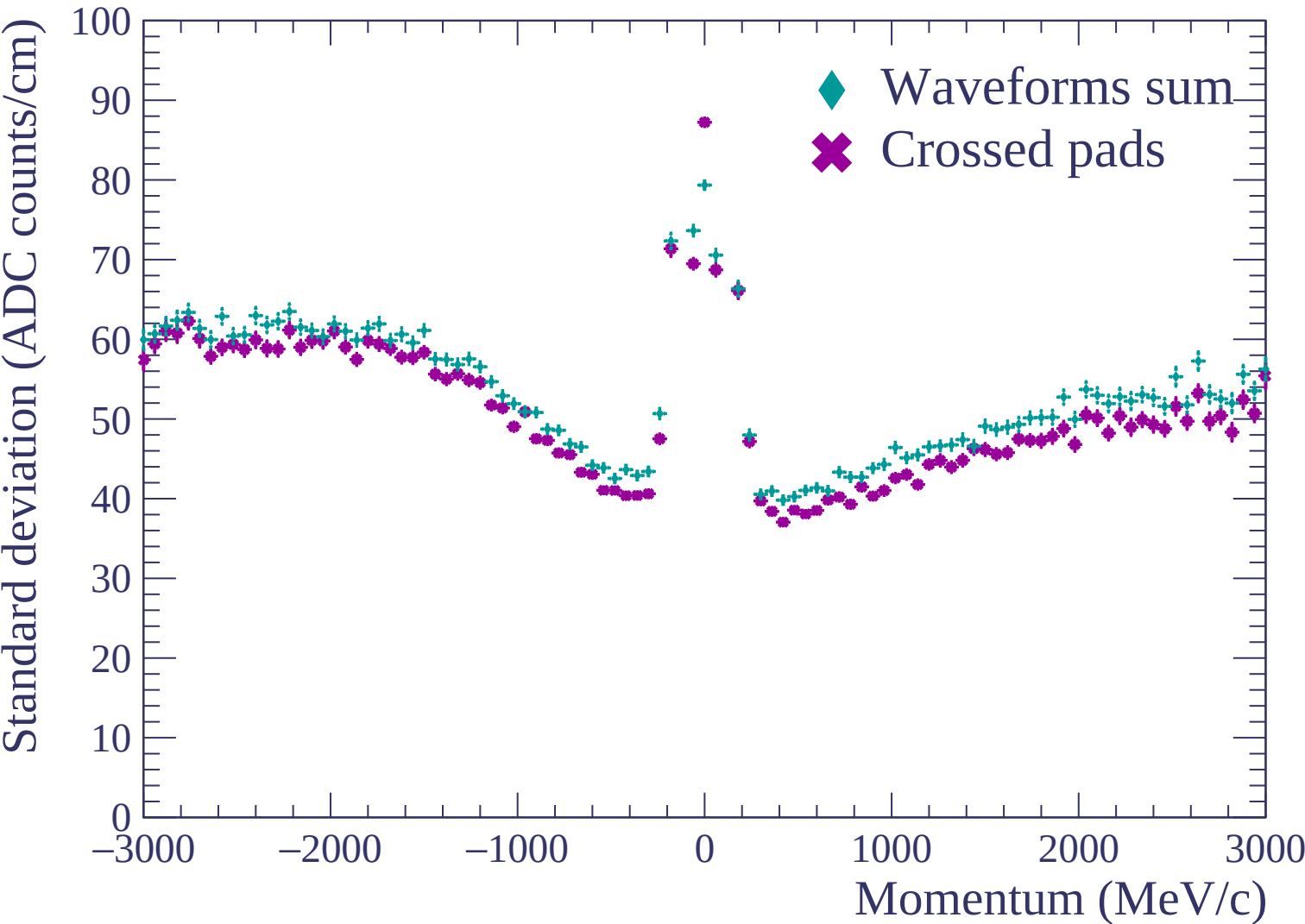


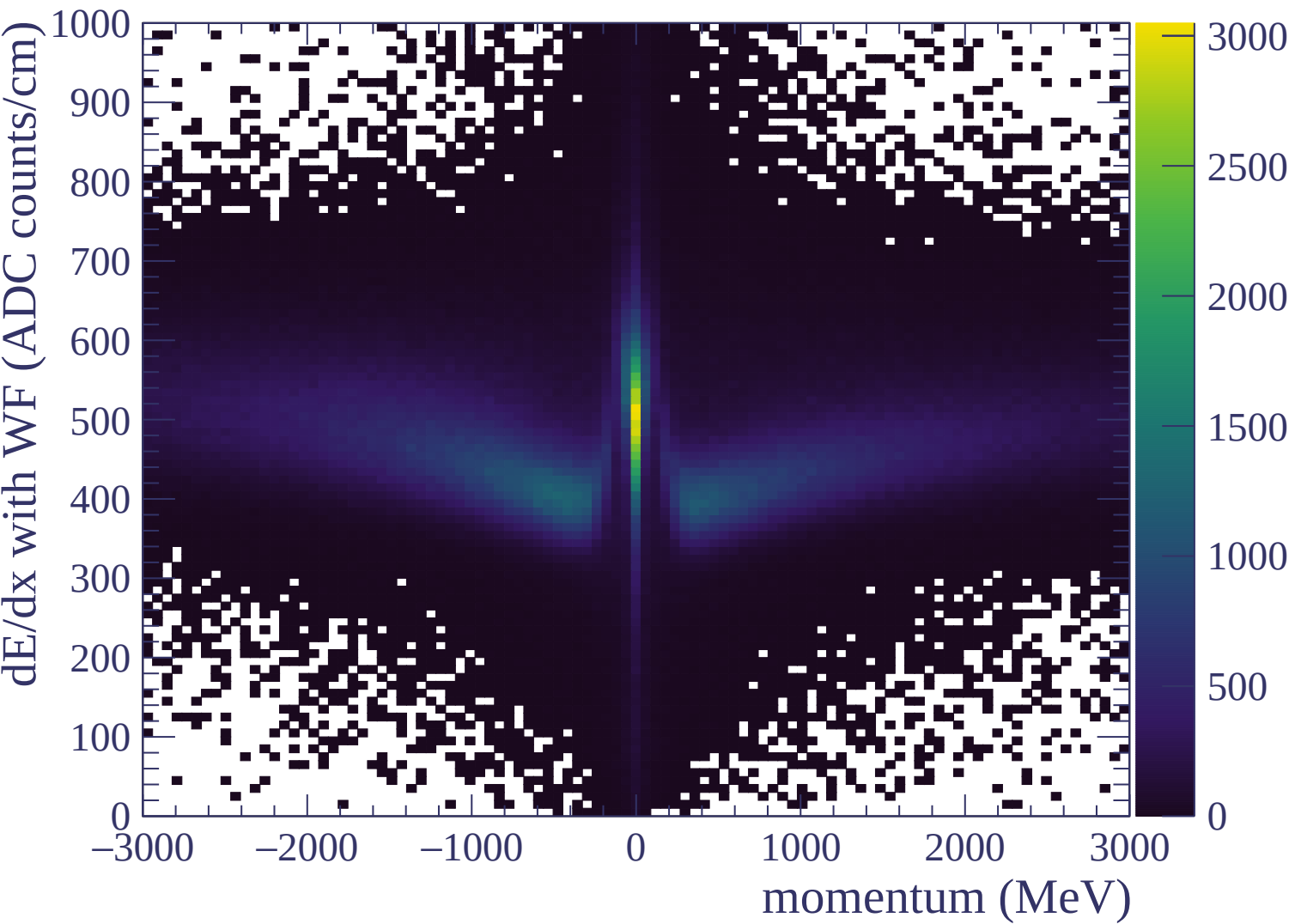


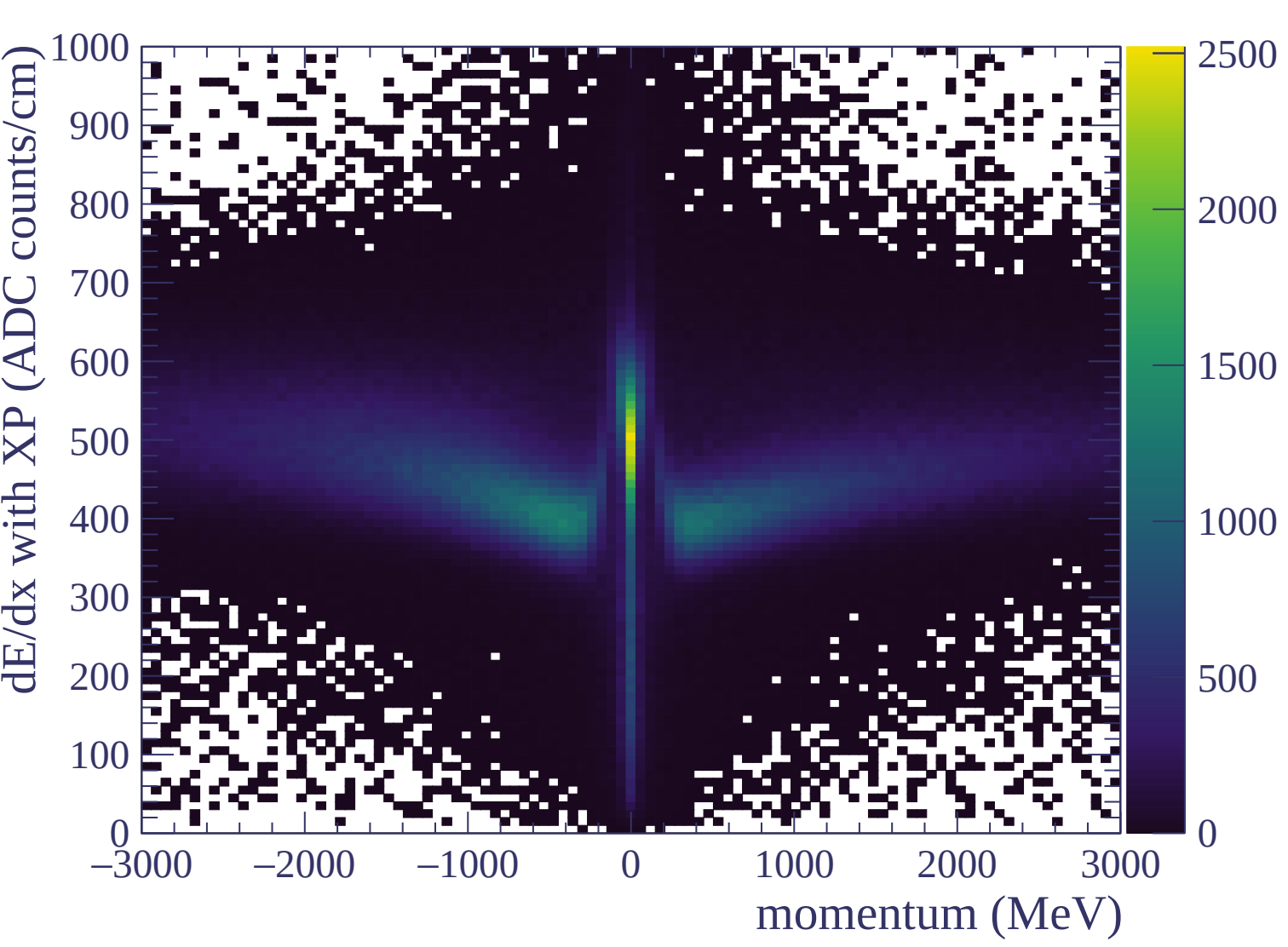
Resolution (%)

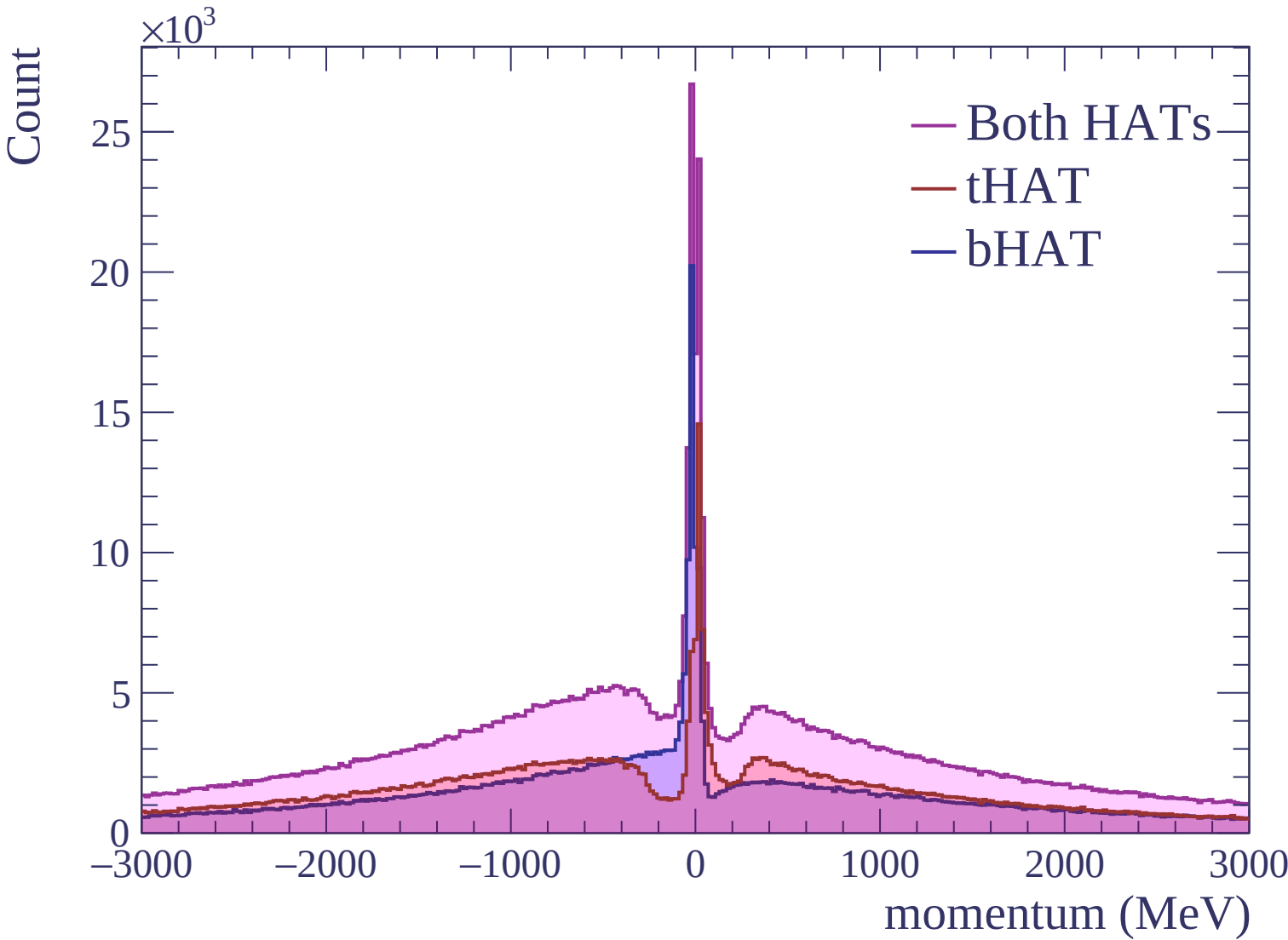






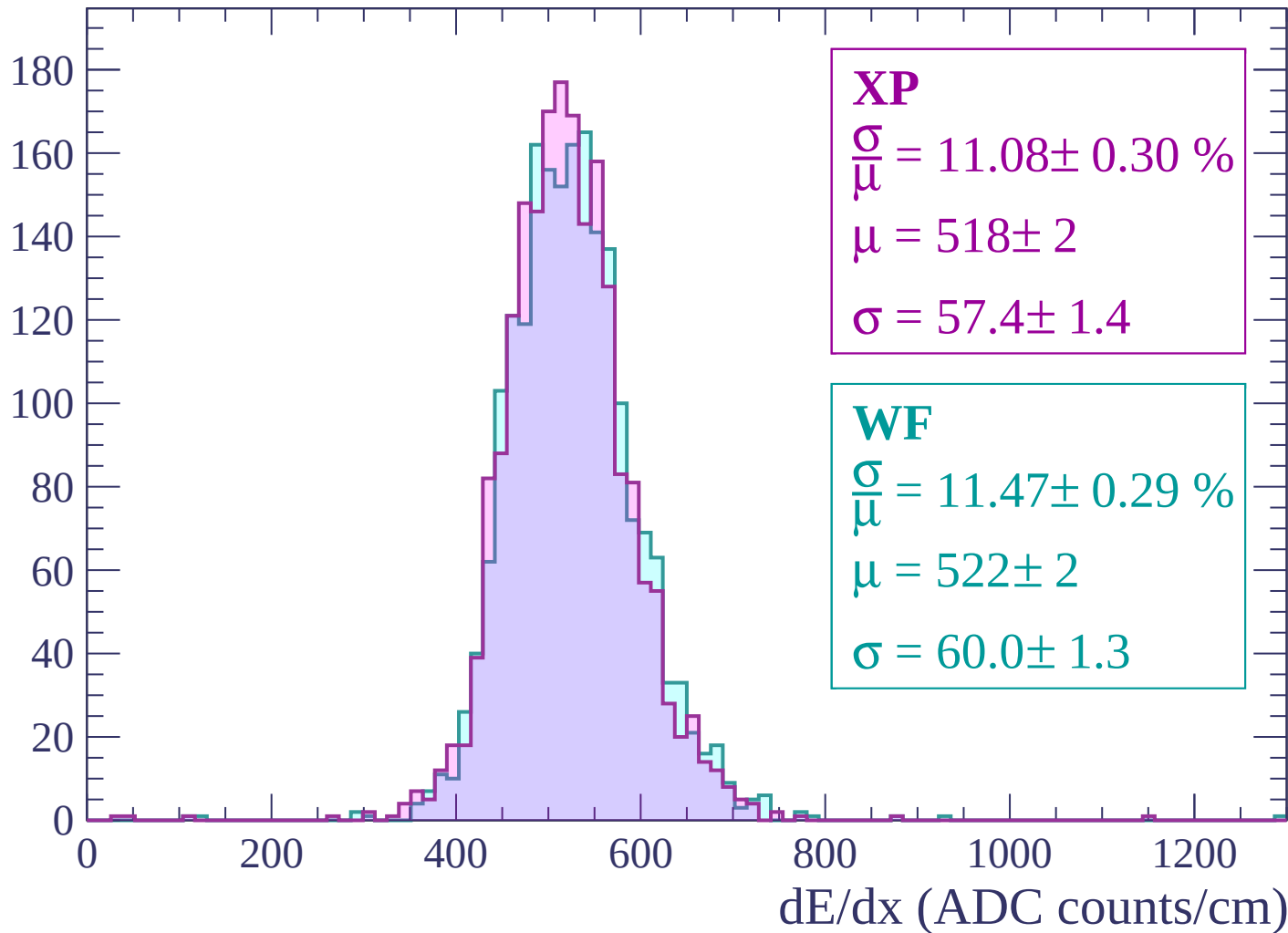






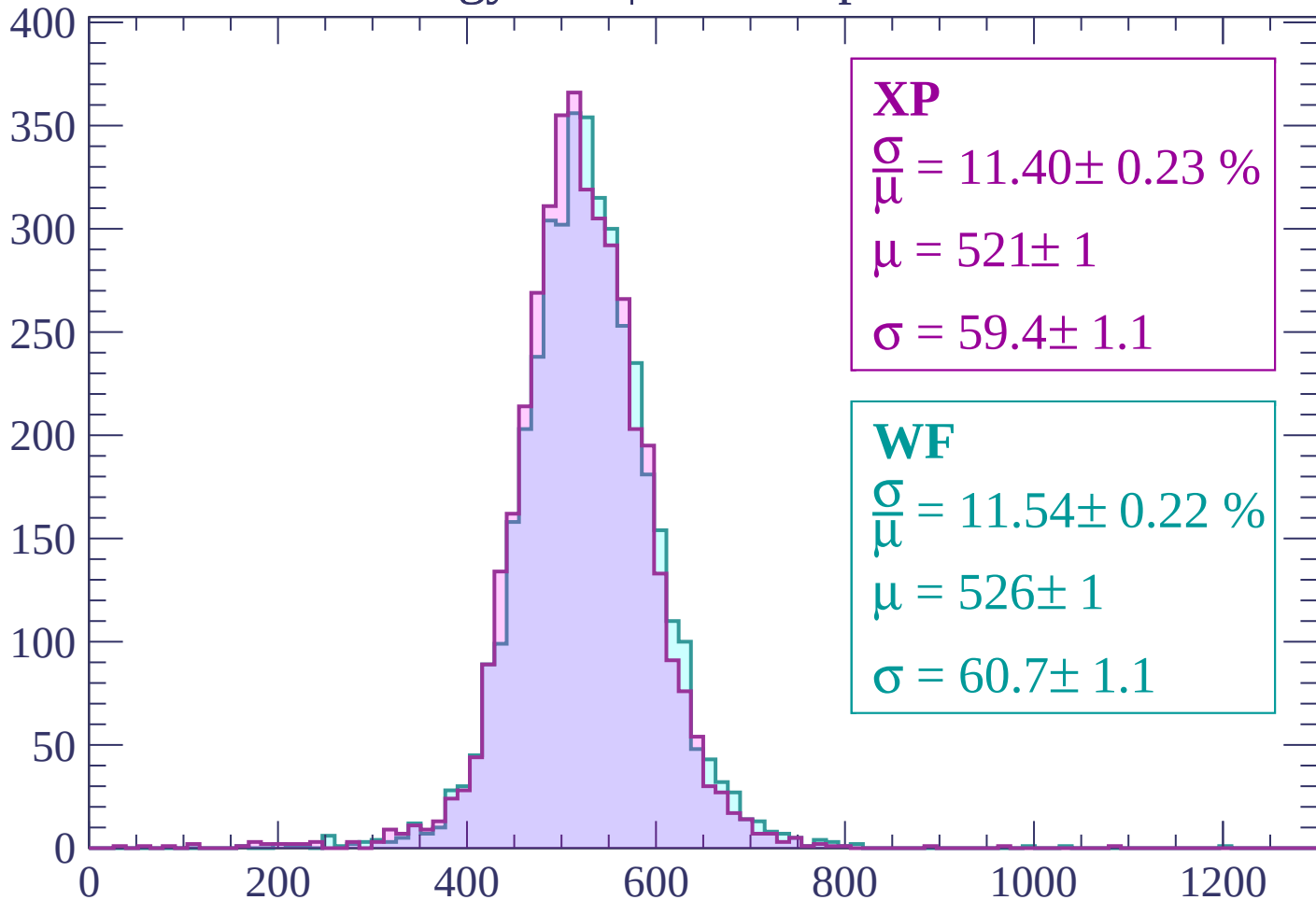
# Energy loss | $-3000 < p < -2940$

Count



Energy loss |  $-2940 < p < -2880$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.40 \pm 0.23 \%$$

$$\mu = 521 \pm 1$$

$$\sigma = 59.4 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 11.54 \pm 0.22 \%$$

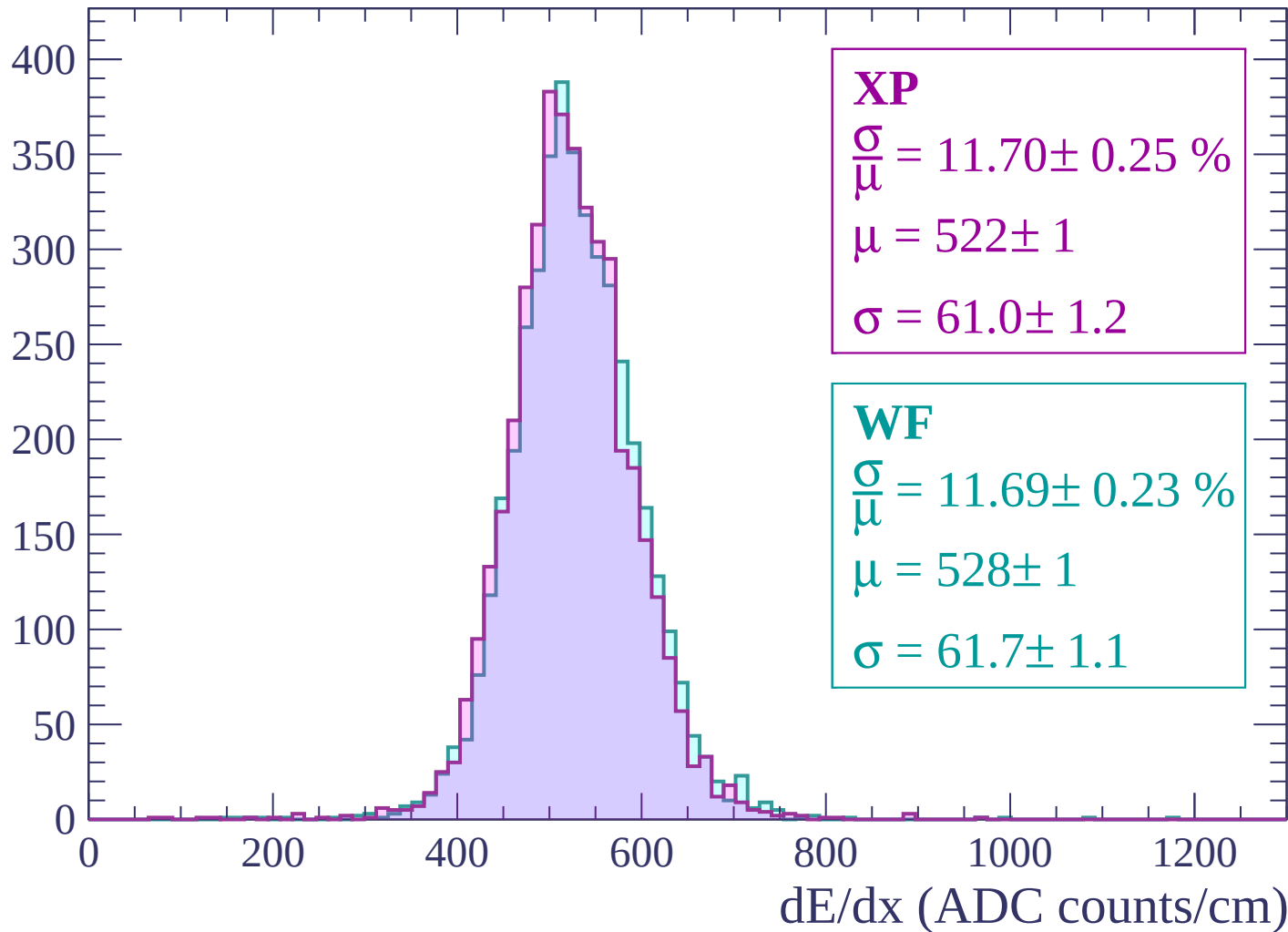
$$\mu = 526 \pm 1$$

$$\sigma = 60.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss |  $-2880 < p < -2820$

Count



# Energy loss | -2820 < p < -2760

Count

400  
350  
300  
250  
200  
150  
100  
50  
0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$$\frac{\sigma}{\mu} = 11.70 \pm 0.25 \%$$

$$\mu = 519 \pm 1$$

$$\sigma = 60.8 \pm 1.2$$

**WF**

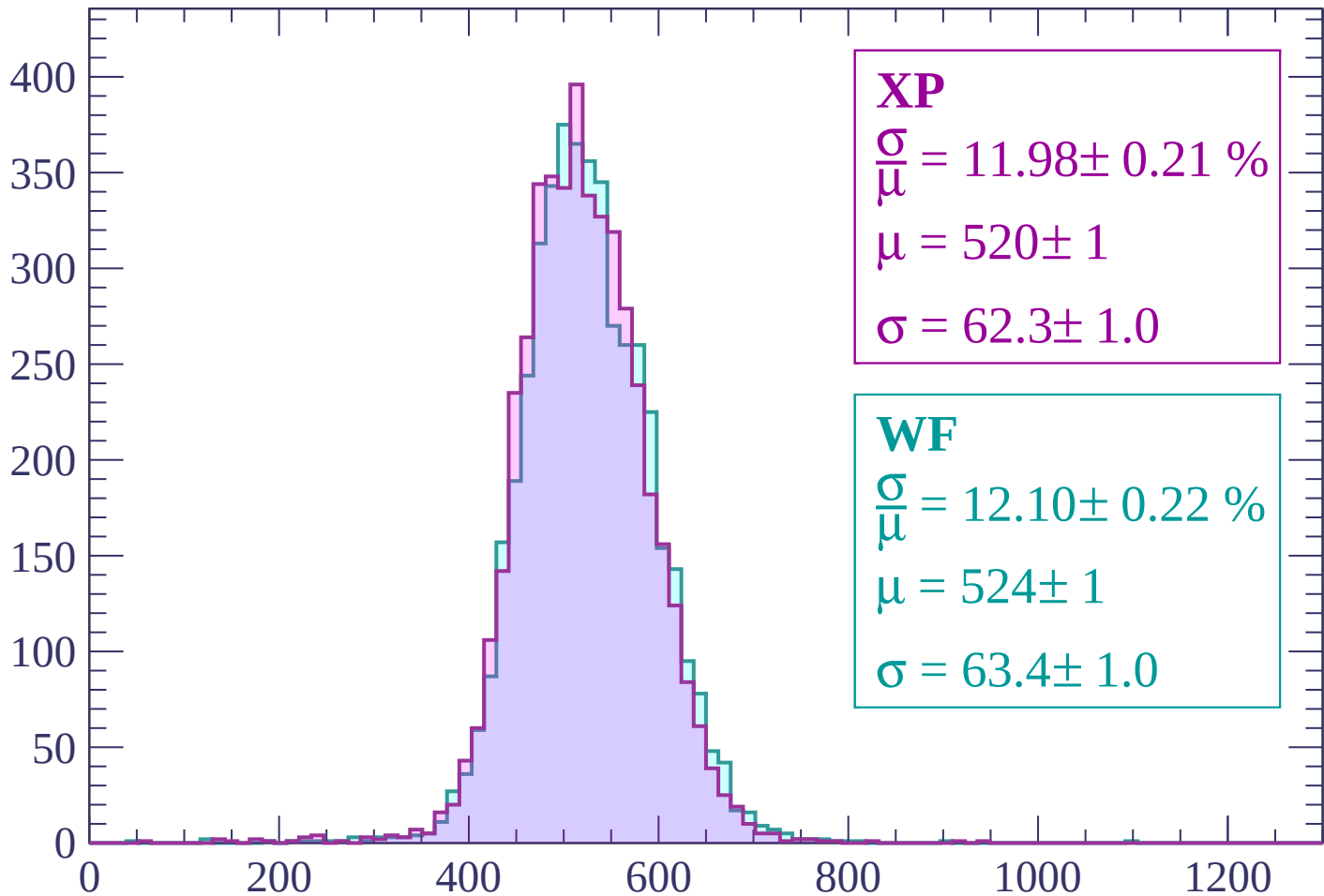
$$\frac{\sigma}{\mu} = 11.91 \pm 0.24 \%$$

$$\mu = 524 \pm 1$$

$$\sigma = 62.4 \pm 1.1$$

# Energy loss | -2760 < p < -2700

Count



**XP**

$$\frac{\sigma}{\mu} = 11.98 \pm 0.21 \%$$

$$\mu = 520 \pm 1$$

$$\sigma = 62.3 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 12.10 \pm 0.22 \%$$

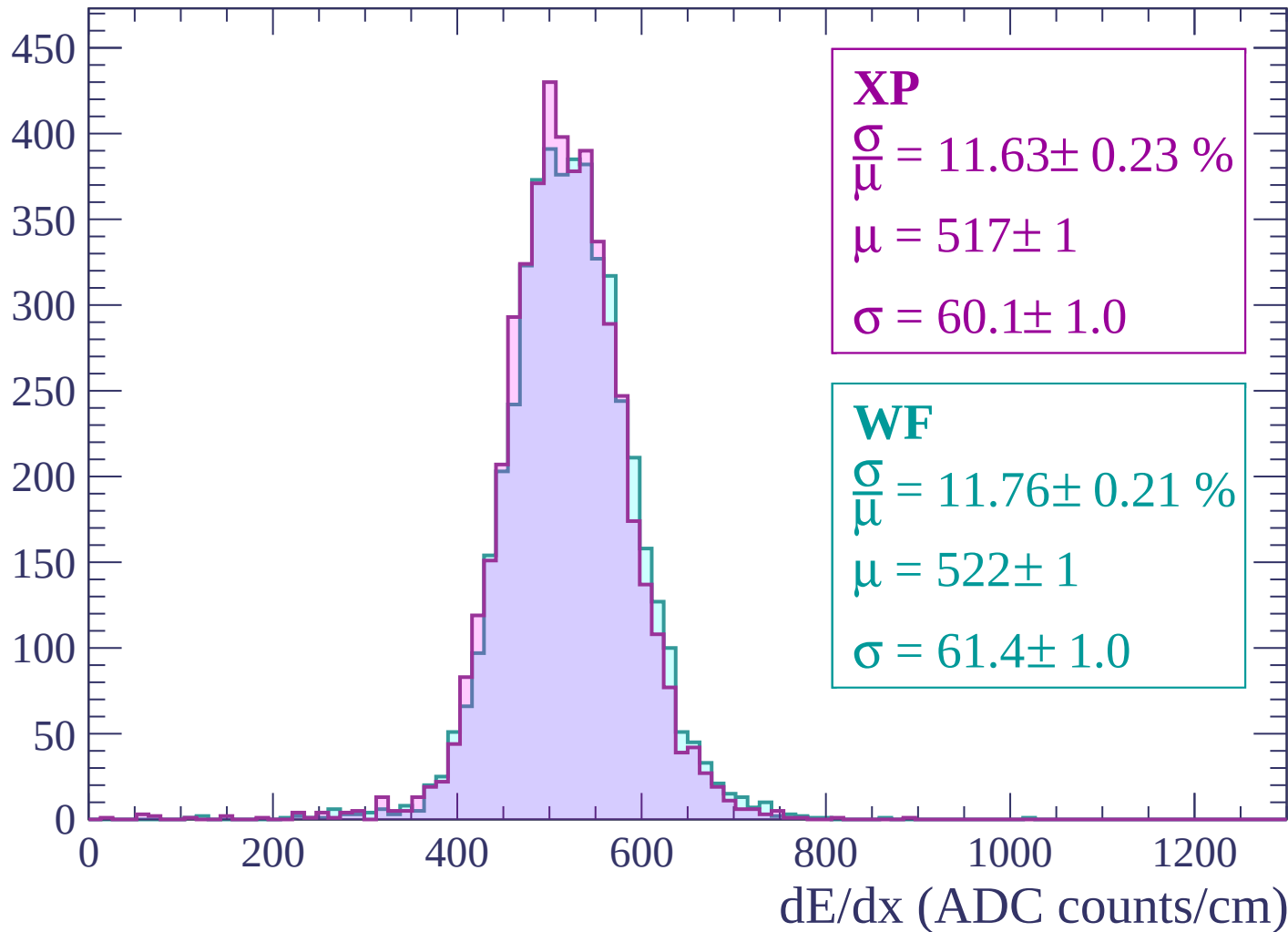
$$\mu = 524 \pm 1$$

$$\sigma = 63.4 \pm 1.0$$

dE/dx (ADC counts/cm)

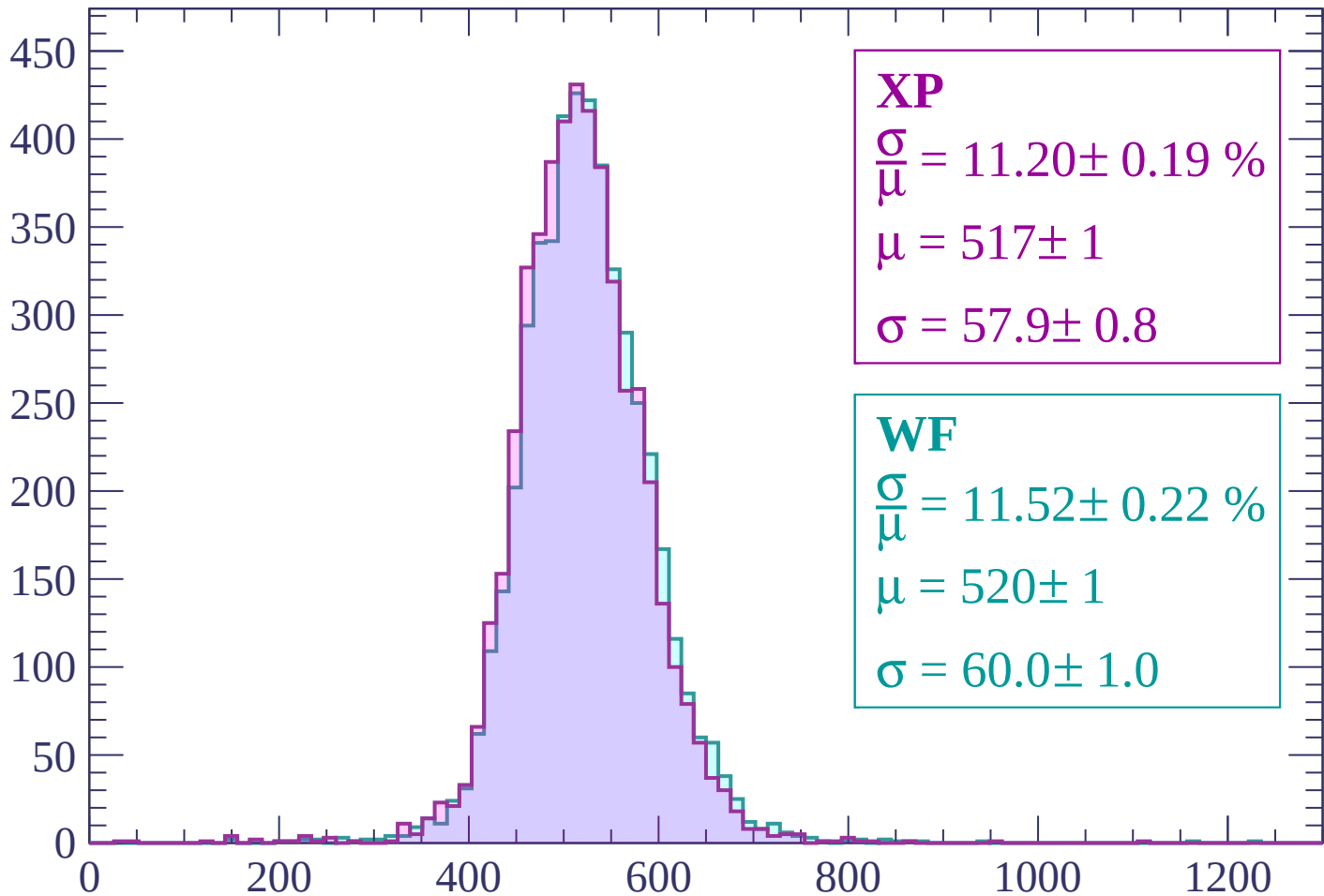
# Energy loss | -2700 < p < -2640

Count



# Energy loss | -2640 < p < -2580

Count



**XP**

$$\frac{\sigma}{\mu} = 11.20 \pm 0.19 \%$$

$$\mu = 517 \pm 1$$

$$\sigma = 57.9 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 11.52 \pm 0.22 \%$$

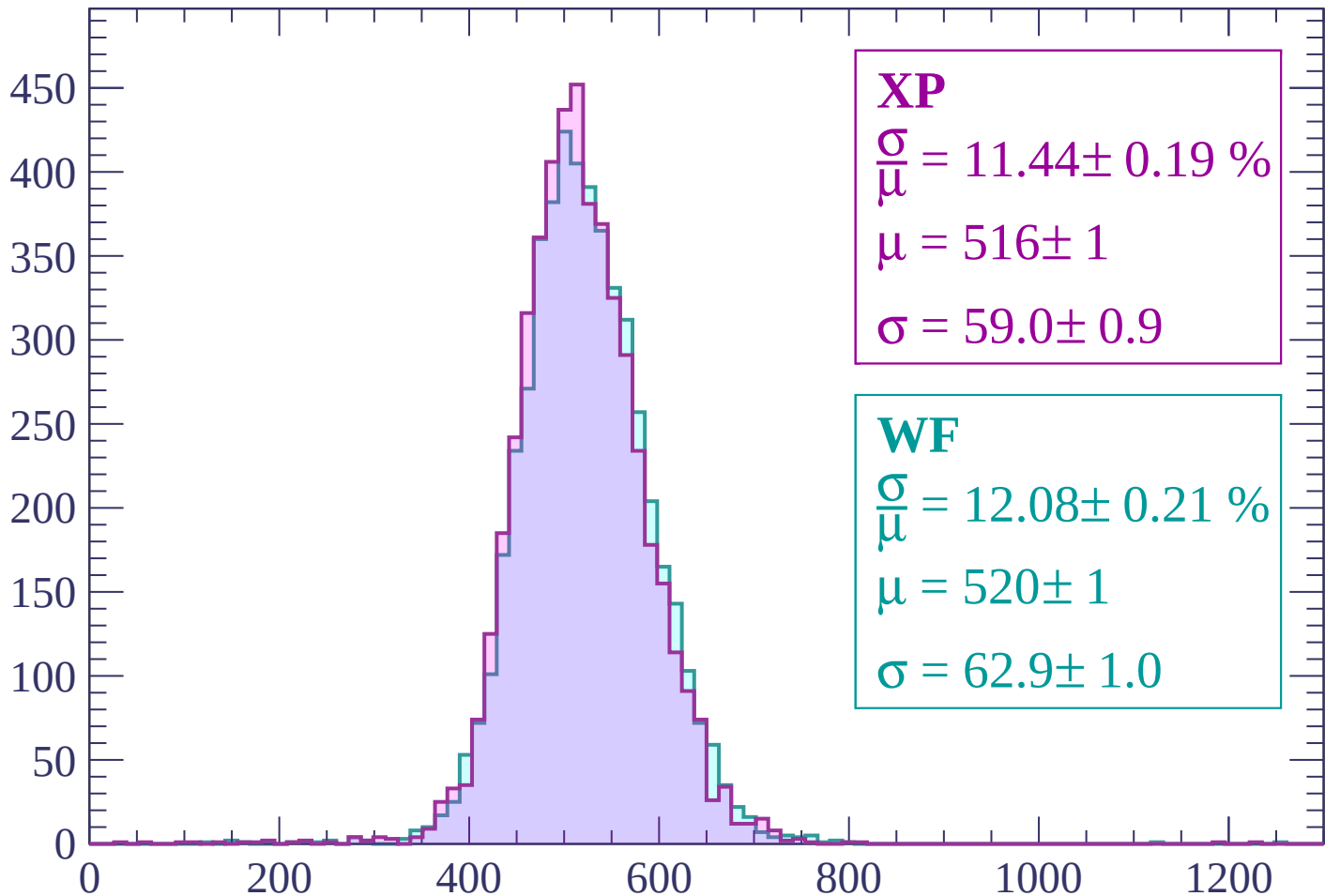
$$\mu = 520 \pm 1$$

$$\sigma = 60.0 \pm 1.0$$

dE/dx (ADC counts/cm)

# Energy loss | -2580 < p < -2520

Count



**XP**

$$\frac{\sigma}{\mu} = 11.44 \pm 0.19 \%$$

$$\mu = 516 \pm 1$$

$$\sigma = 59.0 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 12.08 \pm 0.21 \%$$

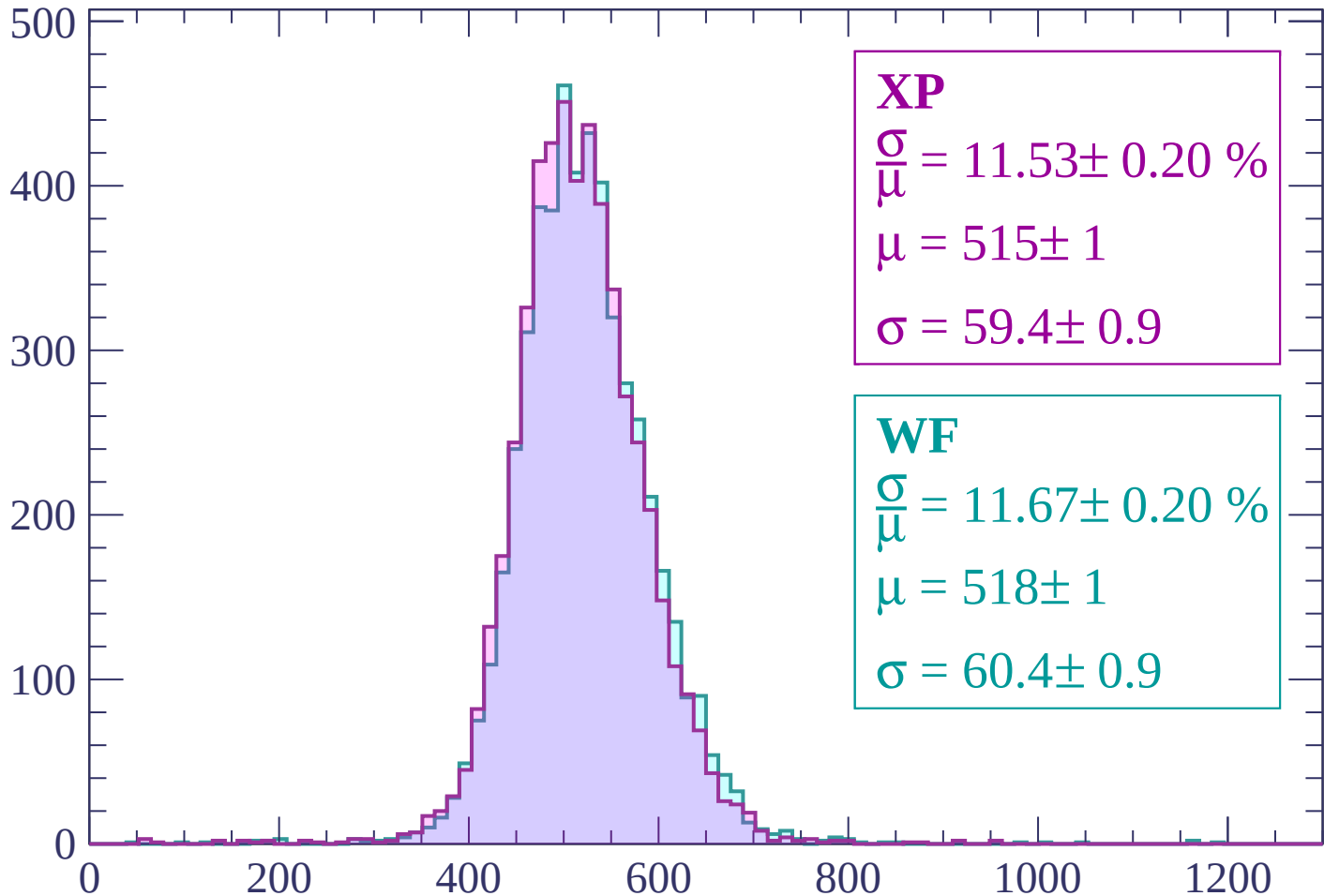
$$\mu = 520 \pm 1$$

$$\sigma = 62.9 \pm 1.0$$

dE/dx (ADC counts/cm)

# Energy loss | -2520 < p < -2460

Count



**XP**

$$\frac{\sigma}{\mu} = 11.53 \pm 0.20 \%$$

$$\mu = 515 \pm 1$$

$$\sigma = 59.4 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 11.67 \pm 0.20 \%$$

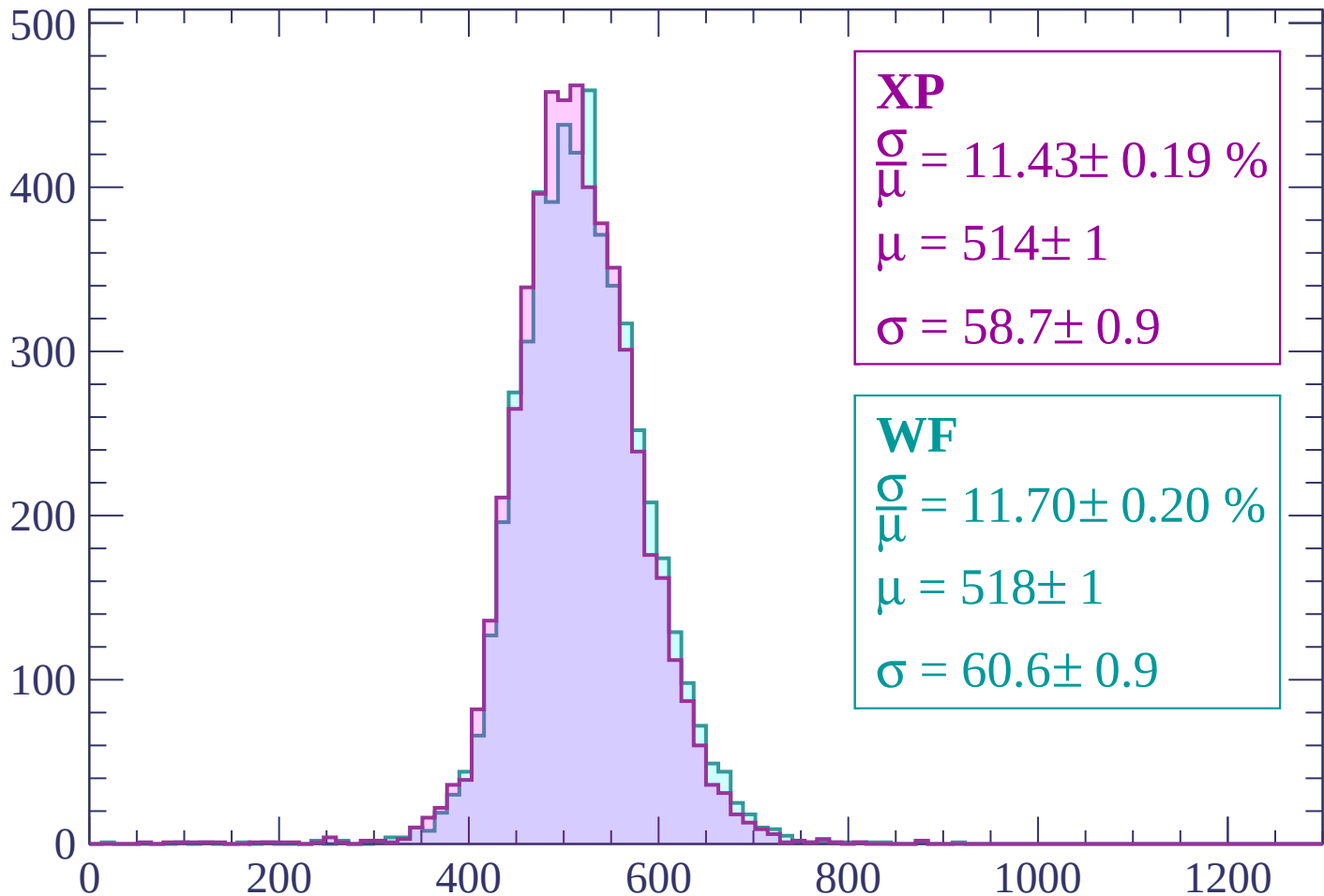
$$\mu = 518 \pm 1$$

$$\sigma = 60.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $-2460 < p < -2400$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.43 \pm 0.19 \%$$

$$\mu = 514 \pm 1$$

$$\sigma = 58.7 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 11.70 \pm 0.20 \%$$

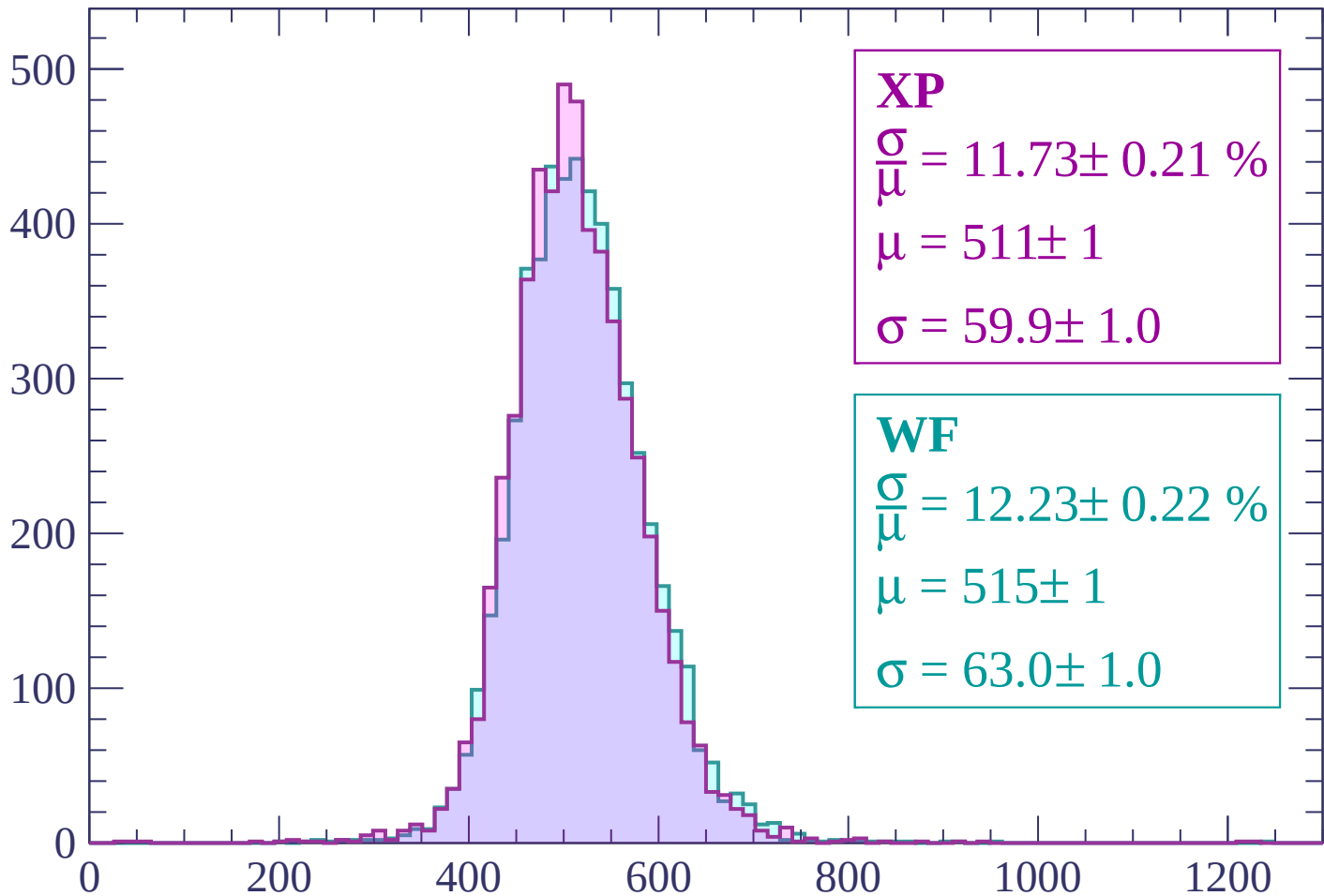
$$\mu = 518 \pm 1$$

$$\sigma = 60.6 \pm 0.9$$

dE/dx (ADC counts/cm)

# Energy loss | -2400 < p < -2340

Count



**XP**

$$\frac{\sigma}{\mu} = 11.73 \pm 0.21 \%$$

$$\mu = 511 \pm 1$$

$$\sigma = 59.9 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 12.23 \pm 0.22 \%$$

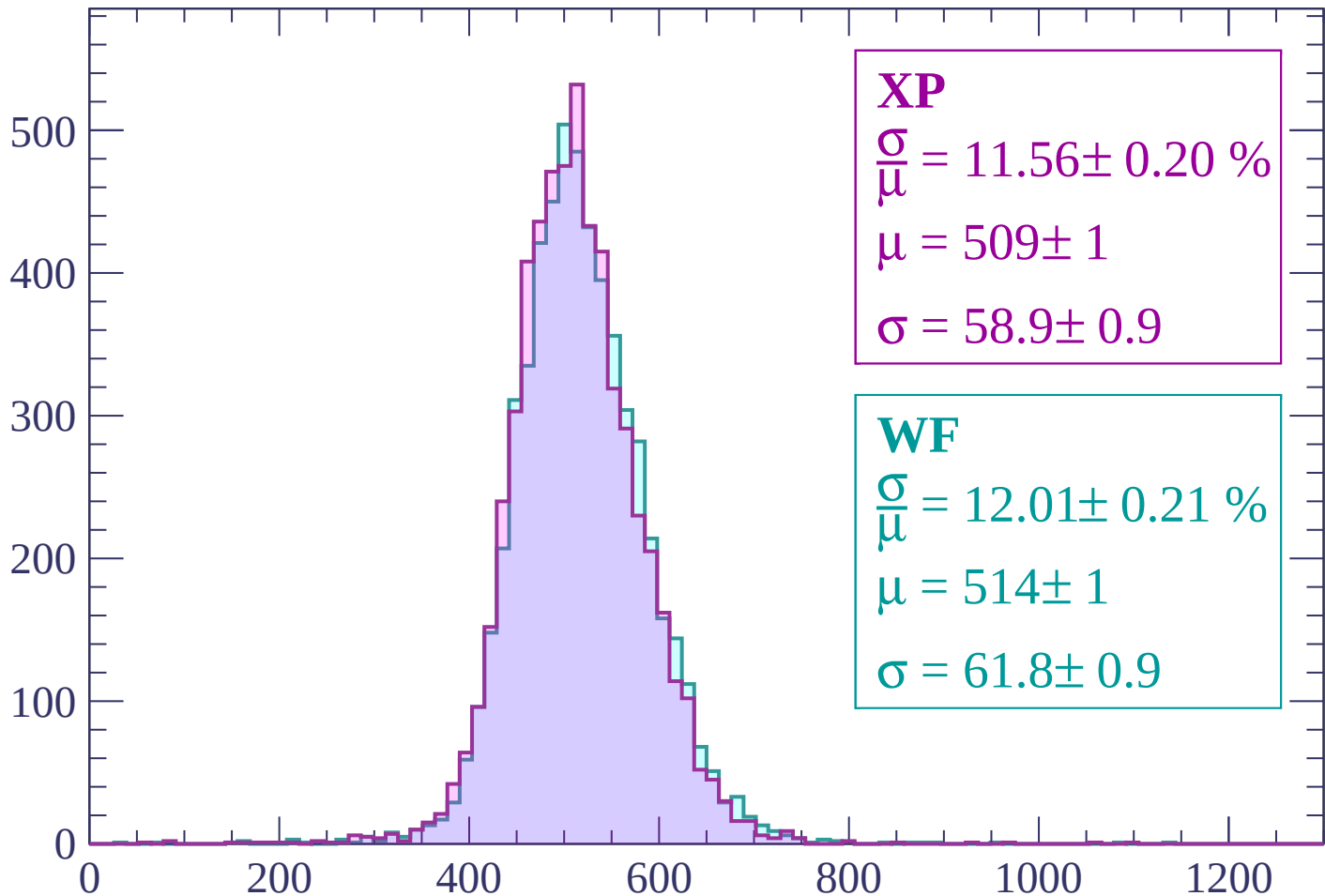
$$\mu = 515 \pm 1$$

$$\sigma = 63.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $-2340 < p < -2280$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.56 \pm 0.20 \%$$

$$\mu = 509 \pm 1$$

$$\sigma = 58.9 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 12.01 \pm 0.21 \%$$

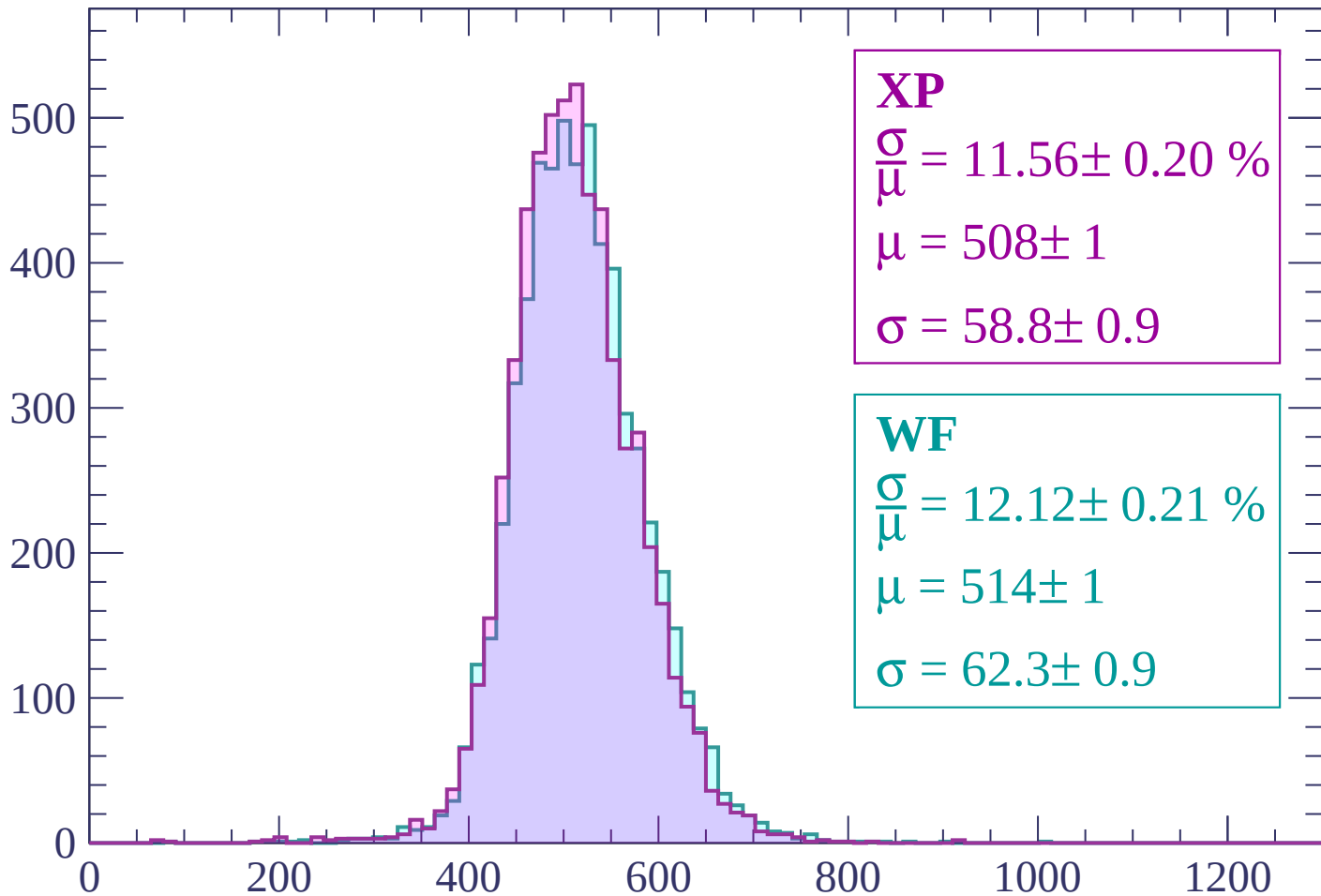
$$\mu = 514 \pm 1$$

$$\sigma = 61.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $-2280 < p < -2220$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.56 \pm 0.20 \%$$

$$\mu = 508 \pm 1$$

$$\sigma = 58.8 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 12.12 \pm 0.21 \%$$

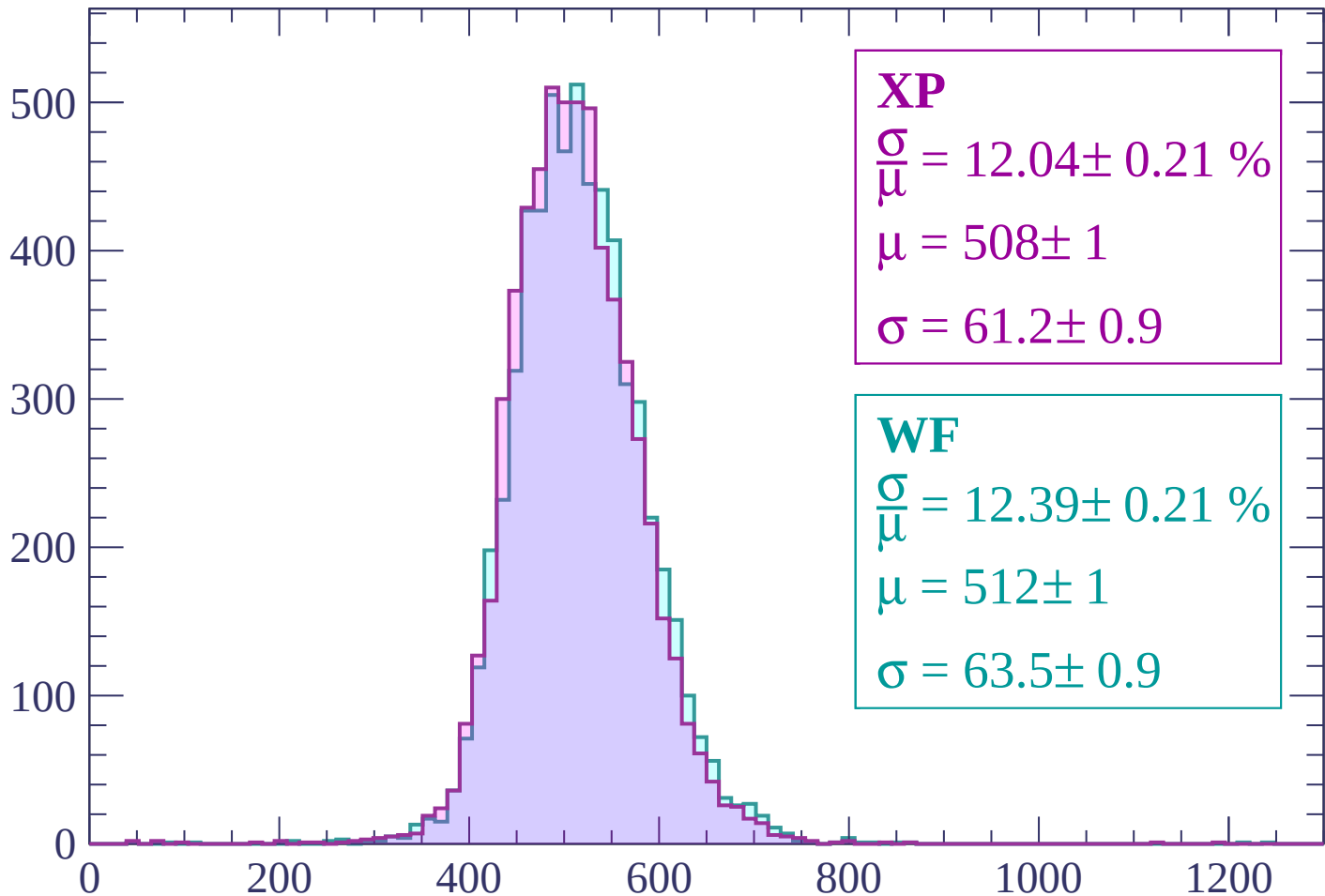
$$\mu = 514 \pm 1$$

$$\sigma = 62.3 \pm 0.9$$

dE/dx (ADC counts/cm)

# Energy loss | -2220 < p < -2160

Count



**XP**

$$\frac{\sigma}{\mu} = 12.04 \pm 0.21 \%$$

$$\mu = 508 \pm 1$$

$$\sigma = 61.2 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 12.39 \pm 0.21 \%$$

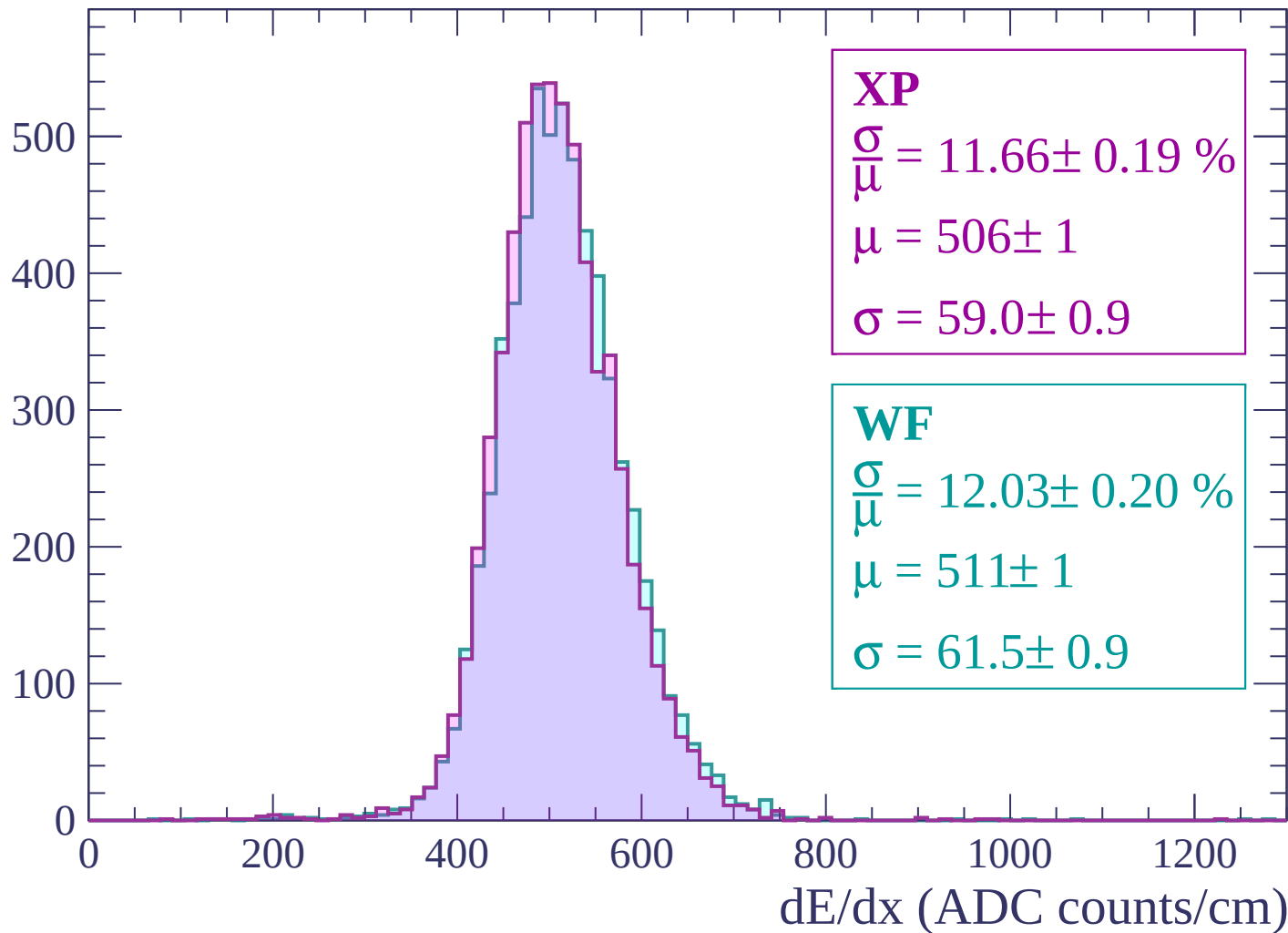
$$\mu = 512 \pm 1$$

$$\sigma = 63.5 \pm 0.9$$

dE/dx (ADC counts/cm)

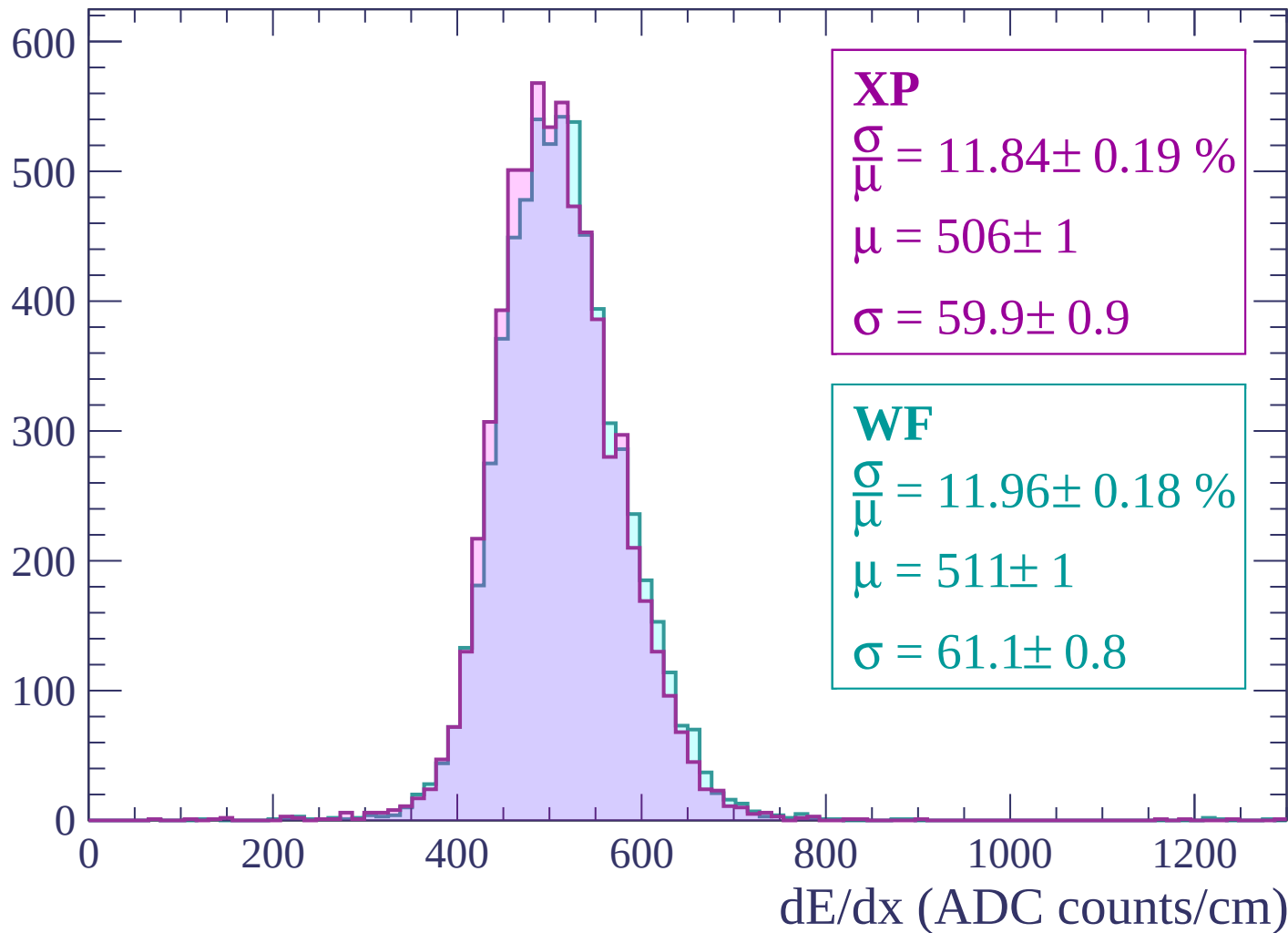
Energy loss |  $-2160 < p < -2100$

Count



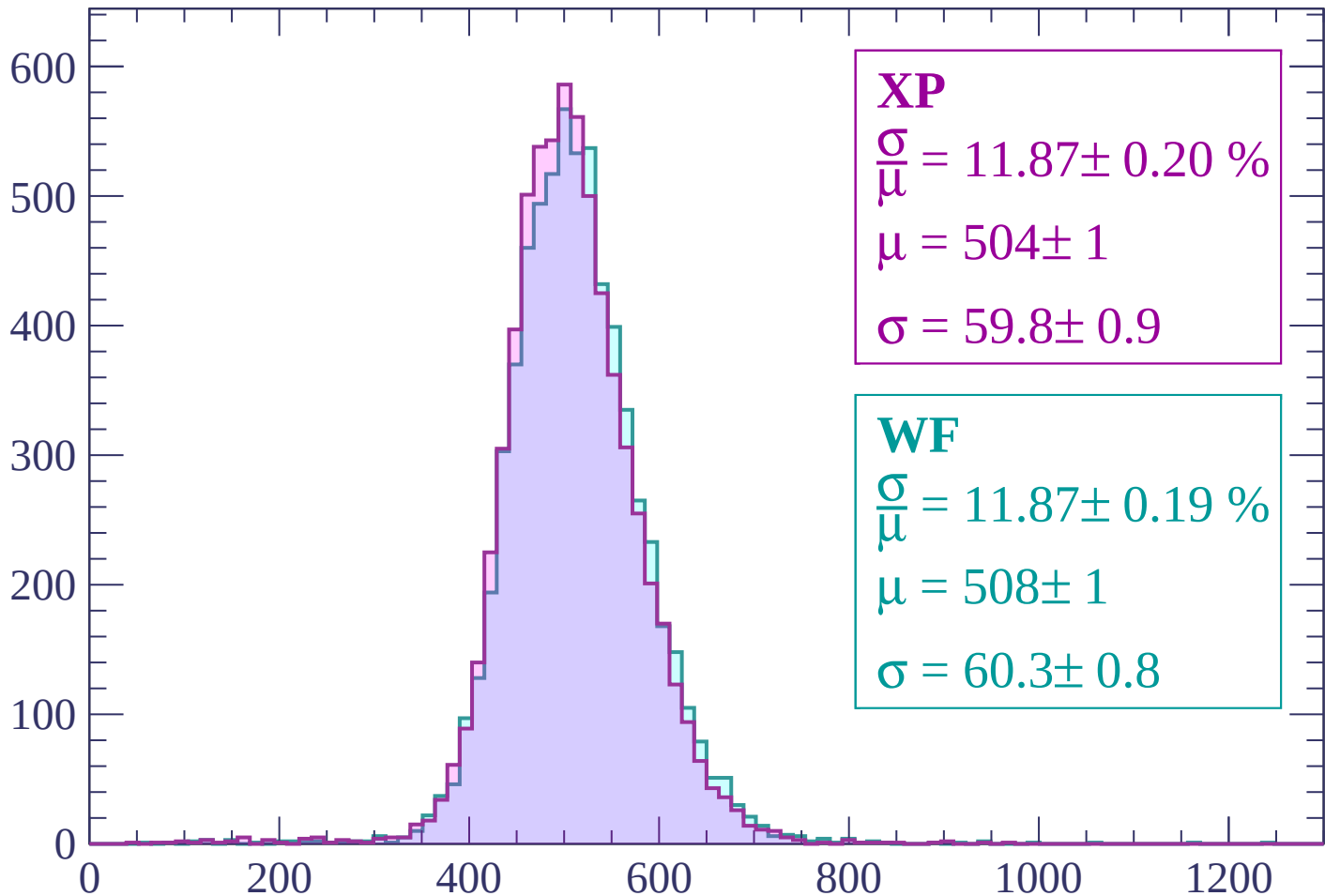
# Energy loss | -2100 < p < -2040

Count



# Energy loss | -2040 < p < -1980

Count



**XP**

$$\frac{\sigma}{\mu} = 11.87 \pm 0.20 \%$$

$$\mu = 504 \pm 1$$

$$\sigma = 59.8 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 11.87 \pm 0.19 \%$$

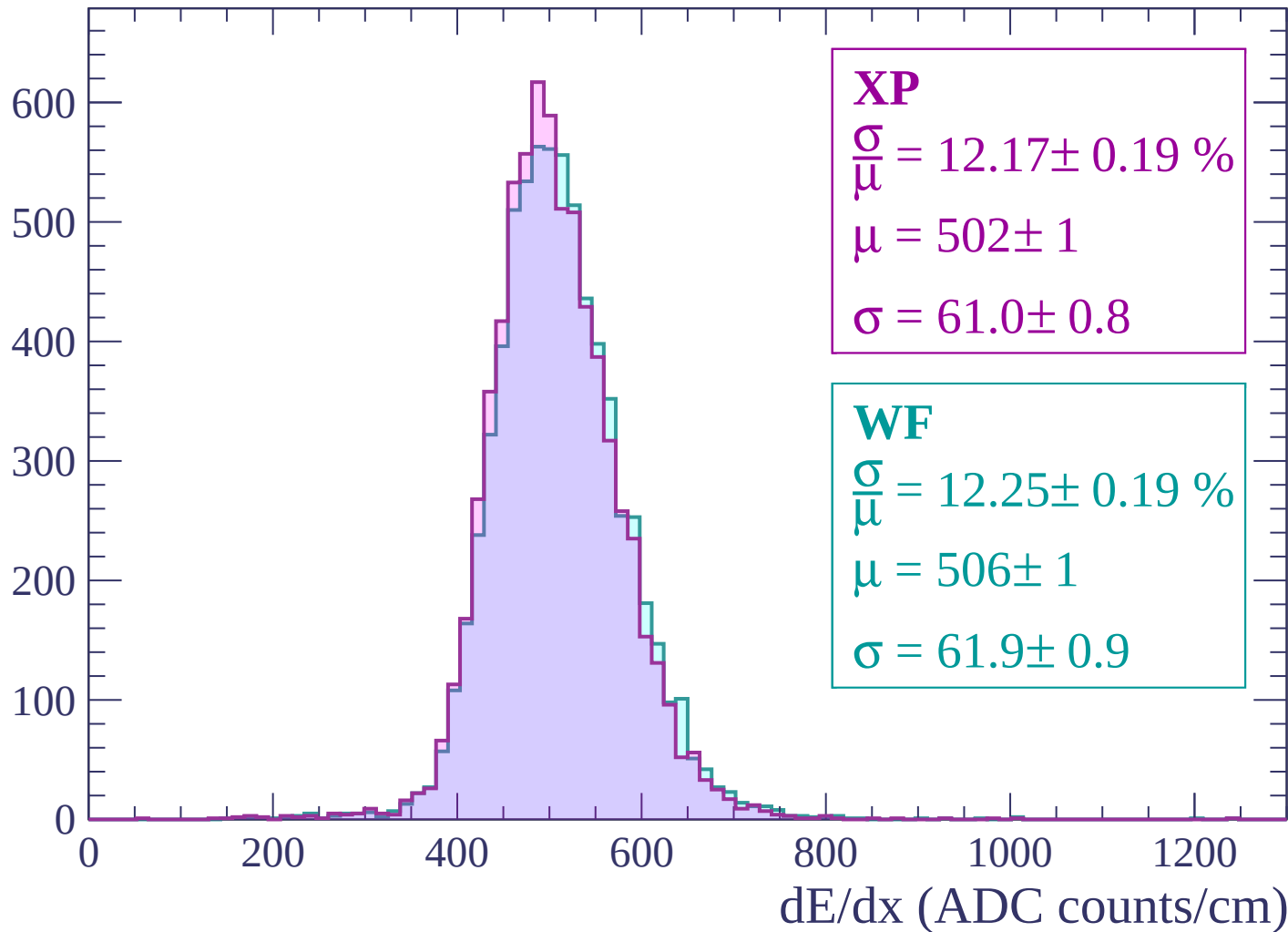
$$\mu = 508 \pm 1$$

$$\sigma = 60.3 \pm 0.8$$

dE/dx (ADC counts/cm)

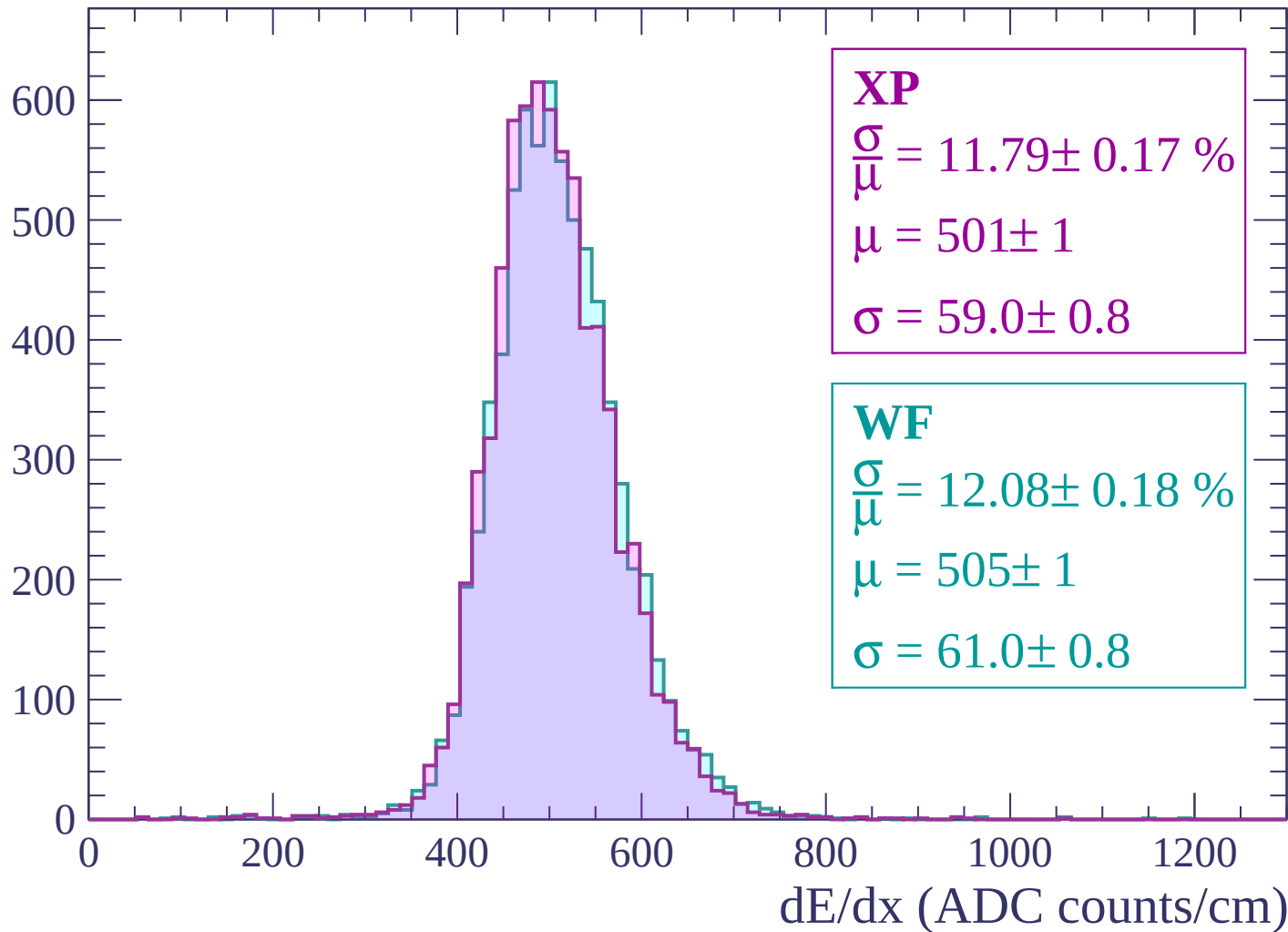
# Energy loss | -1980 < p < -1920

Count



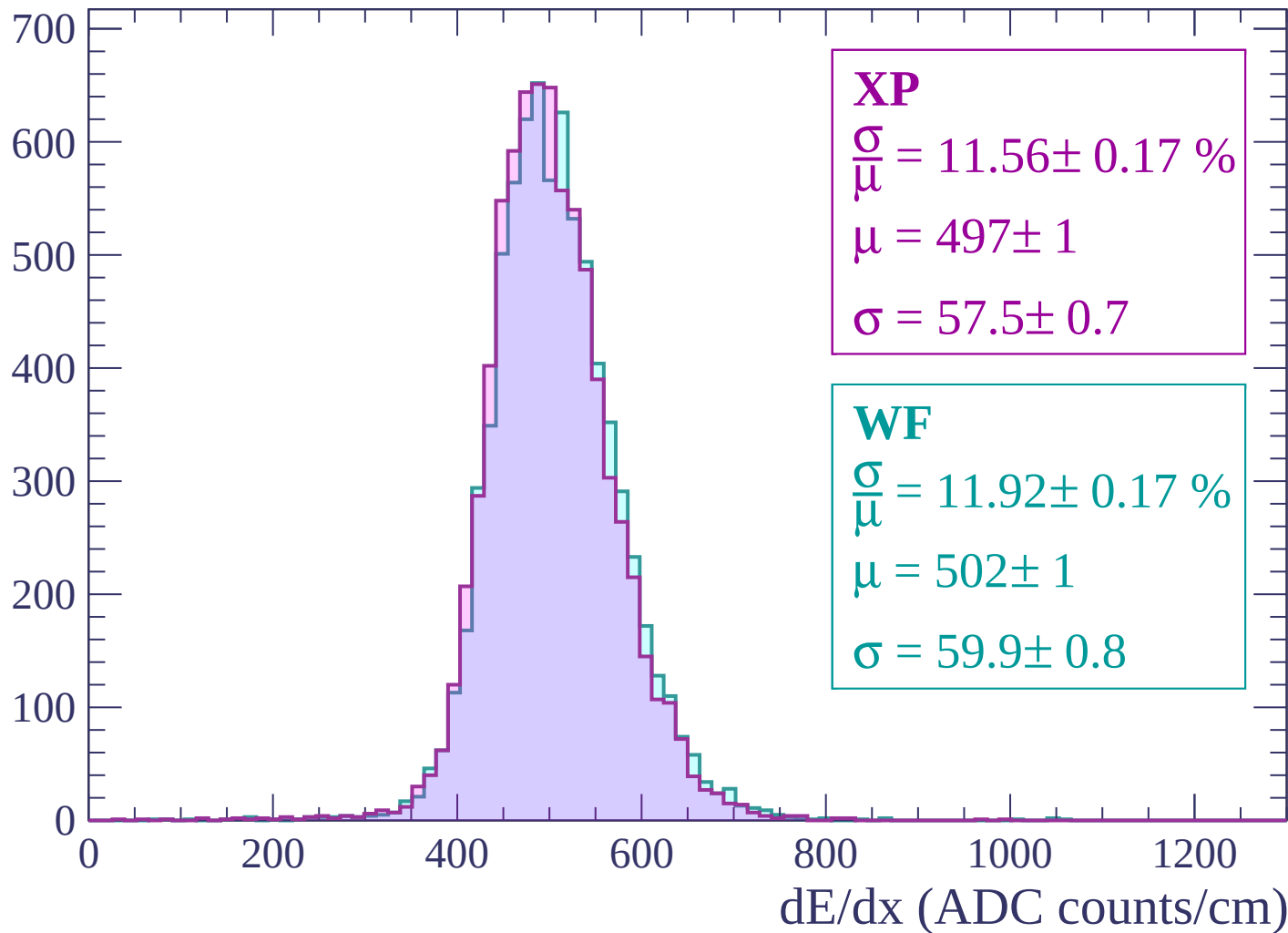
# Energy loss | -1920 < p < -1860

Count

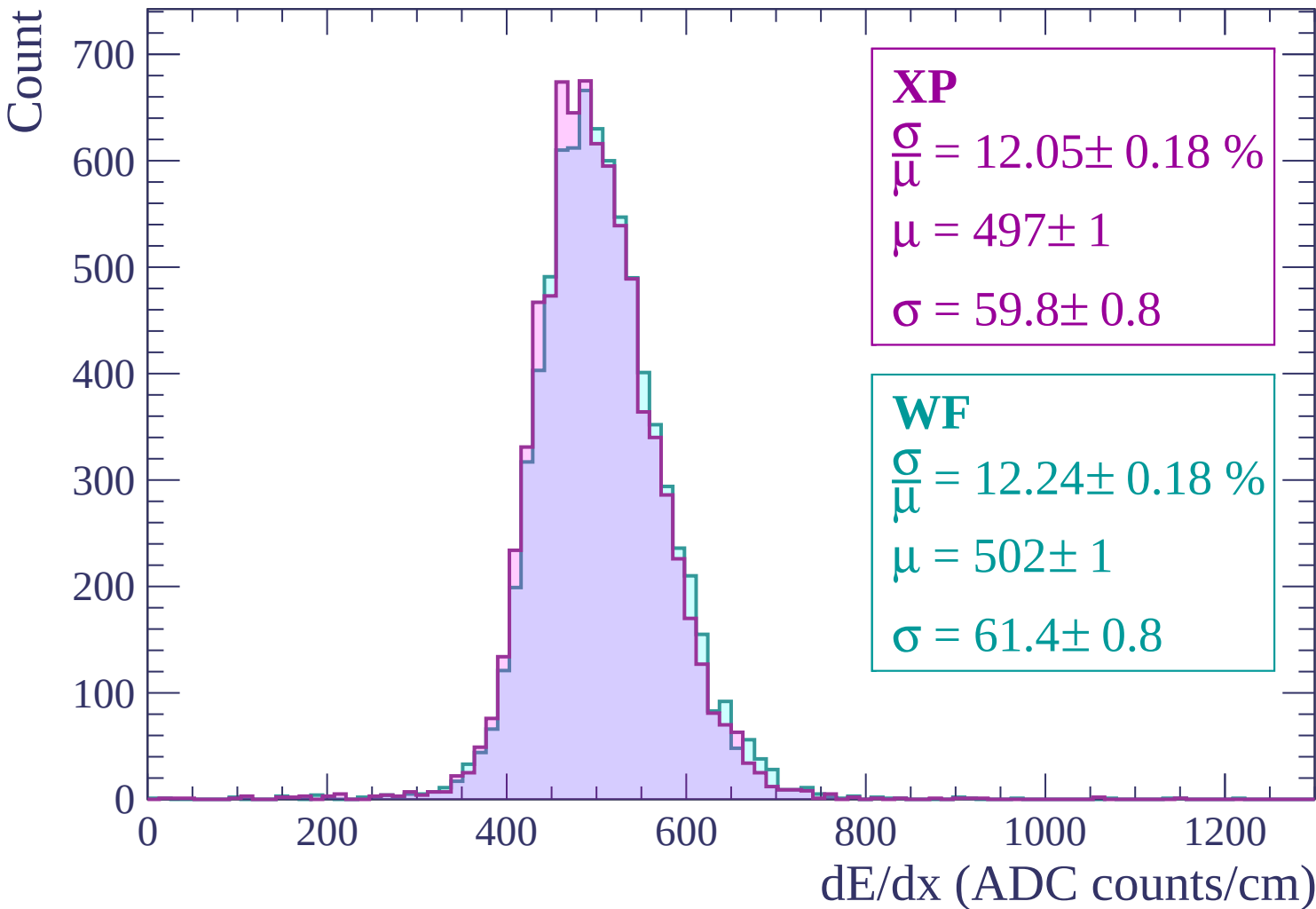


# Energy loss | -1860 < p < -1800

Count

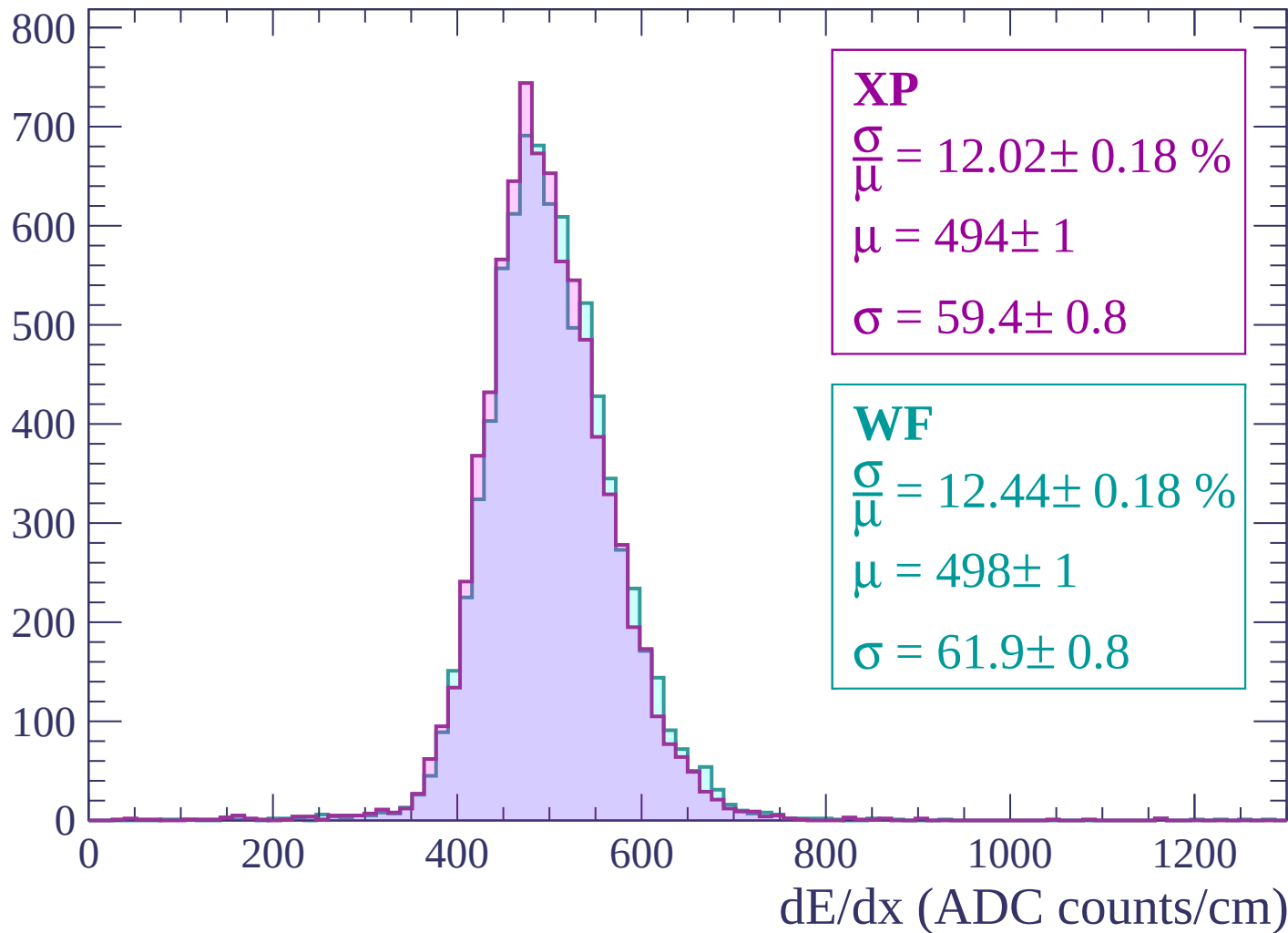


# Energy loss | -1800 < p < -1740



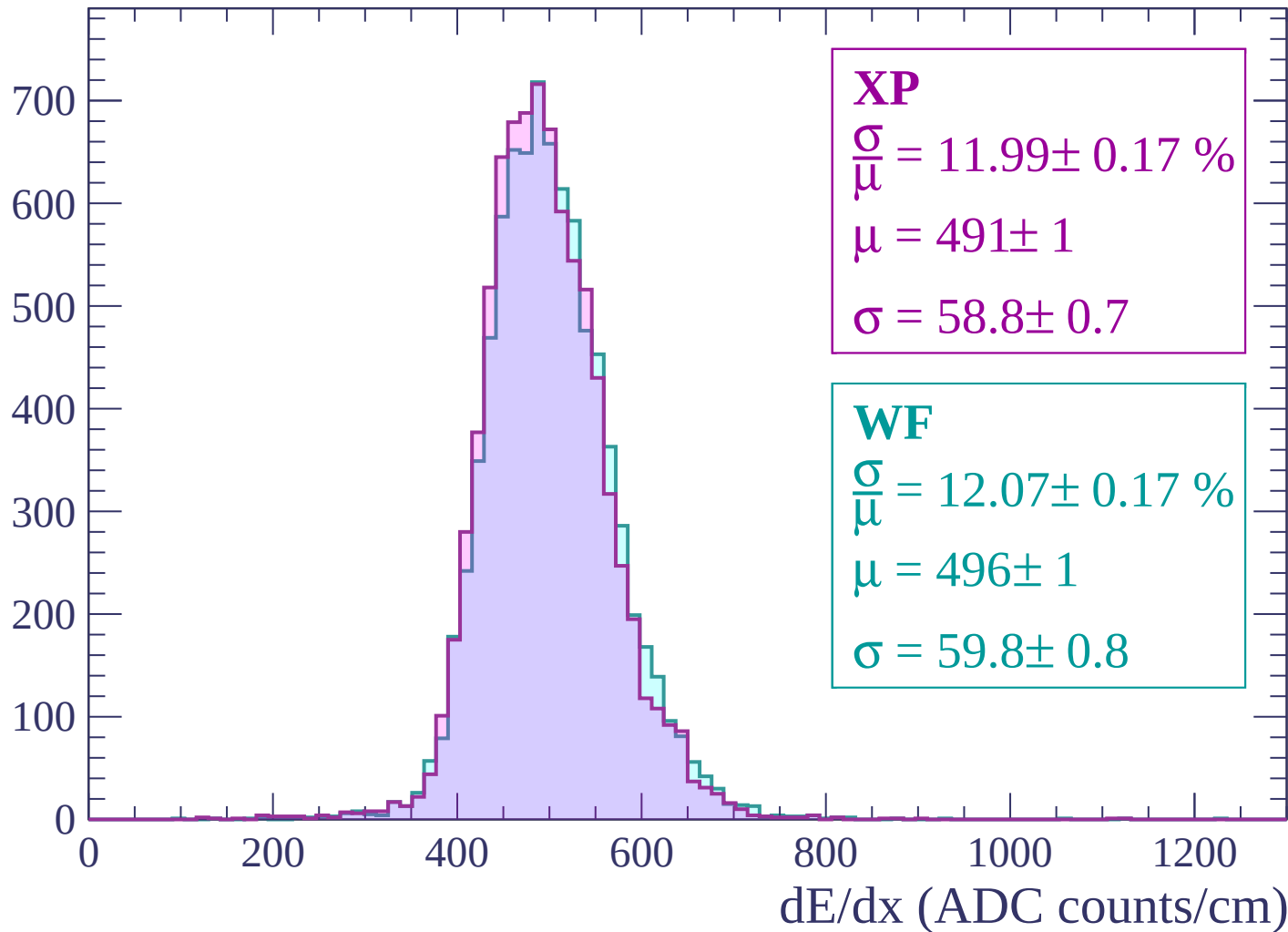
# Energy loss | -1740 < p < -1680

Count



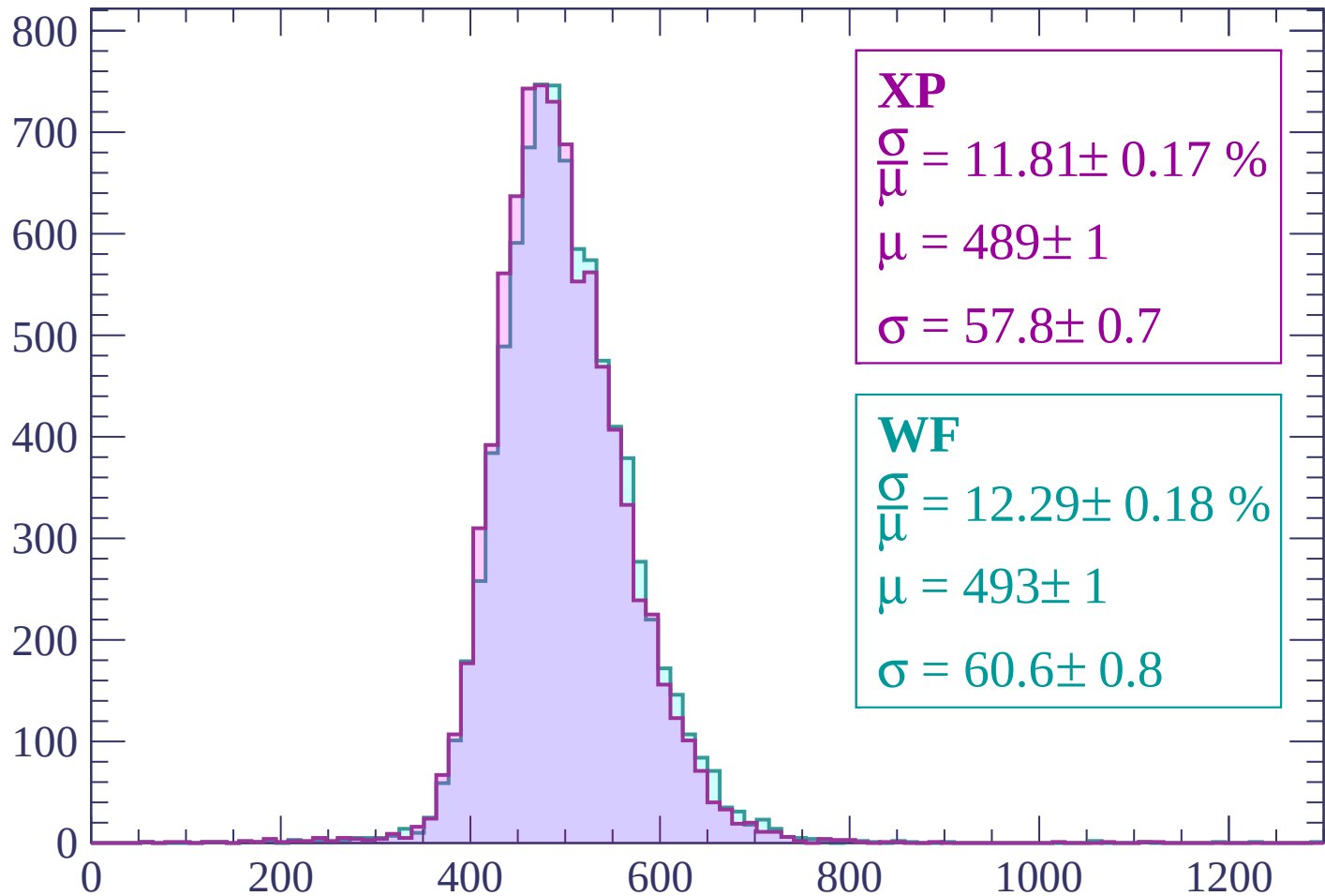
# Energy loss | -1680 < p < -1620

Count



# Energy loss | -1620 < p < -1560

Count



**XP**

$$\frac{\sigma}{\mu} = 11.81 \pm 0.17 \%$$

$$\mu = 489 \pm 1$$

$$\sigma = 57.8 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 12.29 \pm 0.18 \%$$

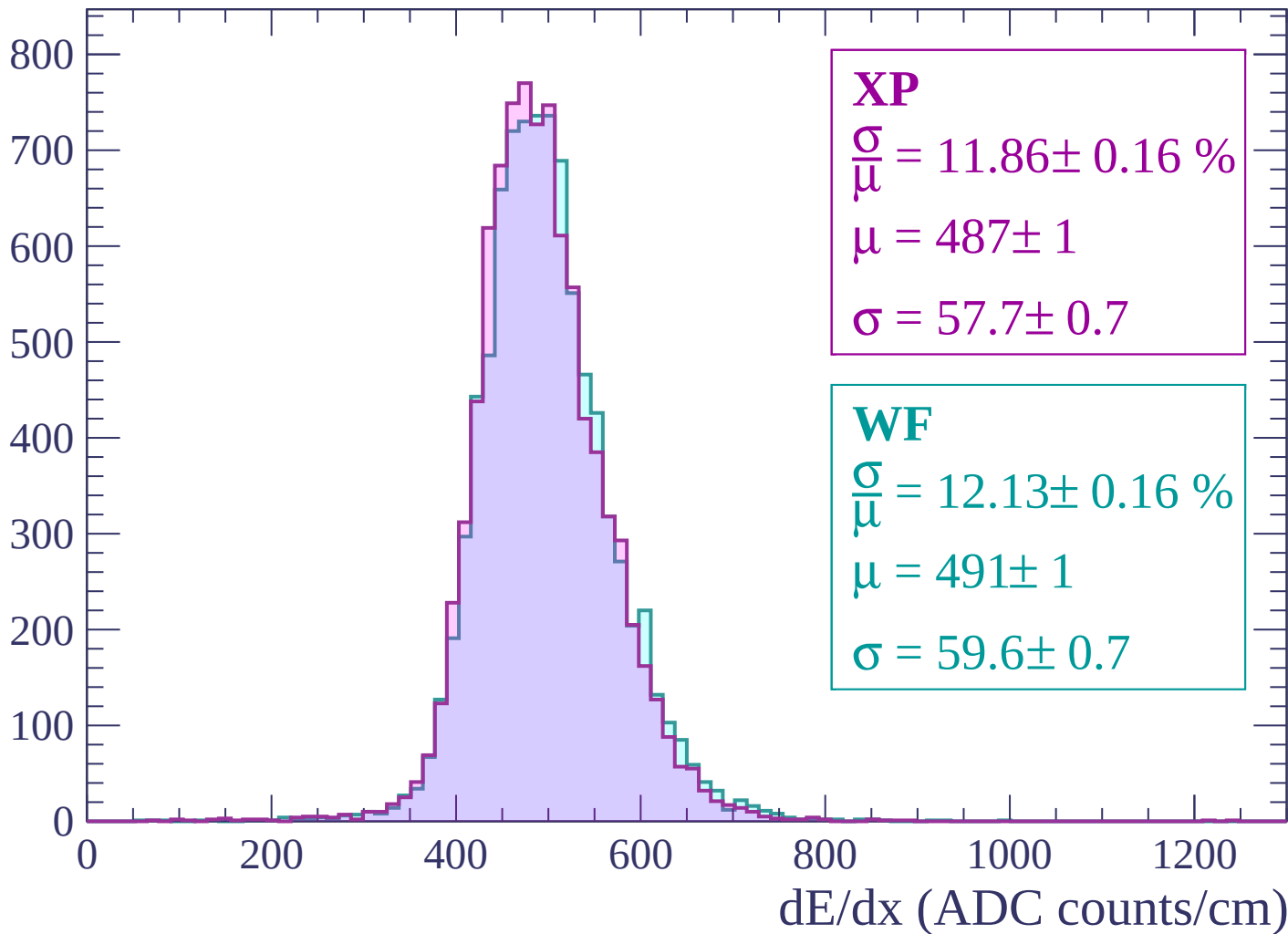
$$\mu = 493 \pm 1$$

$$\sigma = 60.6 \pm 0.8$$

dE/dx (ADC counts/cm)

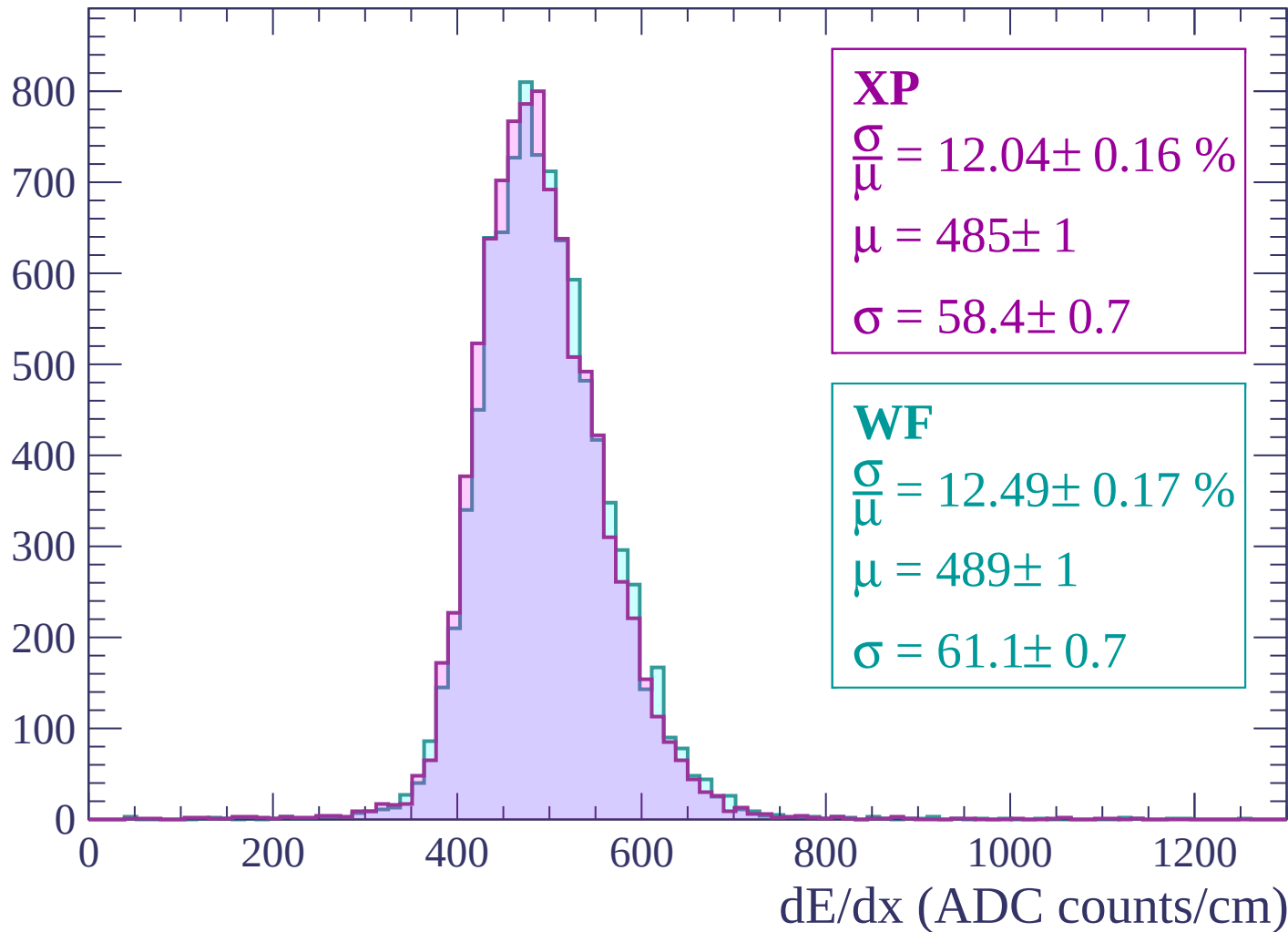
# Energy loss | -1560 < p < -1500

Count



# Energy loss | -1500 < p < -1440

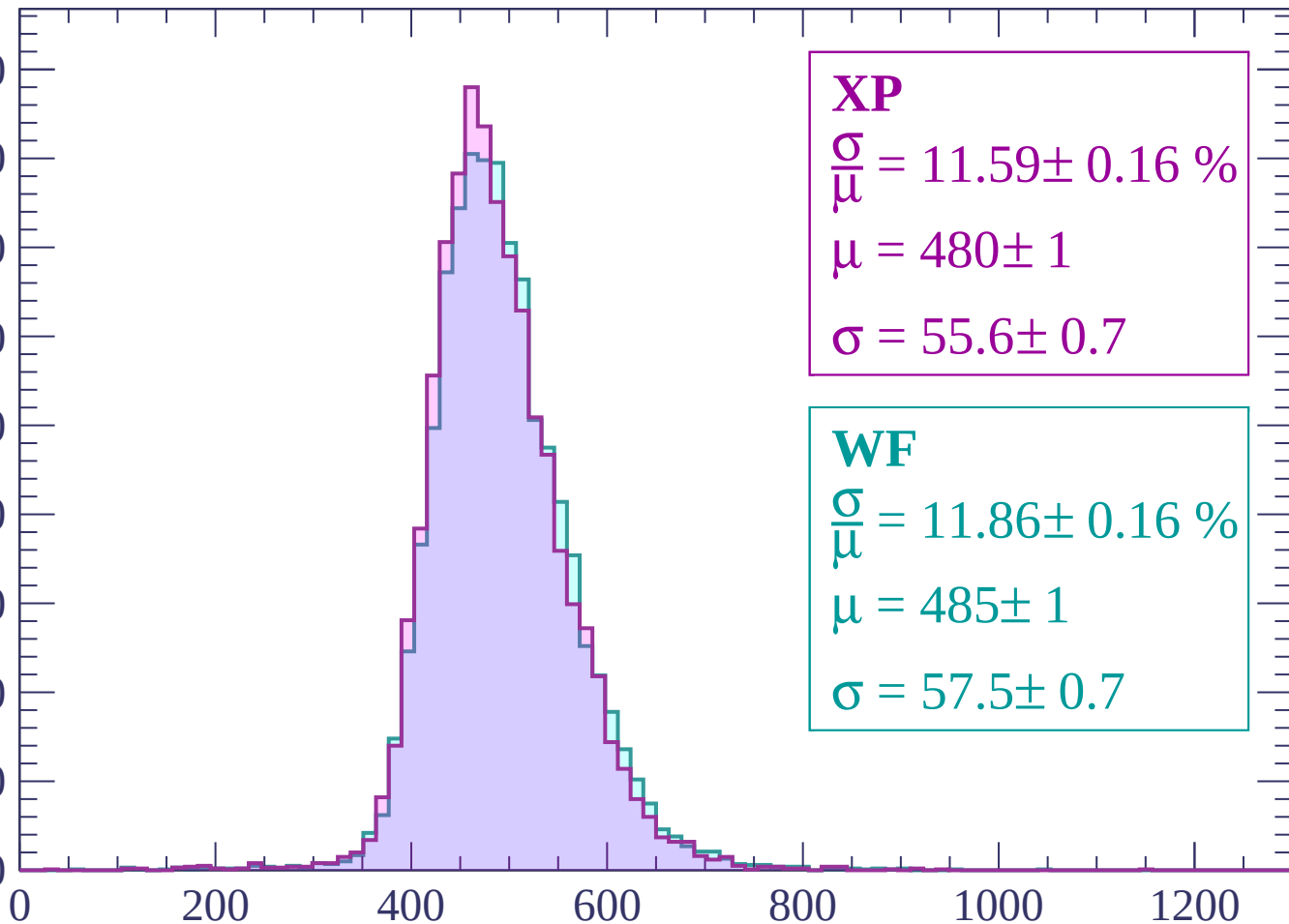
Count



Energy loss |  $-1440 < p < -1380$

Count

900  
800  
700  
600  
500  
400  
300  
200  
100  
0



**XP**

$$\frac{\sigma}{\mu} = 11.59 \pm 0.16 \%$$

$$\mu = 480 \pm 1$$

$$\sigma = 55.6 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 11.86 \pm 0.16 \%$$

$$\mu = 485 \pm 1$$

$$\sigma = 57.5 \pm 0.7$$

dE/dx (ADC counts/cm)

# Energy loss | $-1380 < p < -1320$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$$\frac{\sigma}{\mu} = 11.54 \pm 0.16 \%$$

$$\mu = 477 \pm 1$$

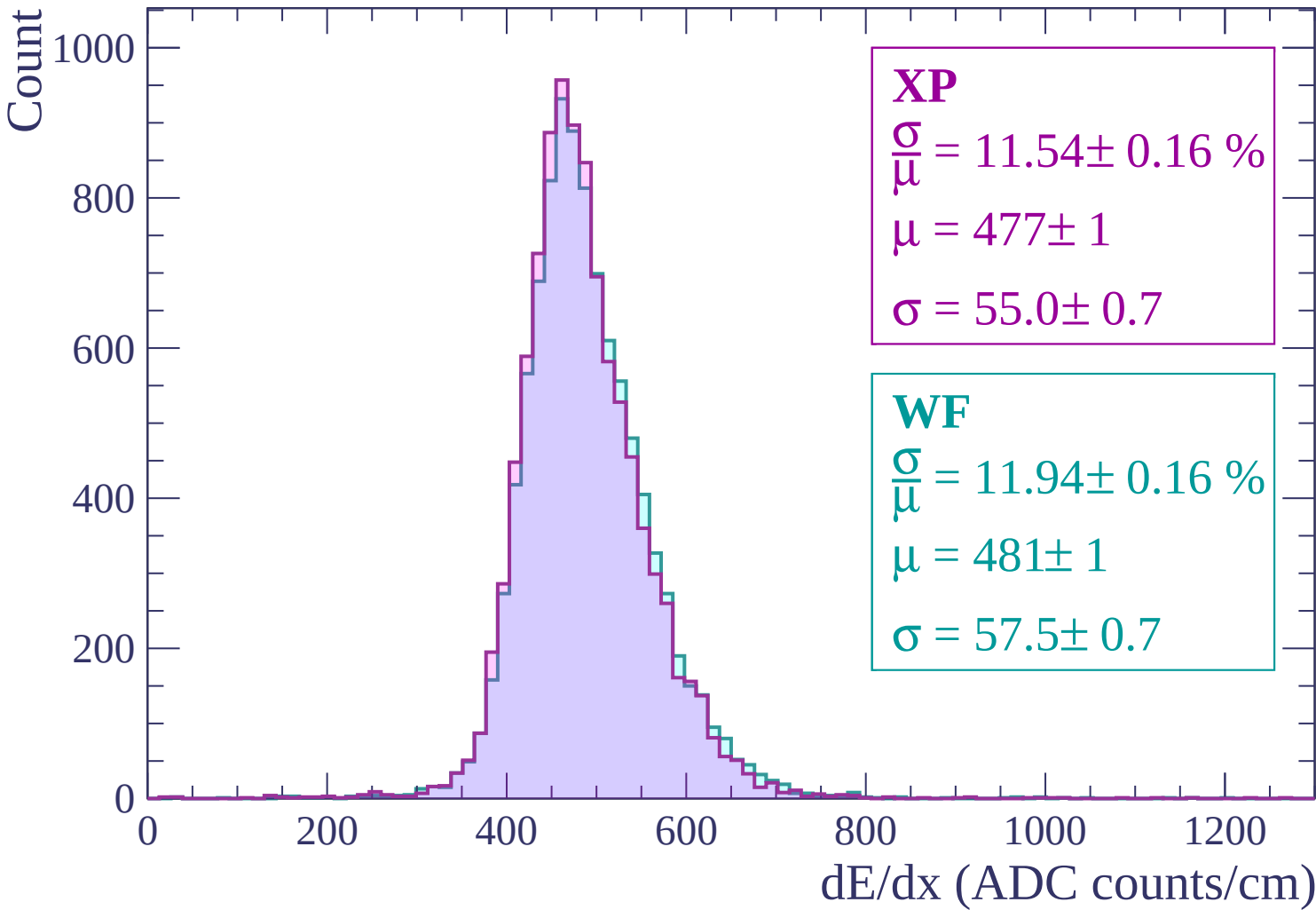
$$\sigma = 55.0 \pm 0.7$$

**WF**

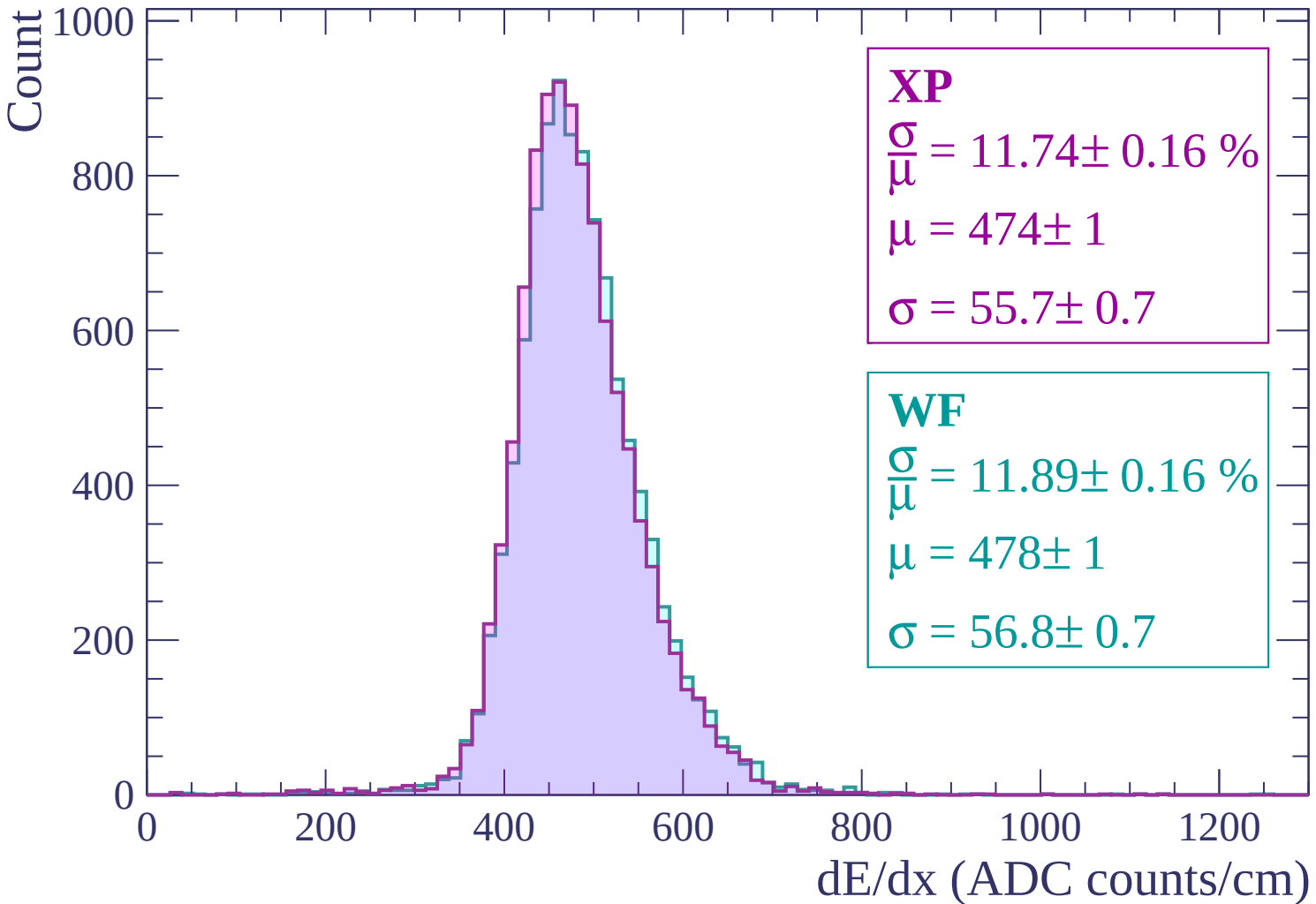
$$\frac{\sigma}{\mu} = 11.94 \pm 0.16 \%$$

$$\mu = 481 \pm 1$$

$$\sigma = 57.5 \pm 0.7$$



# Energy loss | -1320 < p < -1260



# Energy loss | $-1260 < p < -1200$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

**XP**

$$\frac{\sigma}{\mu} = 11.67 \pm 0.16 \%$$

$$\mu = 470 \pm 1$$

$$\sigma = 54.9 \pm 0.7$$

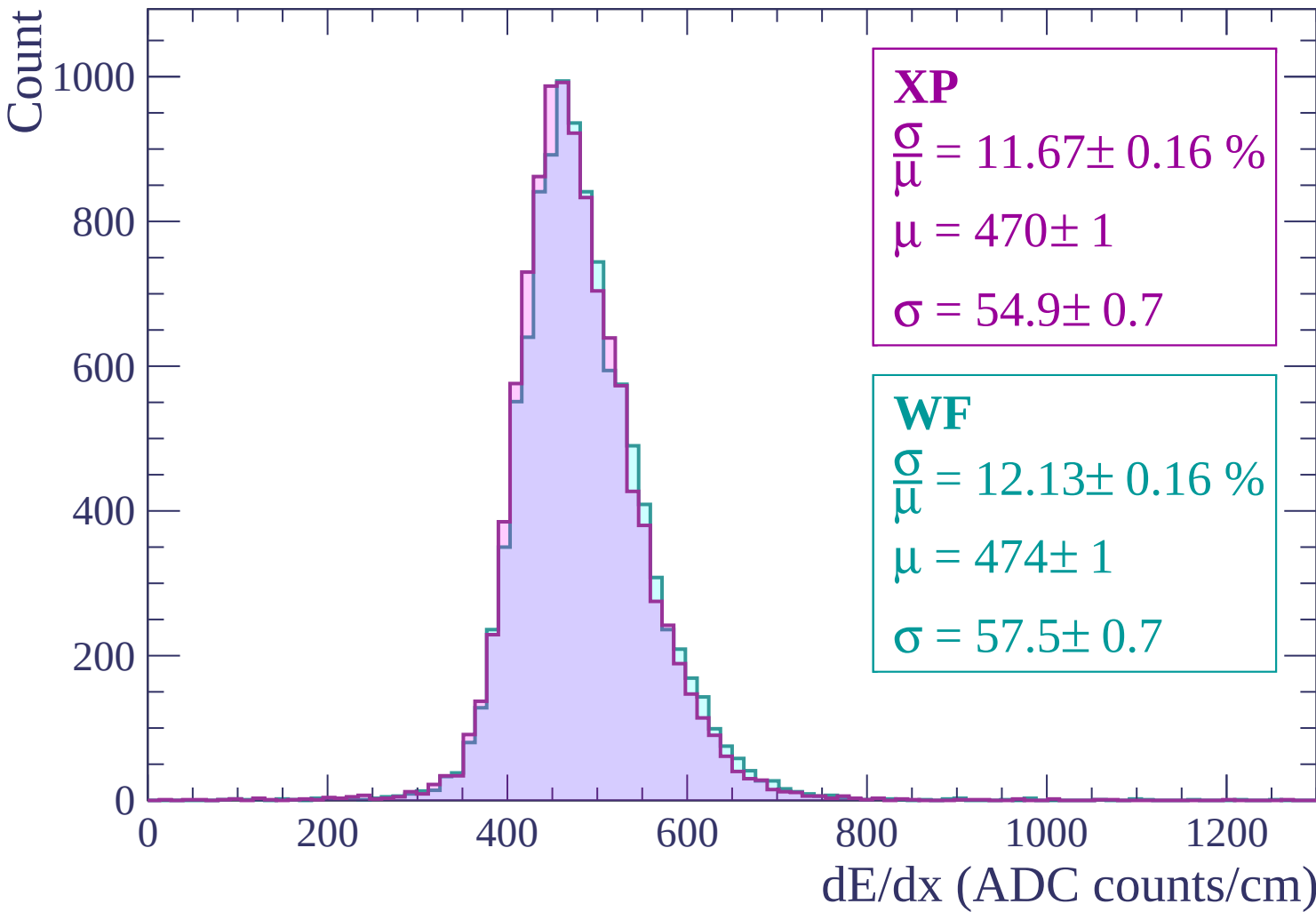
**WF**

$$\frac{\sigma}{\mu} = 12.13 \pm 0.16 \%$$

$$\mu = 474 \pm 1$$

$$\sigma = 57.5 \pm 0.7$$

dE/dx (ADC counts/cm)



# Energy loss | -1200 < p < -1140

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$$\frac{\sigma}{\mu} = 11.71 \pm 0.16 \%$$

$$\mu = 466 \pm 1$$

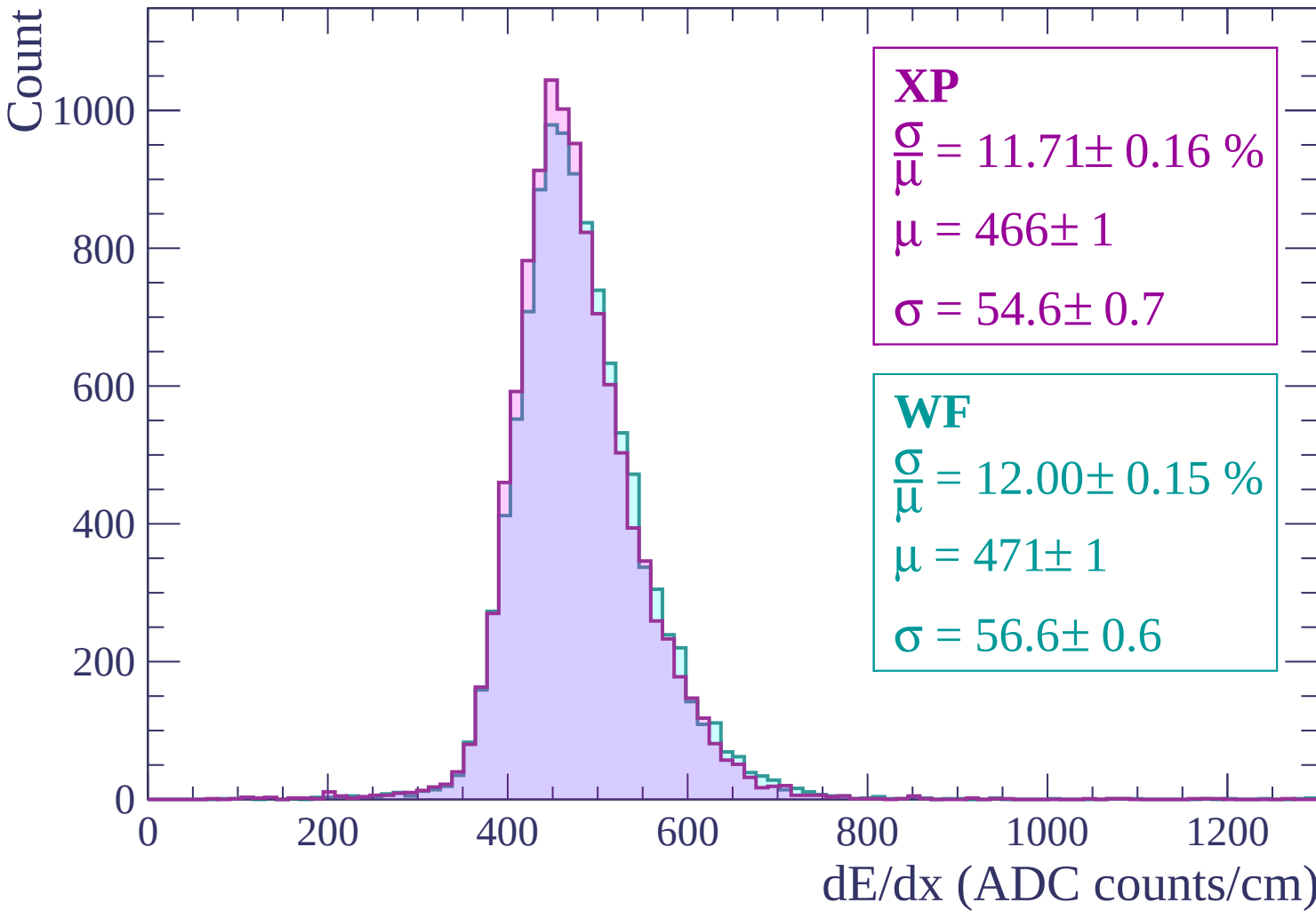
$$\sigma = 54.6 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 12.00 \pm 0.15 \%$$

$$\mu = 471 \pm 1$$

$$\sigma = 56.6 \pm 0.6$$



# Energy loss | -1140 < p < -1080

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$$\frac{\sigma}{\mu} = 11.19 \pm 0.14 \%$$

$$\mu = 462 \pm 1$$

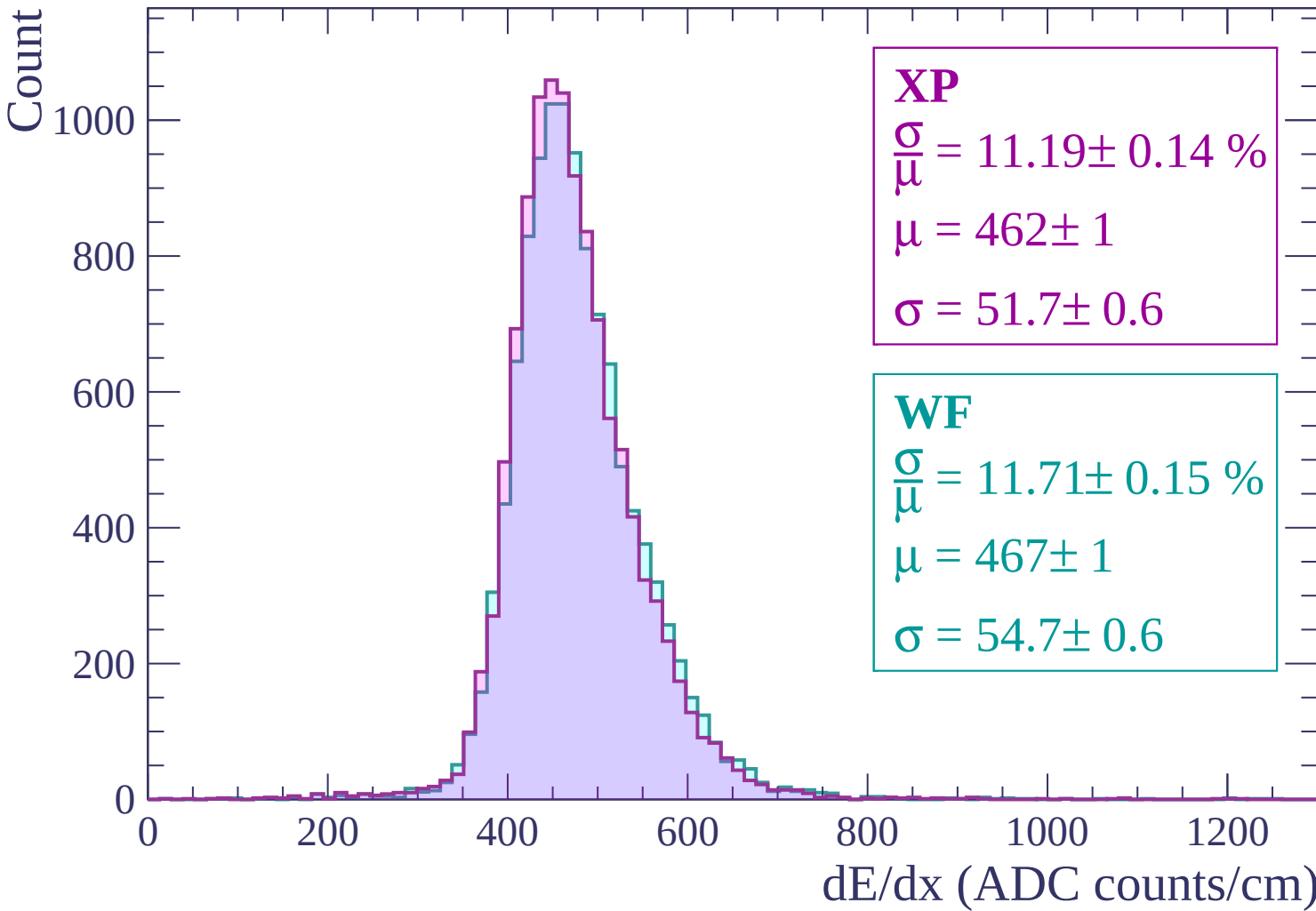
$$\sigma = 51.7 \pm 0.6$$

**WF**

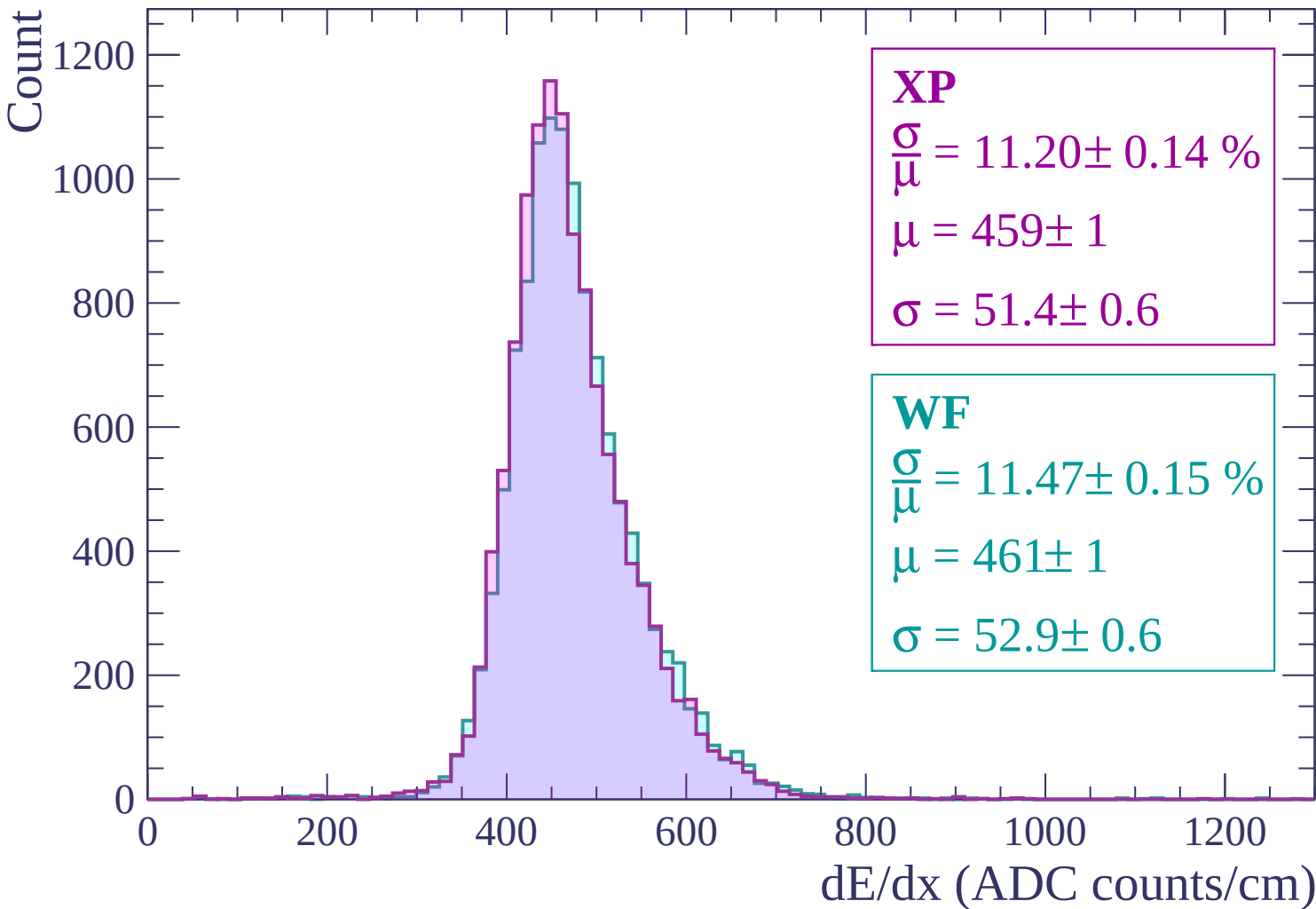
$$\frac{\sigma}{\mu} = 11.71 \pm 0.15 \%$$

$$\mu = 467 \pm 1$$

$$\sigma = 54.7 \pm 0.6$$



# Energy loss | -1080 < p < -1020



Energy loss |  $-1020 < p < -960$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

**XP**

$$\frac{\sigma}{\mu} = 10.85 \pm 0.14 \%$$

$$\mu = 452 \pm 1$$

$$\sigma = 49.0 \pm 0.6$$

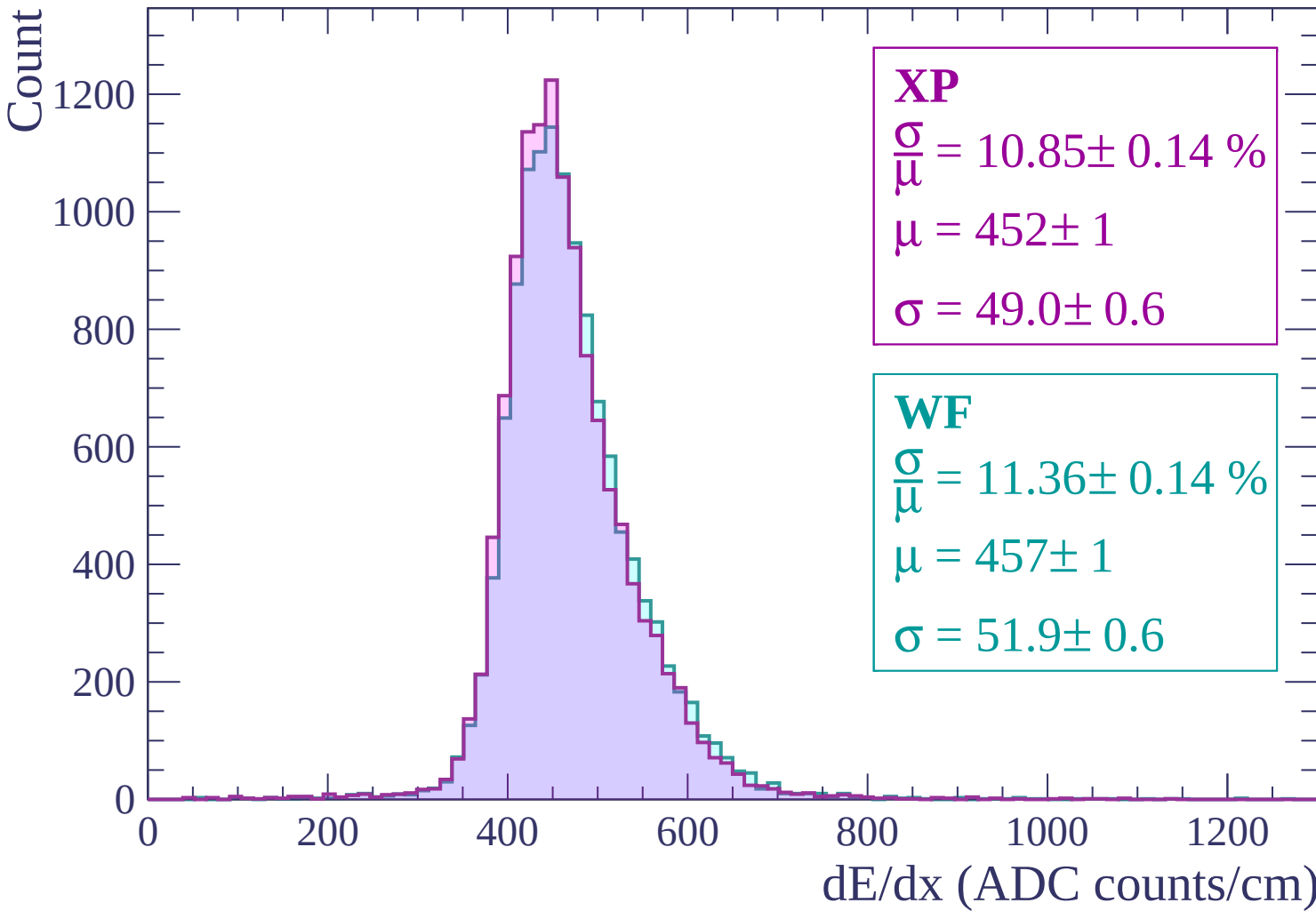
**WF**

$$\frac{\sigma}{\mu} = 11.36 \pm 0.14 \%$$

$$\mu = 457 \pm 1$$

$$\sigma = 51.9 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss |  $-960 < p < -900$

Count

1200

1000

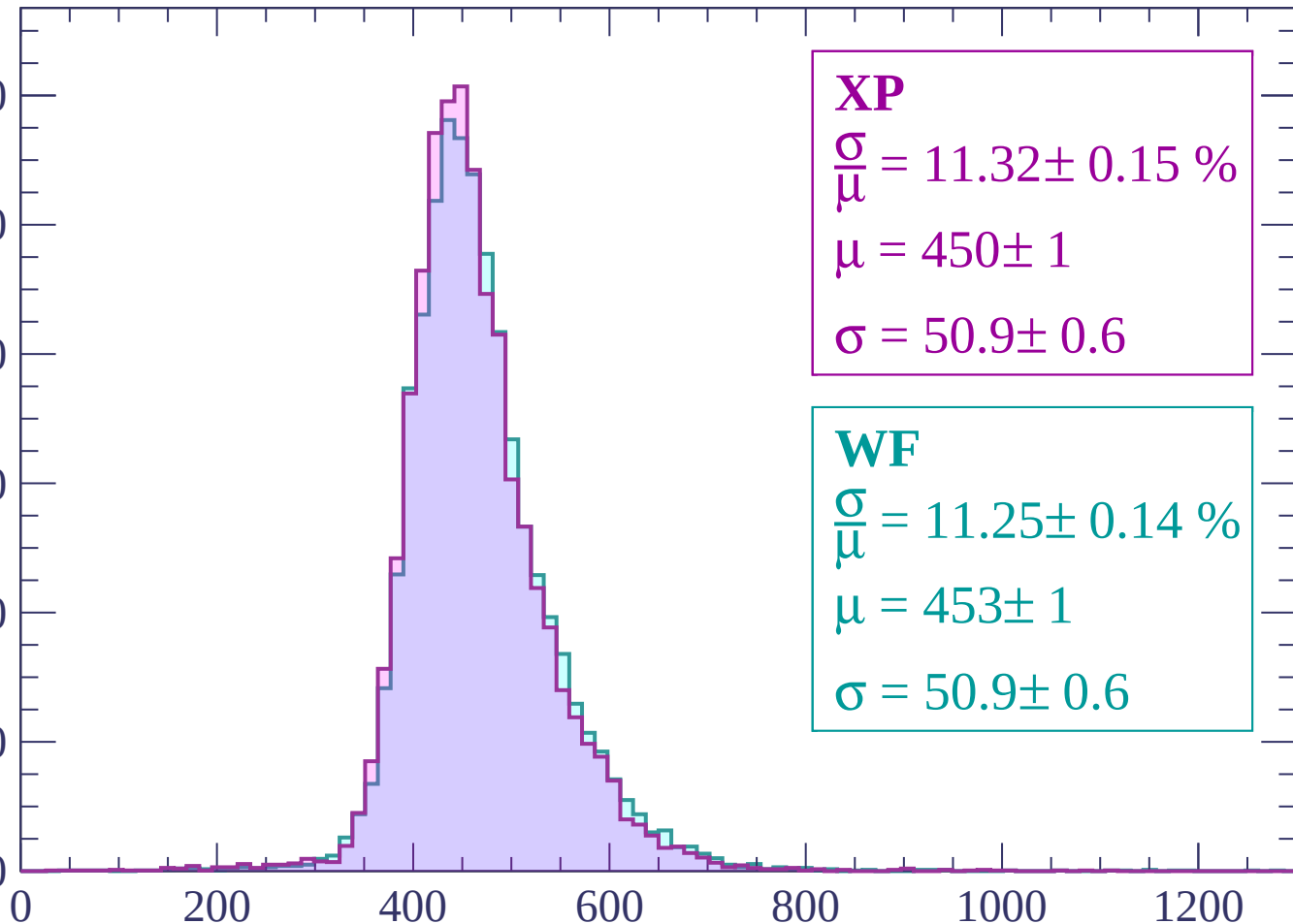
800

600

400

200

0



**XP**

$$\frac{\sigma}{\mu} = 11.32 \pm 0.15 \%$$

$$\mu = 450 \pm 1$$

$$\sigma = 50.9 \pm 0.6$$

**WF**

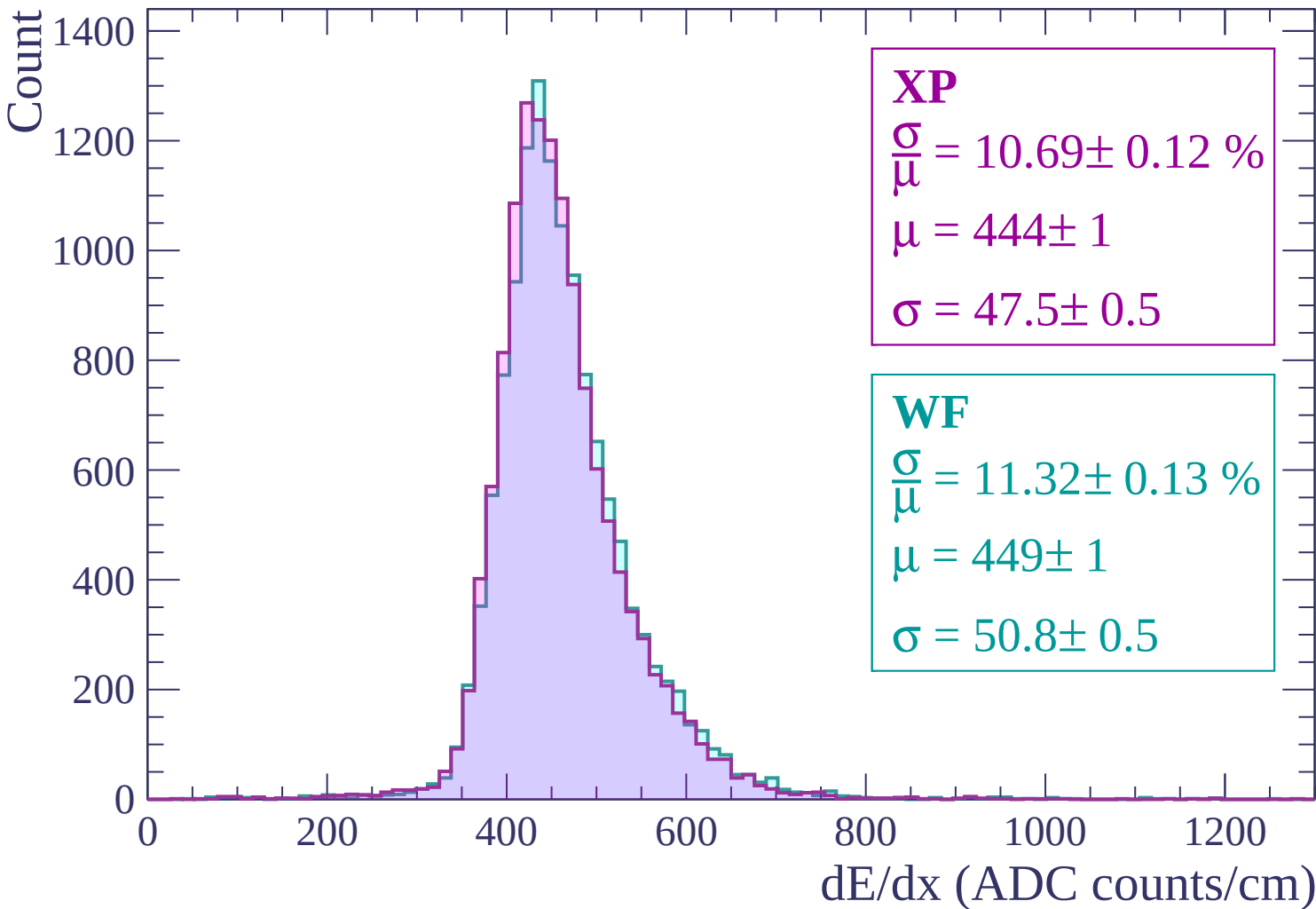
$$\frac{\sigma}{\mu} = 11.25 \pm 0.14 \%$$

$$\mu = 453 \pm 1$$

$$\sigma = 50.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $-900 < p < -840$



Energy loss |  $-840 < p < -780$

Count

1400  
1200  
1000  
800  
600  
400  
200  
0

**XP**

$$\frac{\sigma}{\mu} = 10.75 \pm 0.13 \%$$

$$\mu = 440 \pm 1$$

$$\sigma = 47.3 \pm 0.5$$

**WF**

$$\frac{\sigma}{\mu} = 11.00 \pm 0.13 \%$$

$$\mu = 443 \pm 1$$

$$\sigma = 48.7 \pm 0.5$$

0

200

400

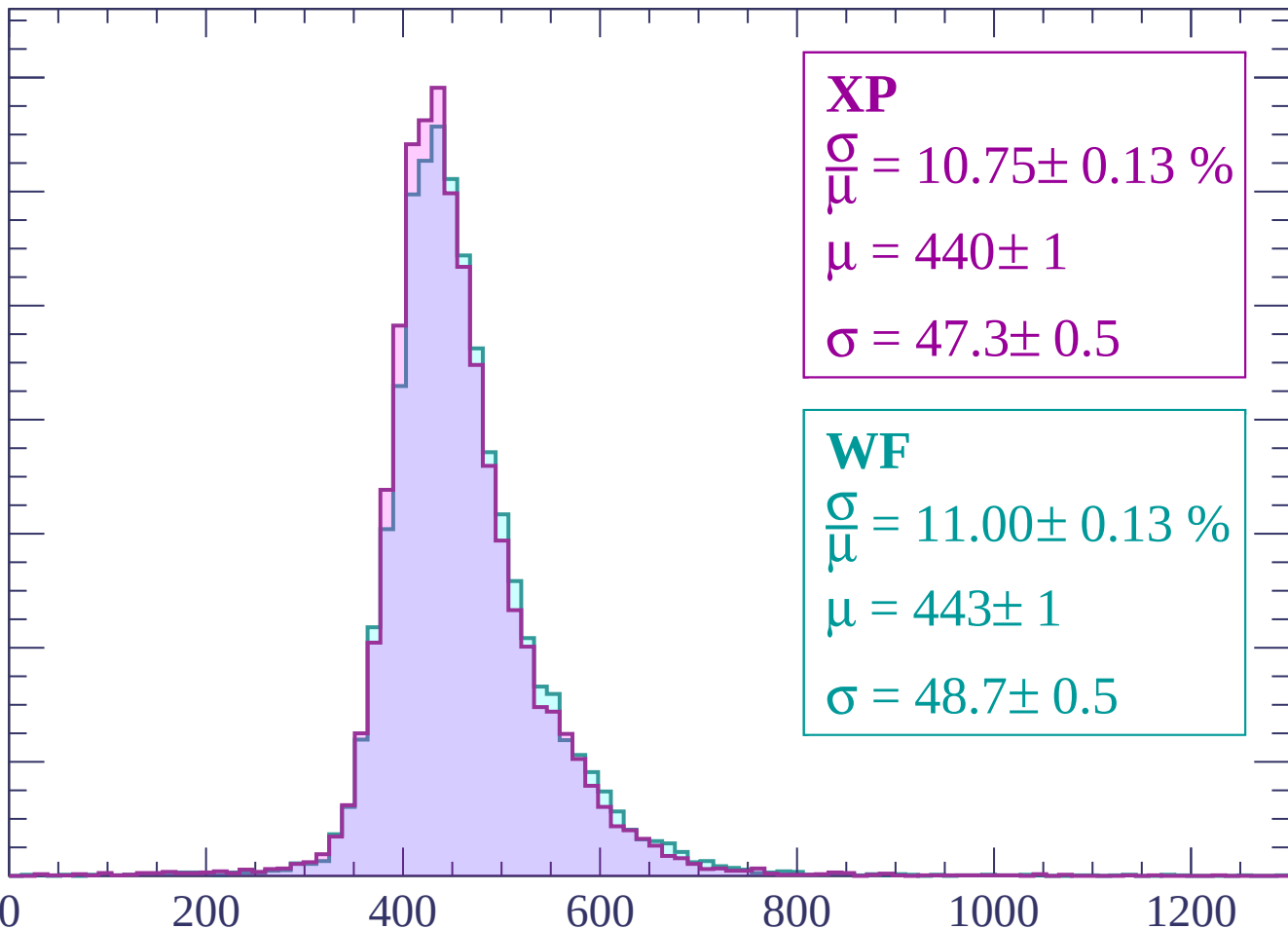
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss |  $-780 < p < -720$

Count

1400  
1200  
1000  
800  
600  
400  
200  
0

**XP**

$$\frac{\sigma}{\mu} = 10.53 \pm 0.12 \%$$

$$\mu = 434 \pm 1$$

$$\sigma = 45.7 \pm 0.5$$

**WF**

$$\frac{\sigma}{\mu} = 11.09 \pm 0.13 \%$$

$$\mu = 438 \pm 1$$

$$\sigma = 48.6 \pm 0.5$$

0

200

400

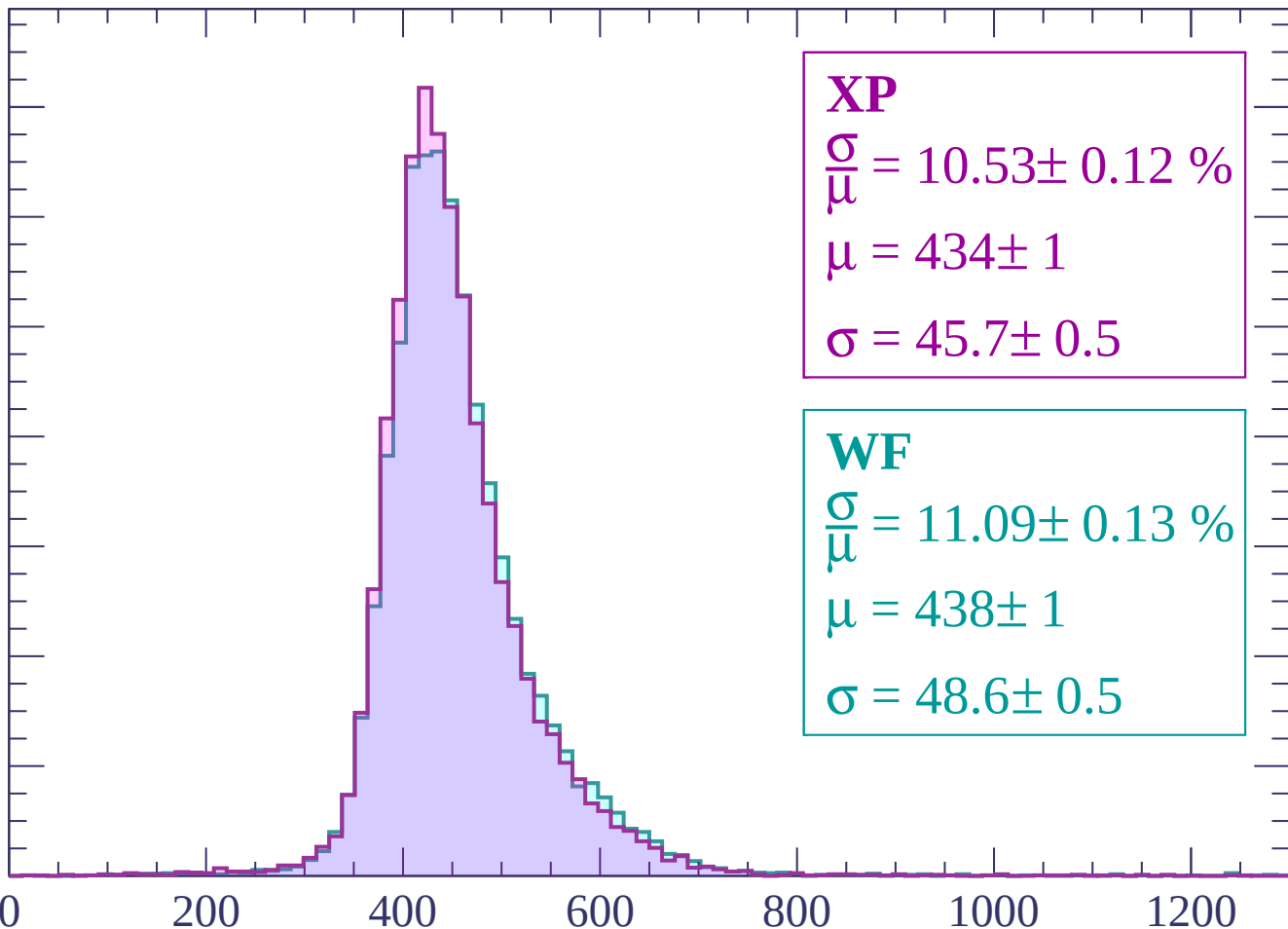
600

800

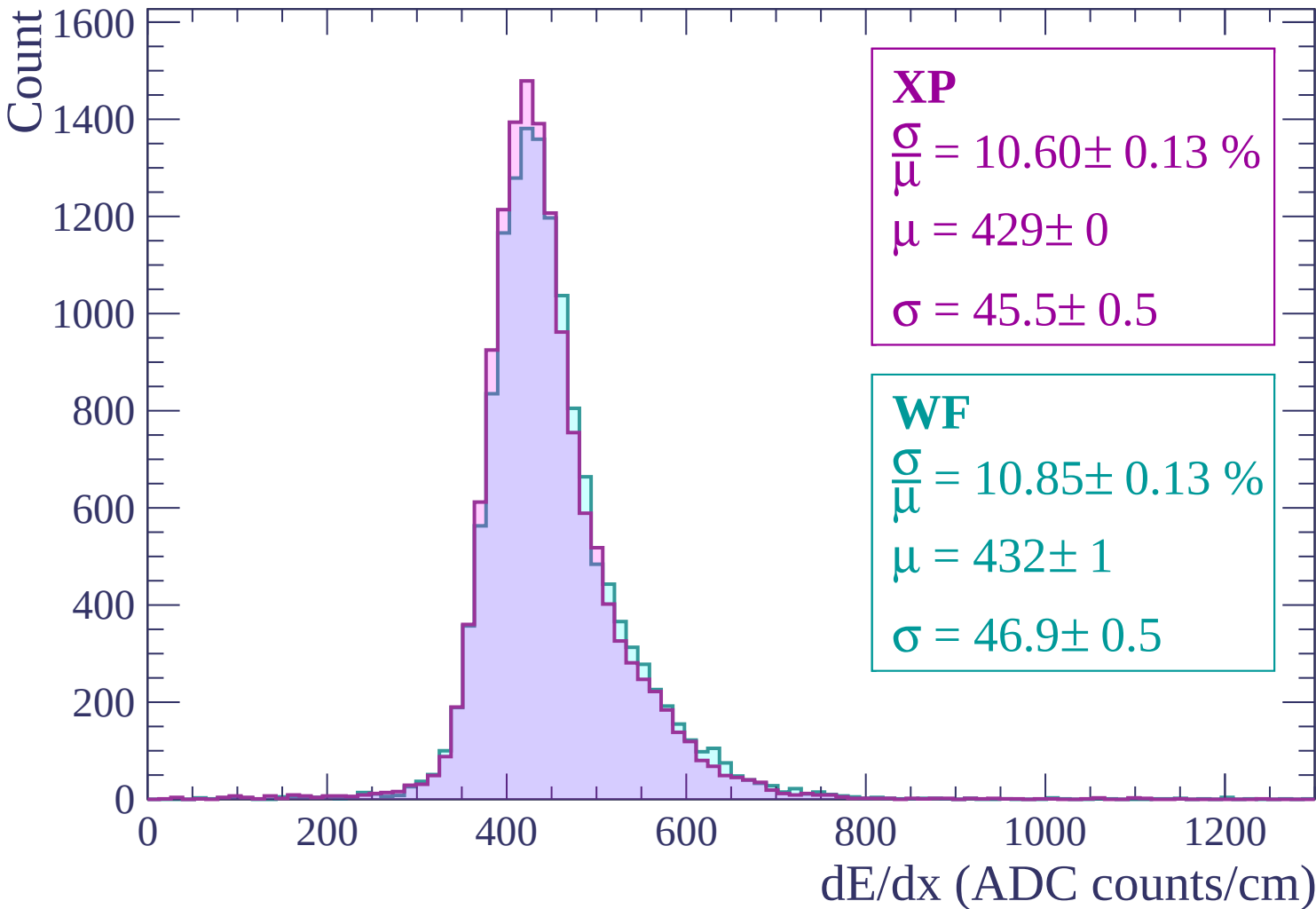
1000

1200

dE/dx (ADC counts/cm)



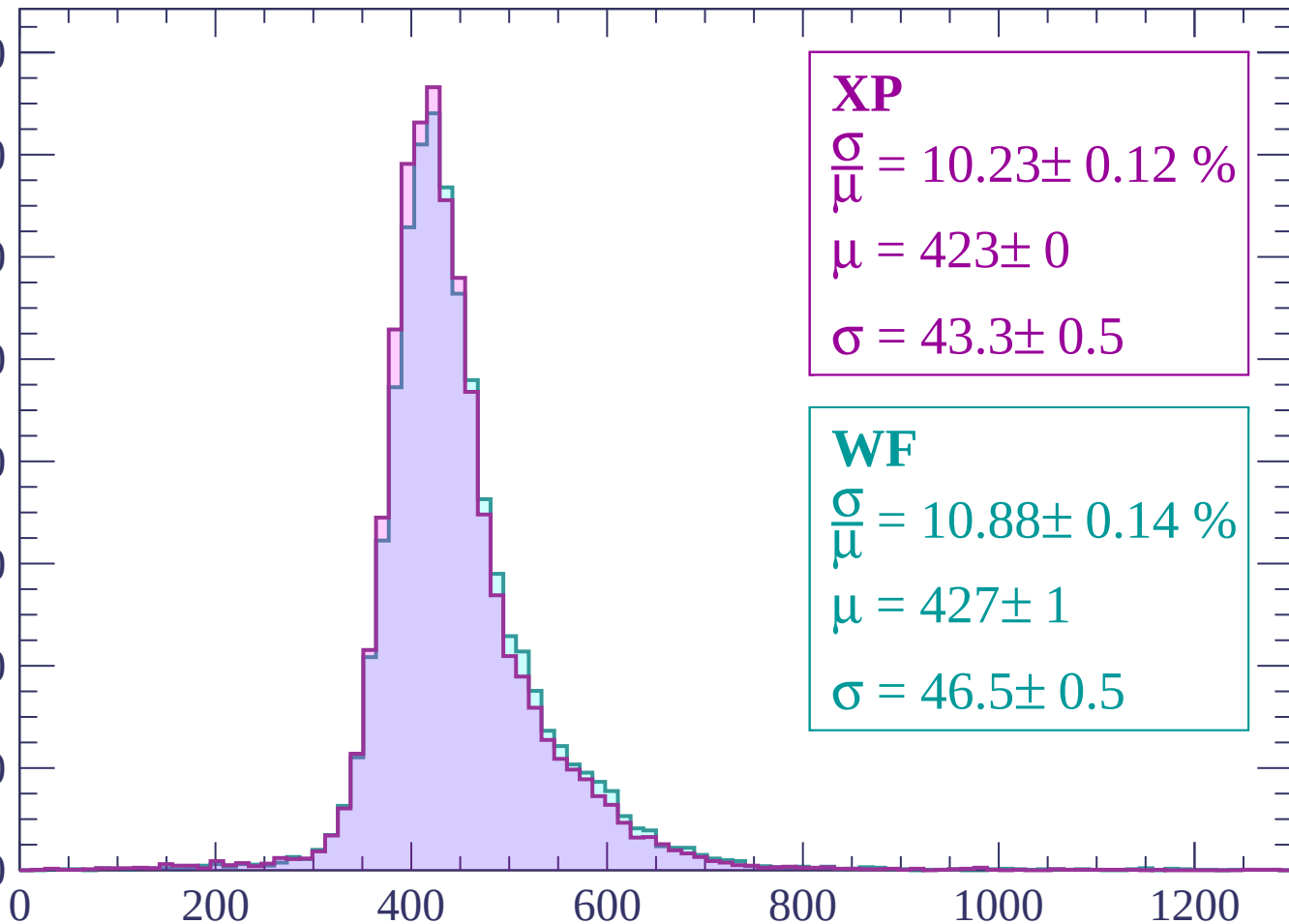
# Energy loss | -720 < p < -660



# Energy loss | $-660 < p < -600$

Count

1600  
1400  
1200  
1000  
800  
600  
400  
200  
0



**XP**

$$\frac{\sigma}{\mu} = 10.23 \pm 0.12 \%$$

$$\mu = 423 \pm 0$$

$$\sigma = 43.3 \pm 0.5$$

**WF**

$$\frac{\sigma}{\mu} = 10.88 \pm 0.14 \%$$

$$\mu = 427 \pm 1$$

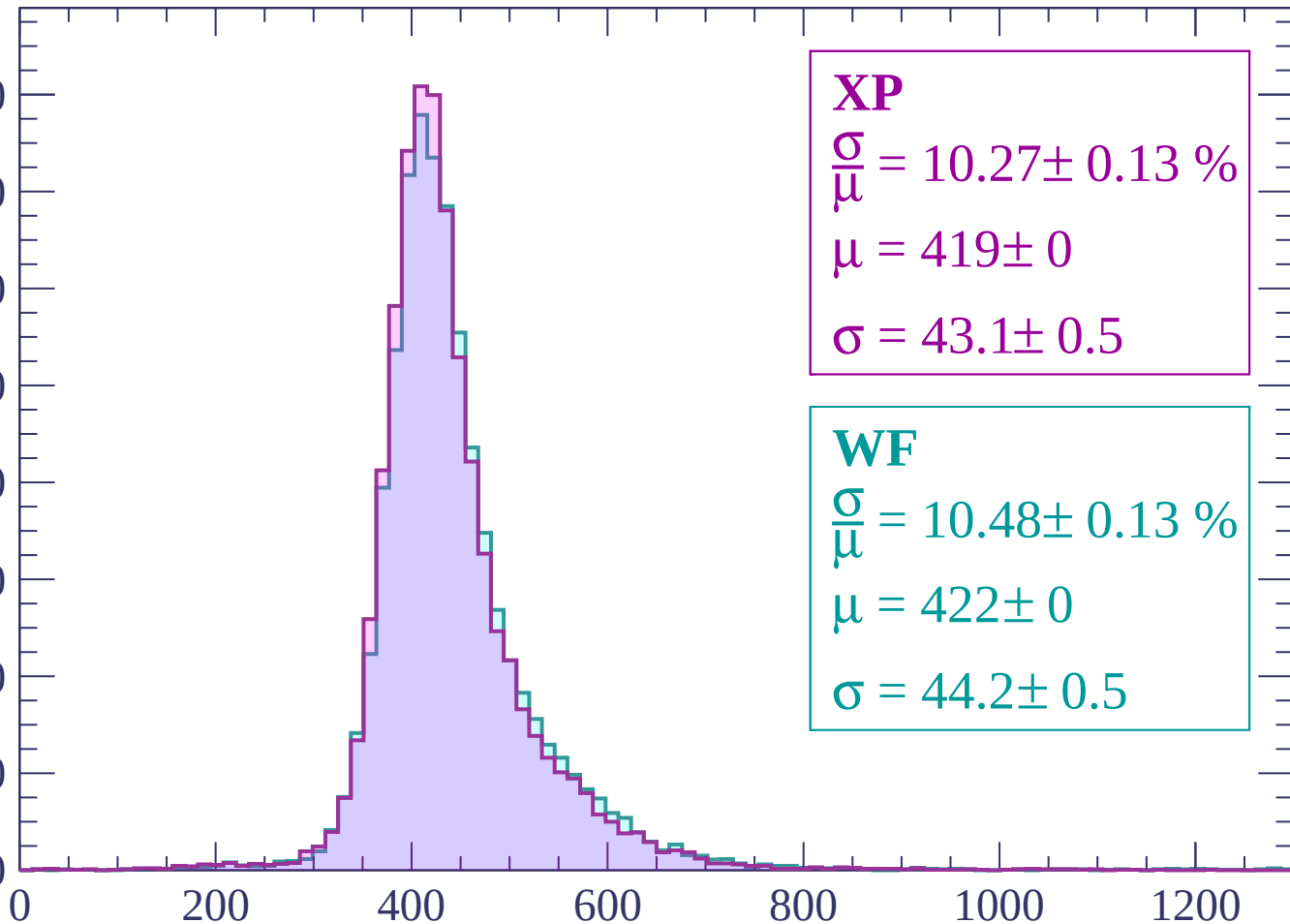
$$\sigma = 46.5 \pm 0.5$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-600 < p < -540$

Count

1600  
1400  
1200  
1000  
800  
600  
400  
200  
0



**XP**

$$\frac{\sigma}{\mu} = 10.27 \pm 0.13 \%$$

$$\mu = 419 \pm 0$$

$$\sigma = 43.1 \pm 0.5$$

**WF**

$$\frac{\sigma}{\mu} = 10.48 \pm 0.13 \%$$

$$\mu = 422 \pm 0$$

$$\sigma = 44.2 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss |  $-540 < p < -480$

Count

1800  
1600  
1400  
1200  
1000  
800  
600  
400  
200  
0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$$\frac{\sigma}{\mu} = 9.93 \pm 0.11 \%$$

$$\mu = 413 \pm 0$$

$$\sigma = 41.1 \pm 0.4$$

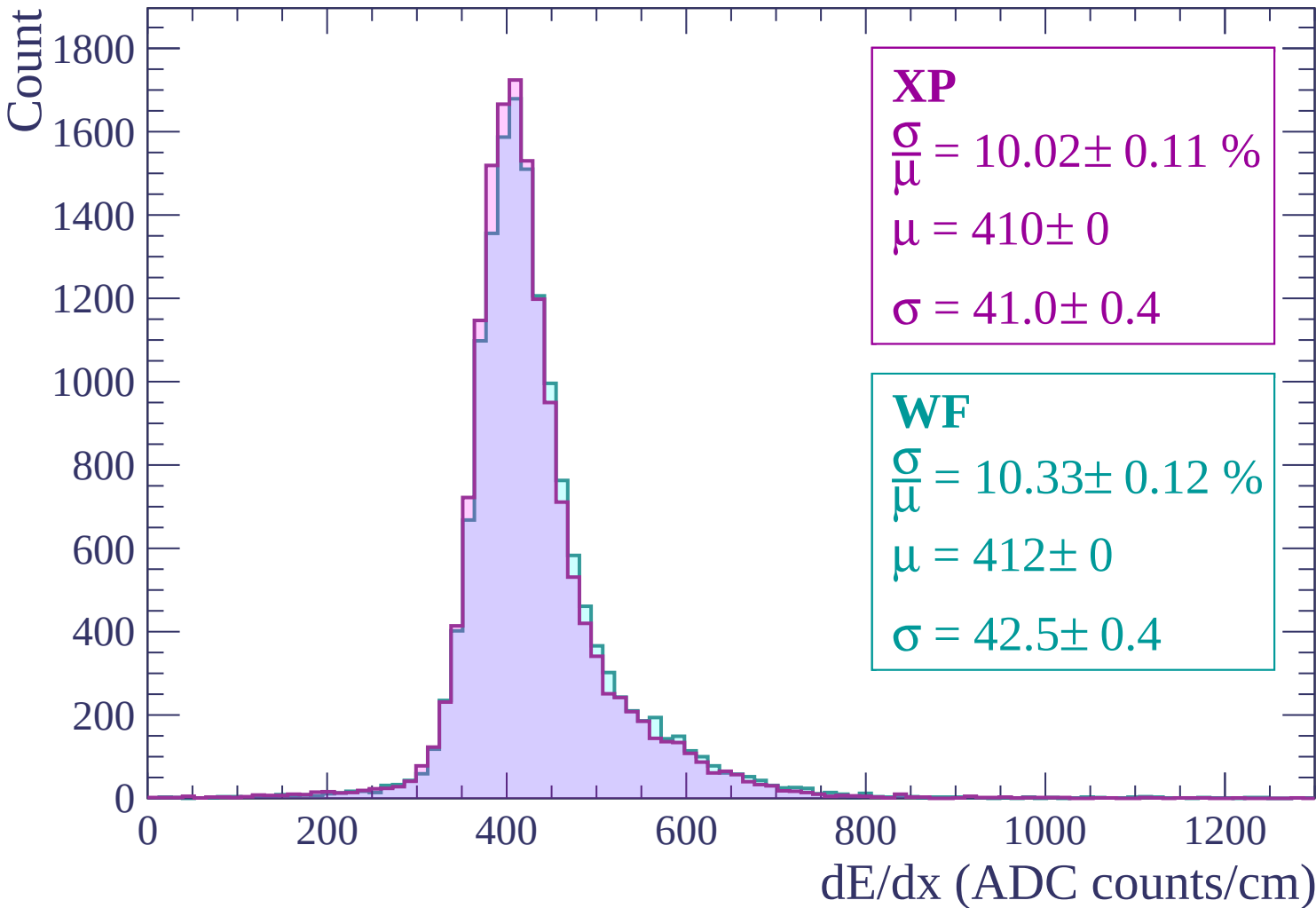
**WF**

$$\frac{\sigma}{\mu} = 10.52 \pm 0.12 \%$$

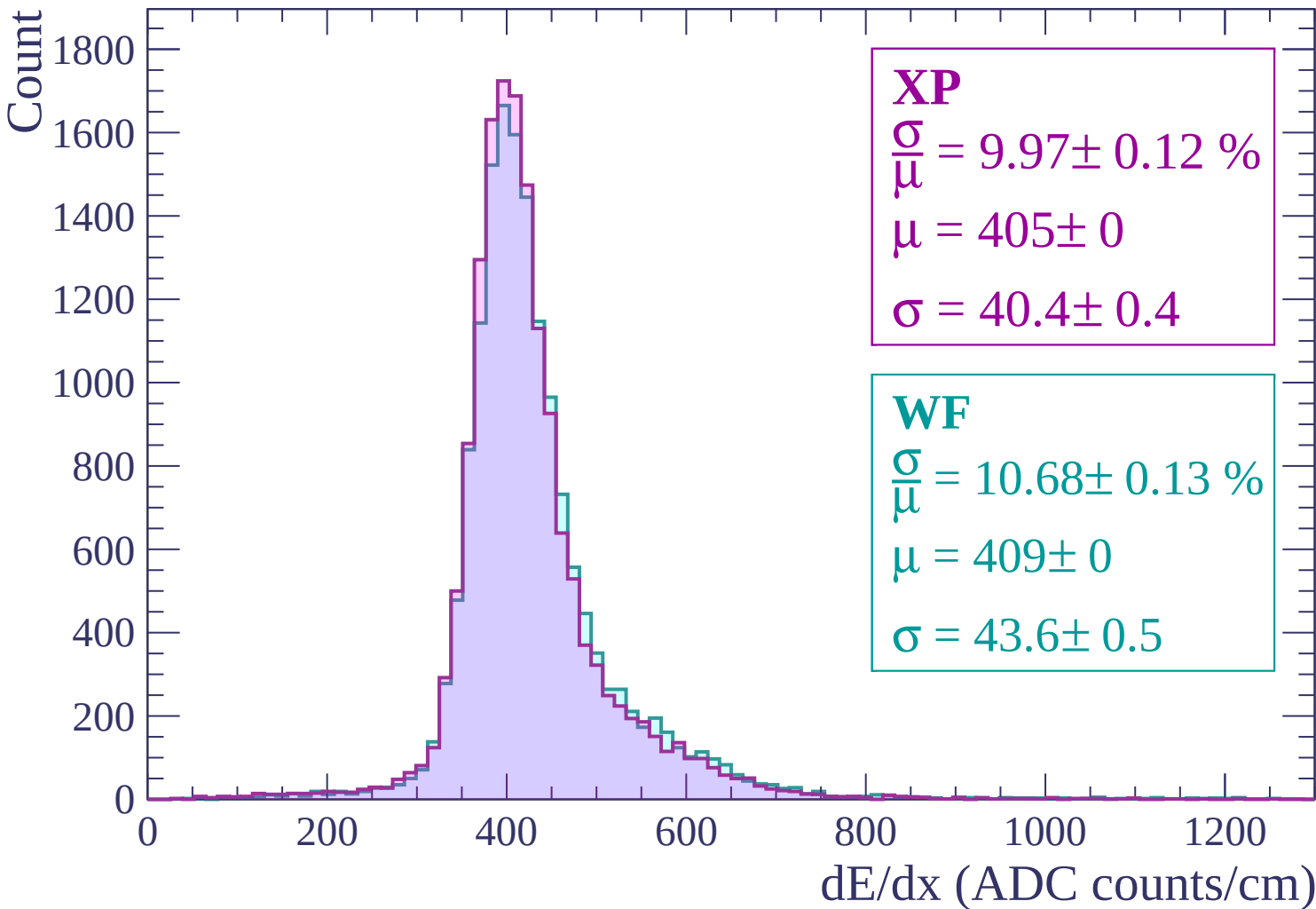
$$\mu = 417 \pm 0$$

$$\sigma = 43.9 \pm 0.4$$

Energy loss |  $-480 < p < -420$



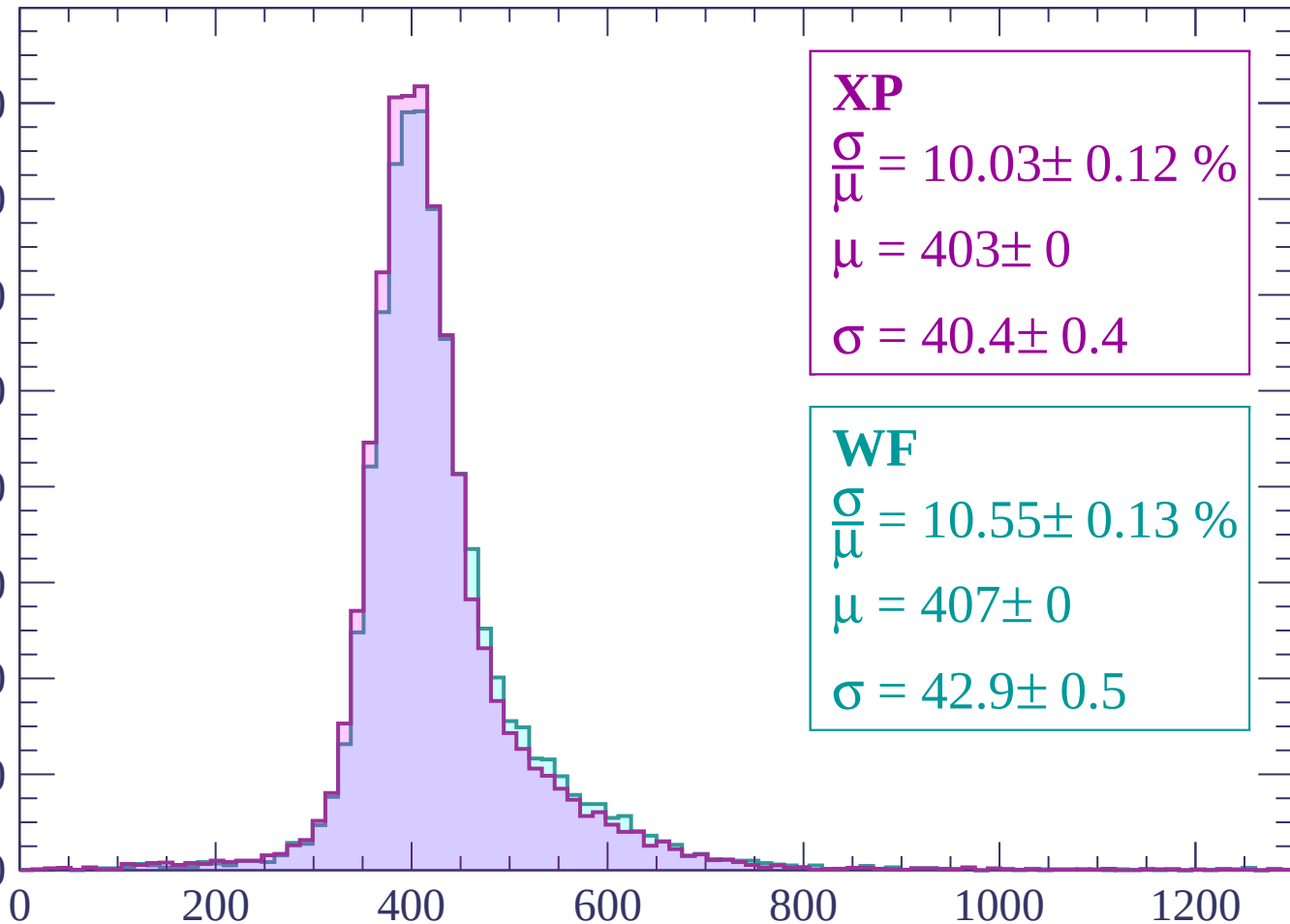
Energy loss |  $-420 < p < -360$



# Energy loss | -360 < p < -300

Count

1600  
1400  
1200  
1000  
800  
600  
400  
200  
0



**XP**

$$\frac{\sigma}{\mu} = 10.03 \pm 0.12 \%$$

$$\mu = 403 \pm 0$$

$$\sigma = 40.4 \pm 0.4$$

**WF**

$$\frac{\sigma}{\mu} = 10.55 \pm 0.13 \%$$

$$\mu = 407 \pm 0$$

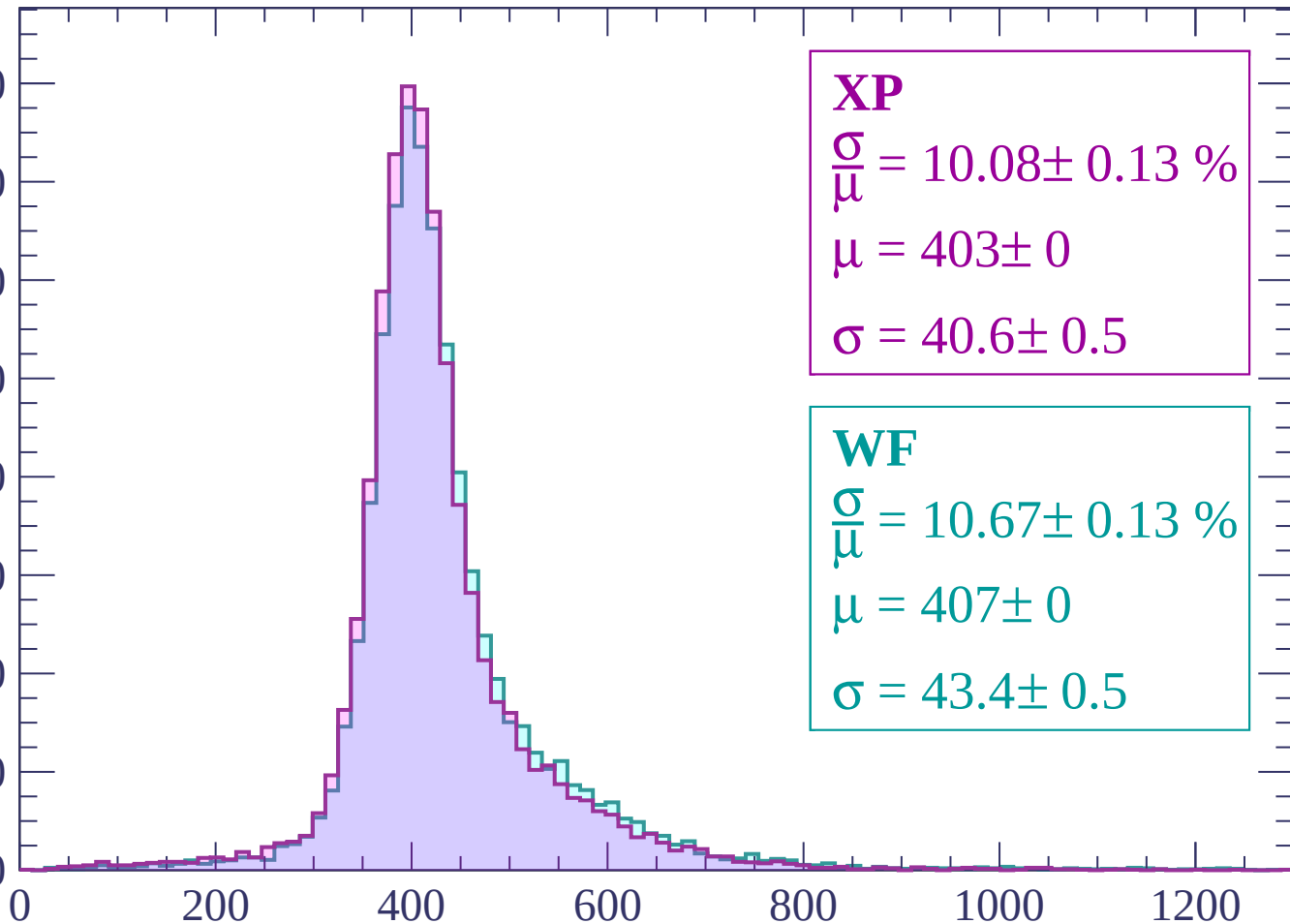
$$\sigma = 42.9 \pm 0.5$$

dE/dx (ADC counts/cm)

# Energy loss | $-300 < p < -240$

Count

1600  
1400  
1200  
1000  
800  
600  
400  
200  
0



**XP**

$$\frac{\sigma}{\mu} = 10.08 \pm 0.13 \%$$

$$\mu = 403 \pm 0$$

$$\sigma = 40.6 \pm 0.5$$

**WF**

$$\frac{\sigma}{\mu} = 10.67 \pm 0.13 \%$$

$$\mu = 407 \pm 0$$

$$\sigma = 43.4 \pm 0.5$$

dE/dx (ADC counts/cm)

# Energy loss | -240 < p < -180

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

**XP**

$$\frac{\sigma}{\mu} = 11.46 \pm 0.16 \%$$

$$\mu = 415 \pm 1$$

$$\sigma = 47.5 \pm 0.6$$

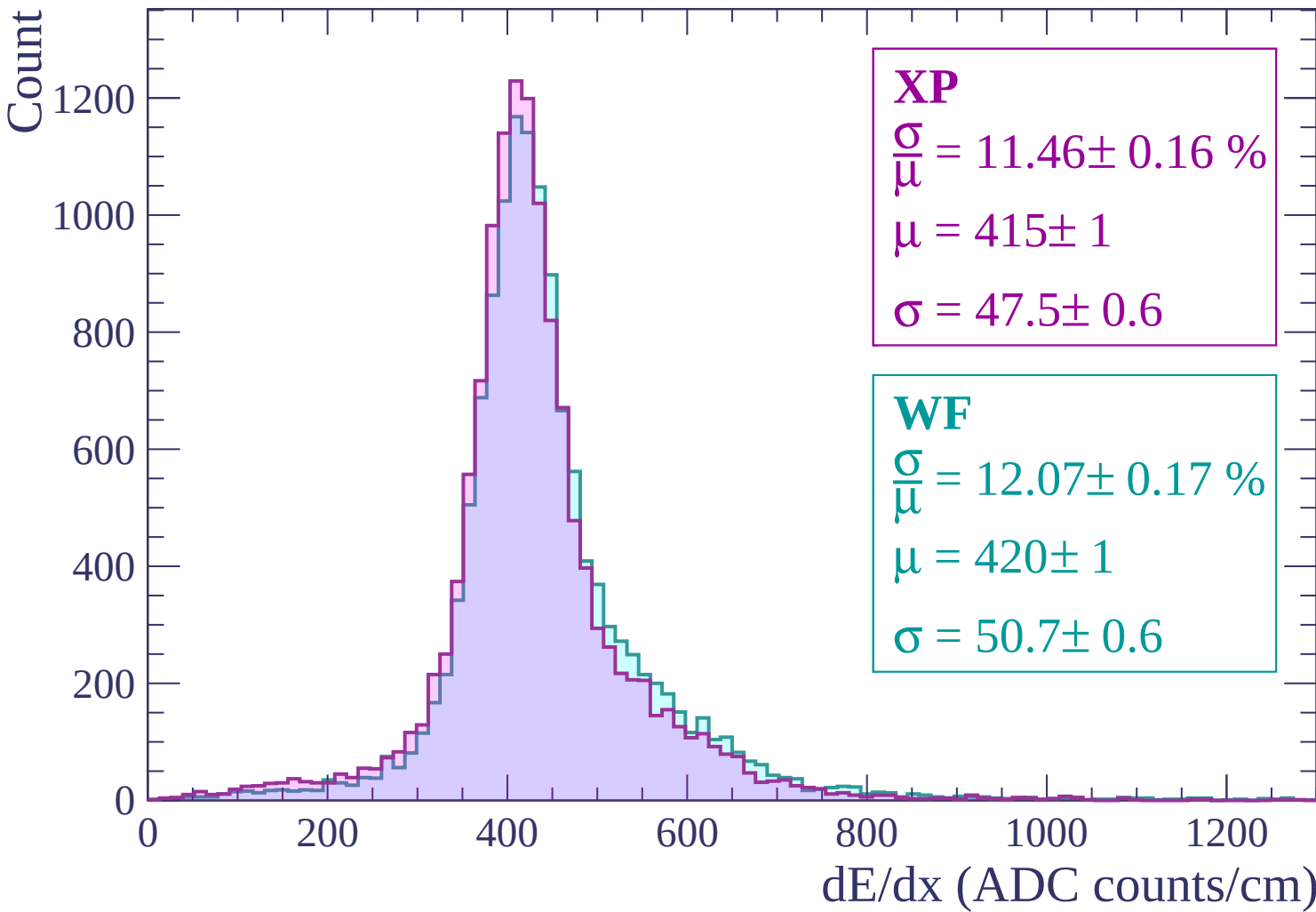
**WF**

$$\frac{\sigma}{\mu} = 12.07 \pm 0.17 \%$$

$$\mu = 420 \pm 1$$

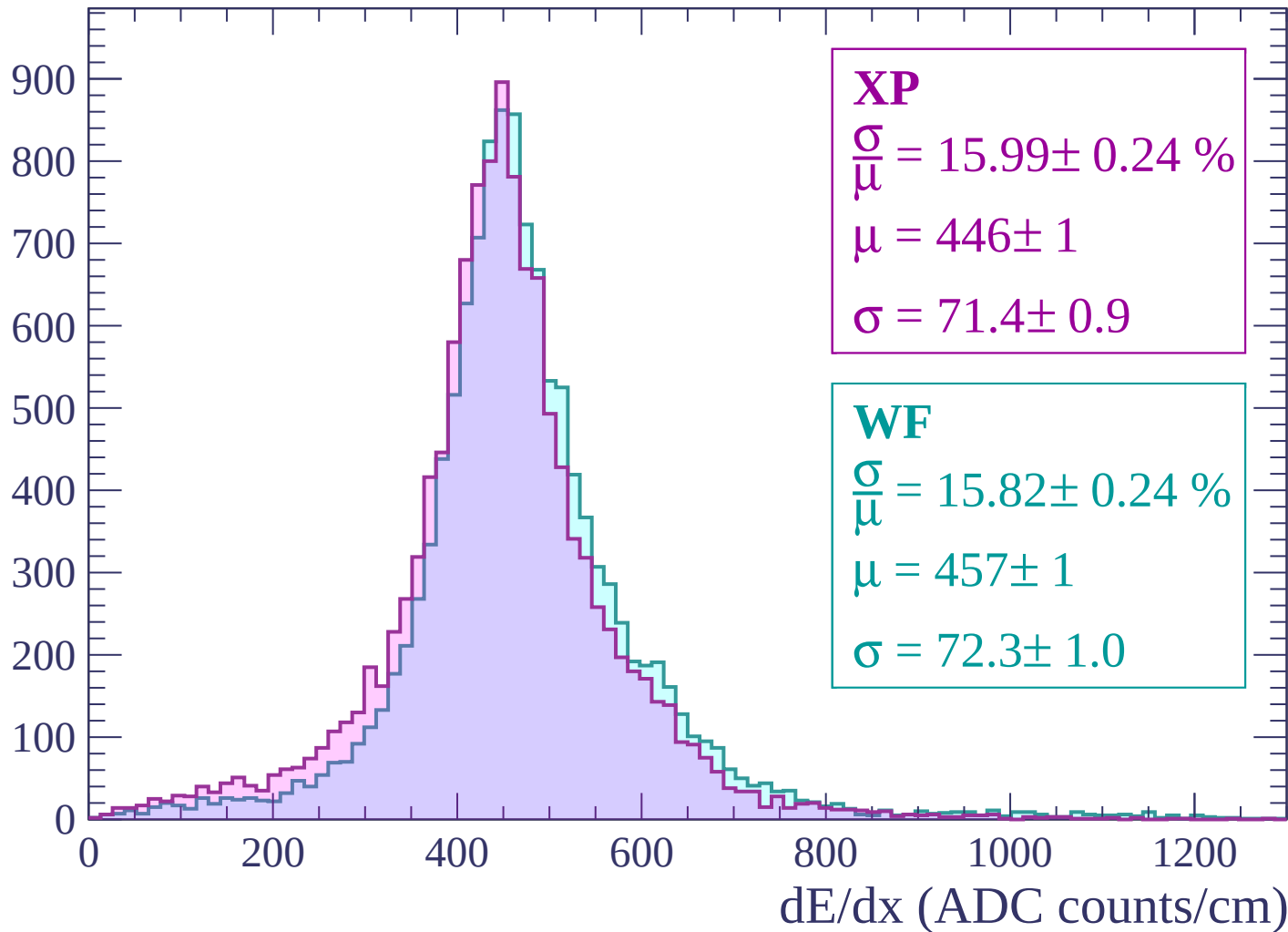
$$\sigma = 50.7 \pm 0.6$$

dE/dx (ADC counts/cm)



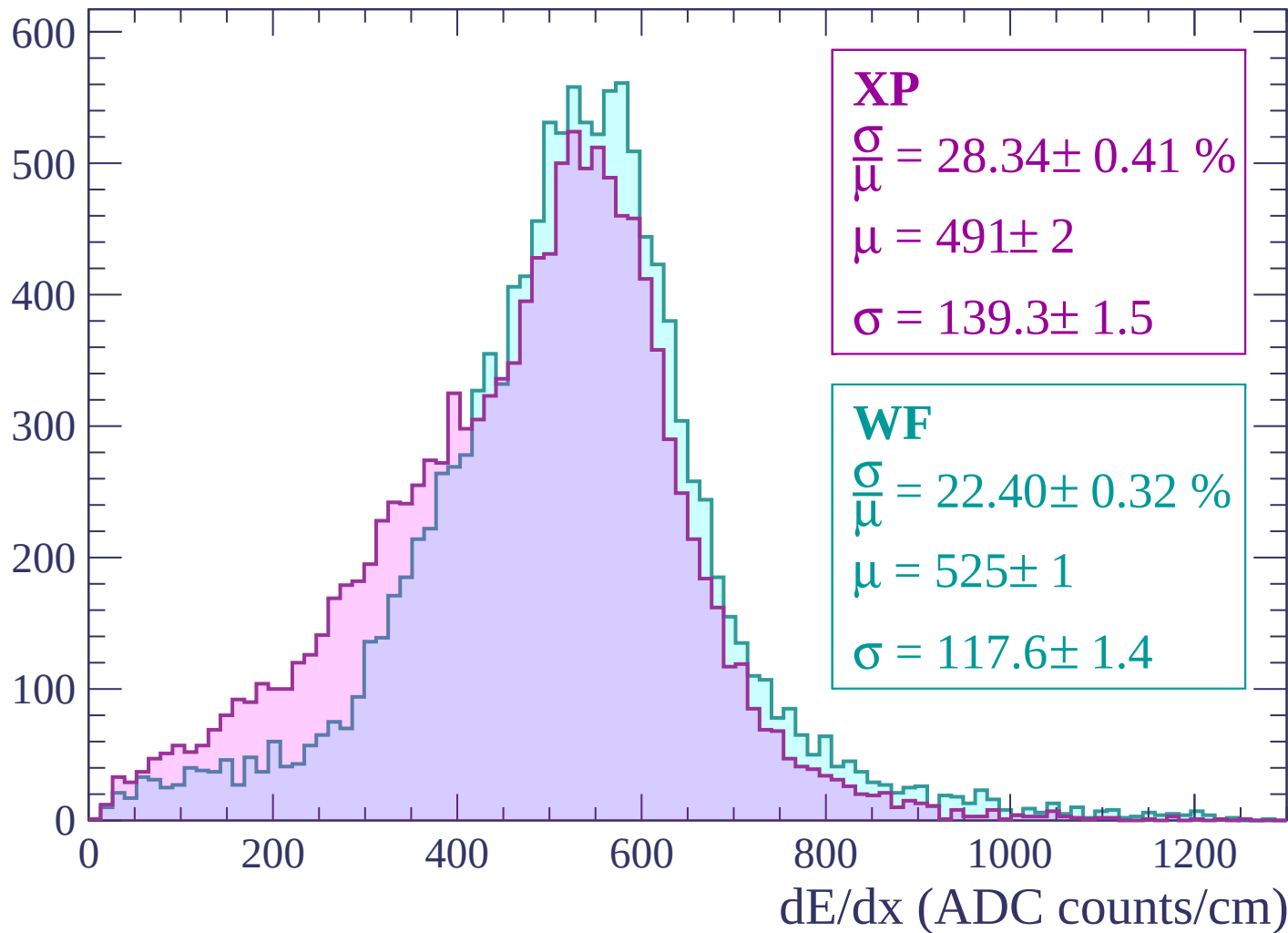
# Energy loss | -180 < p < -120

Count



Energy loss |  $-120 < p < -60$

Count



Energy loss |  $-60 < p < 0$

Count

1600  
1400  
1200  
1000  
800  
600  
400  
200  
0

**XP**

$$\frac{\sigma}{\mu} = 12.78 \pm 0.13 \%$$

$$\mu = 544 \pm 1$$

$$\sigma = 69.5 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 13.46 \pm 0.13 \%$$

$$\mu = 547 \pm 1$$

$$\sigma = 73.6 \pm 0.7$$

0

200

400

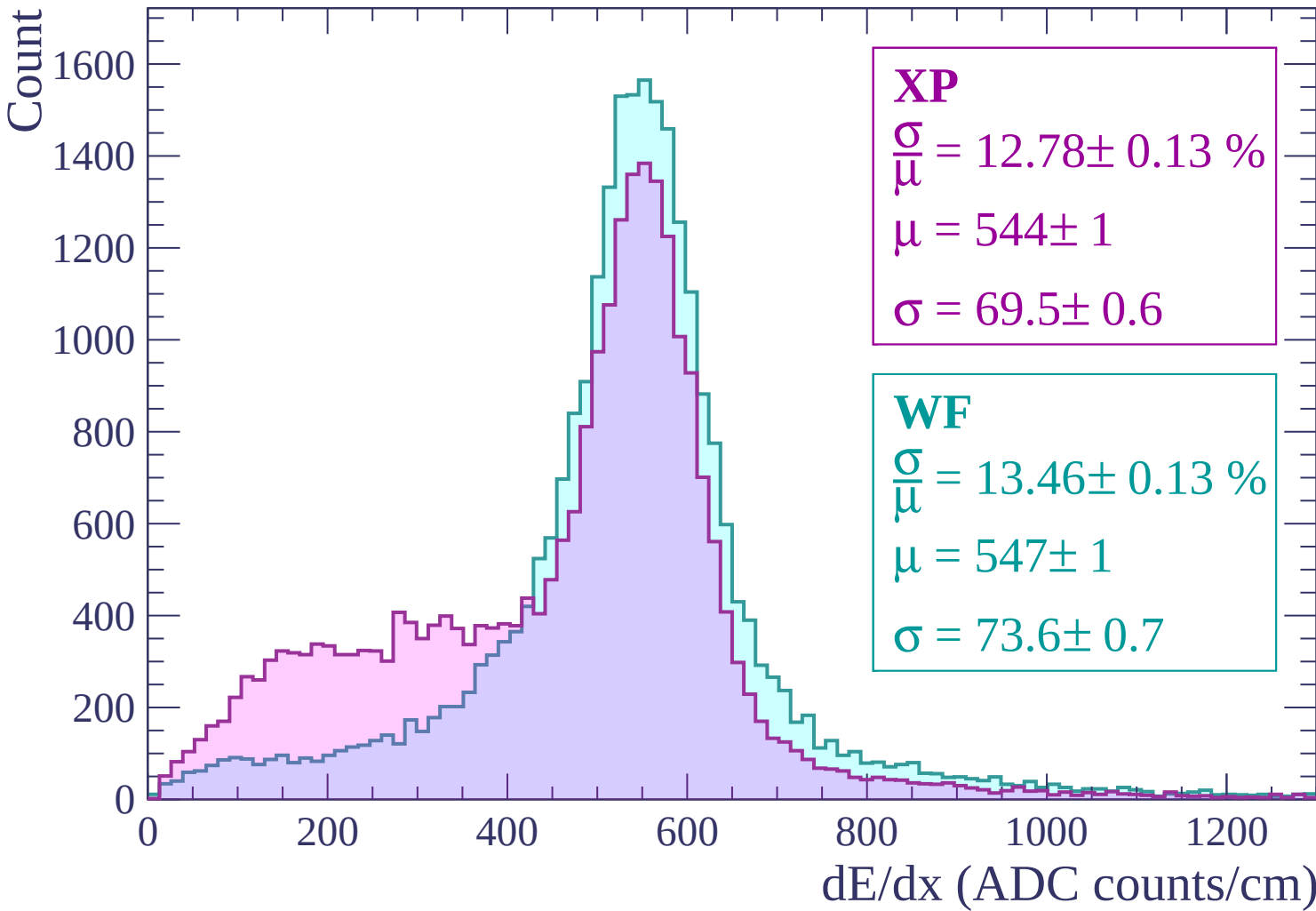
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss |  $0 < p < 60$

Count

4000  
3500  
3000  
2500  
2000  
1500  
1000  
500  
0

**XP**

$$\frac{\sigma}{\mu} = 18.00 \pm 0.12 \%$$

$$\mu = 485 \pm 0$$

$$\sigma = 87.2 \pm 0.5$$

**WF**

$$\frac{\sigma}{\mu} = 15.85 \pm 0.10 \%$$

$$\mu = 501 \pm 0$$

$$\sigma = 79.3 \pm 0.4$$

0

200

400

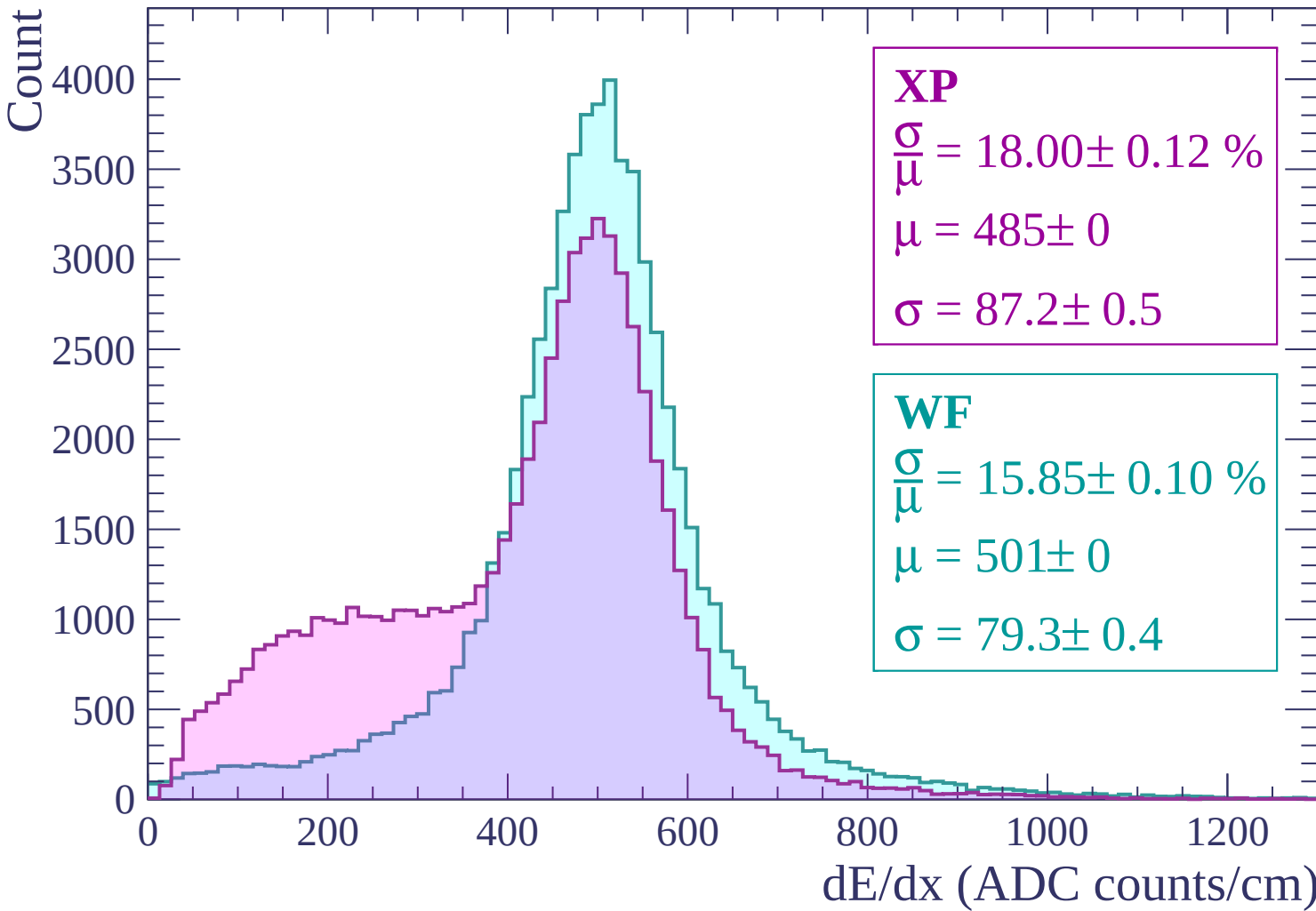
600

800

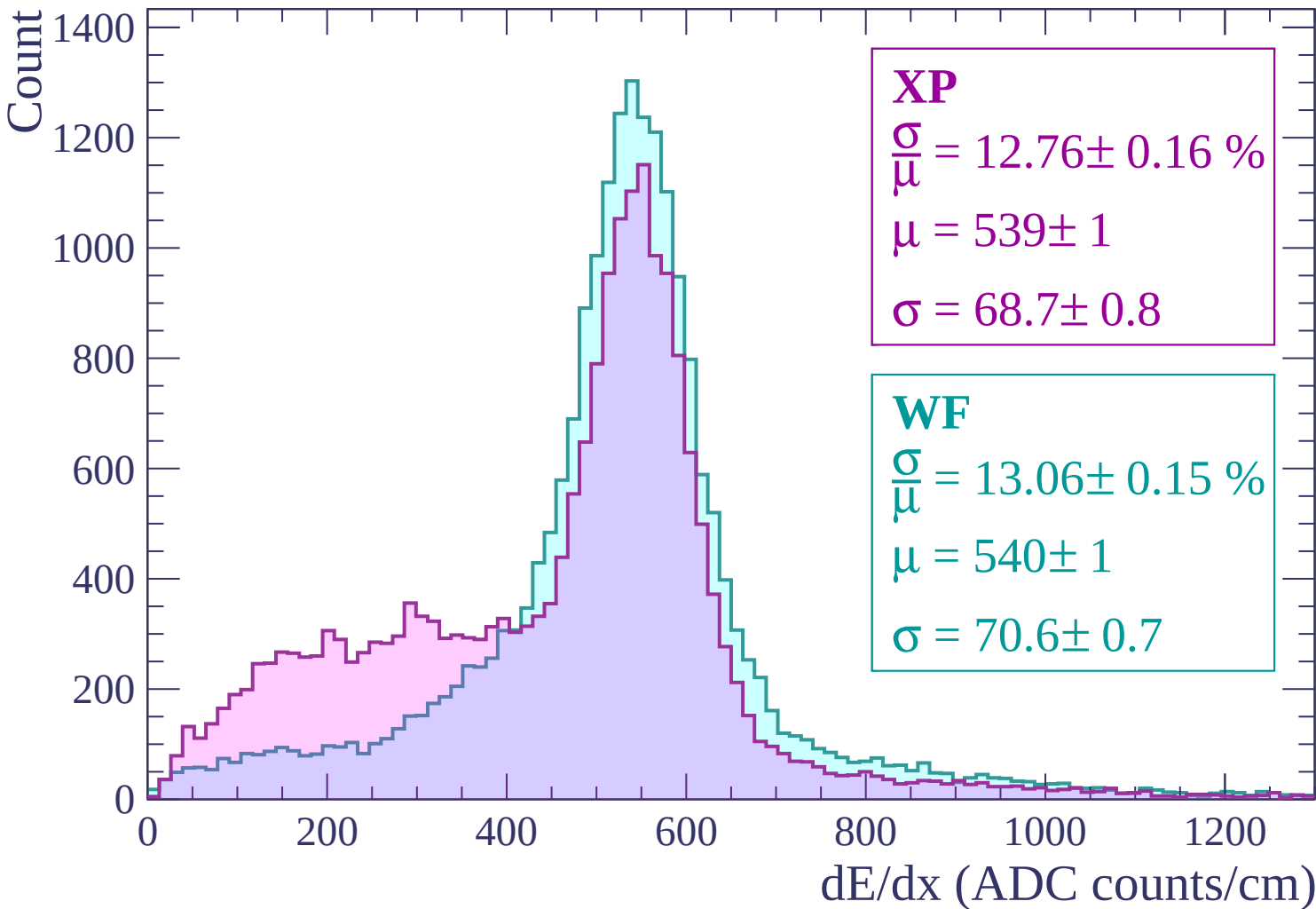
1000

1200

dE/dx (ADC counts/cm)

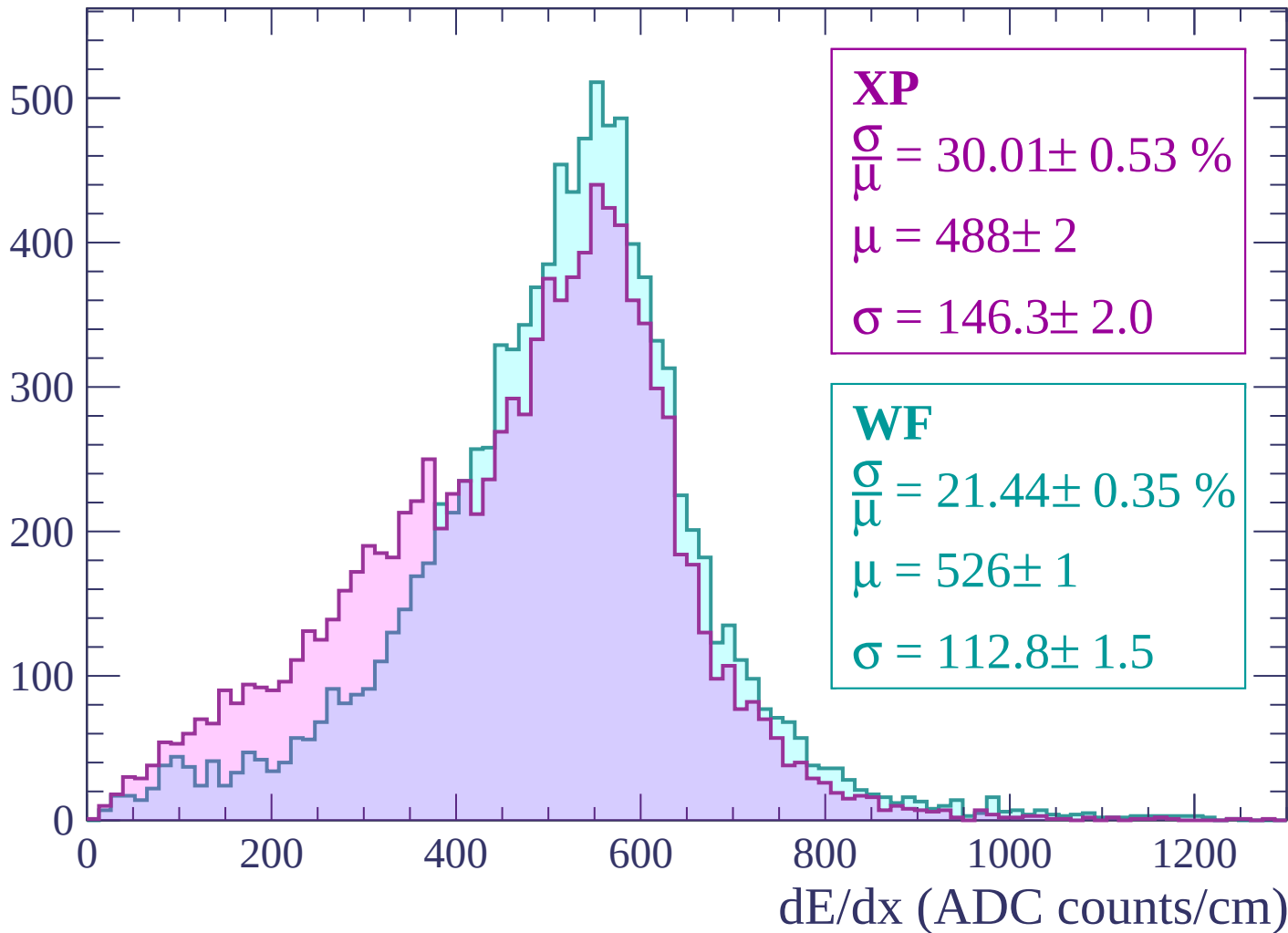


# Energy loss | $60 < p < 120$



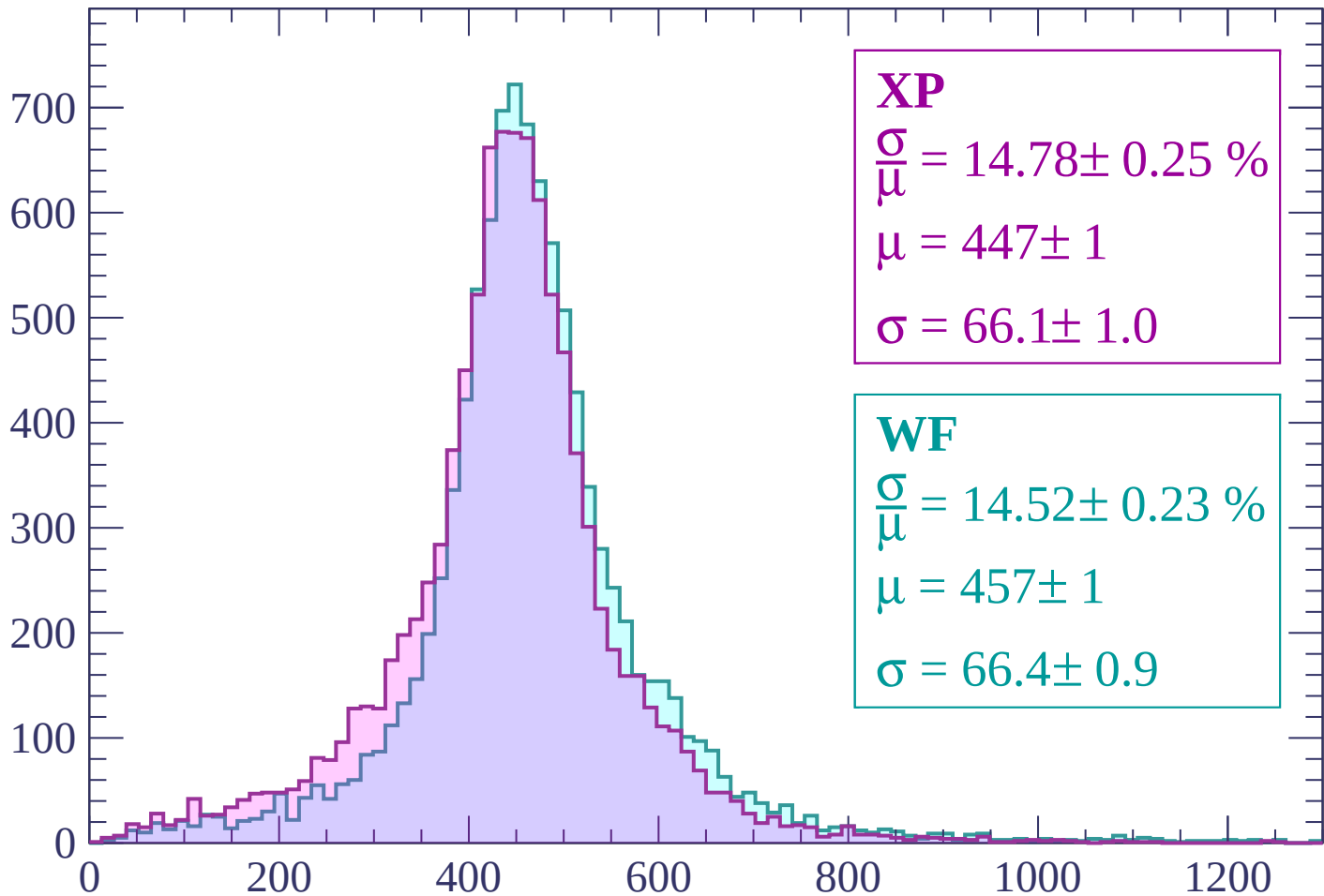
Energy loss |  $120 < p < 180$

Count



Energy loss |  $180 < p < 240$

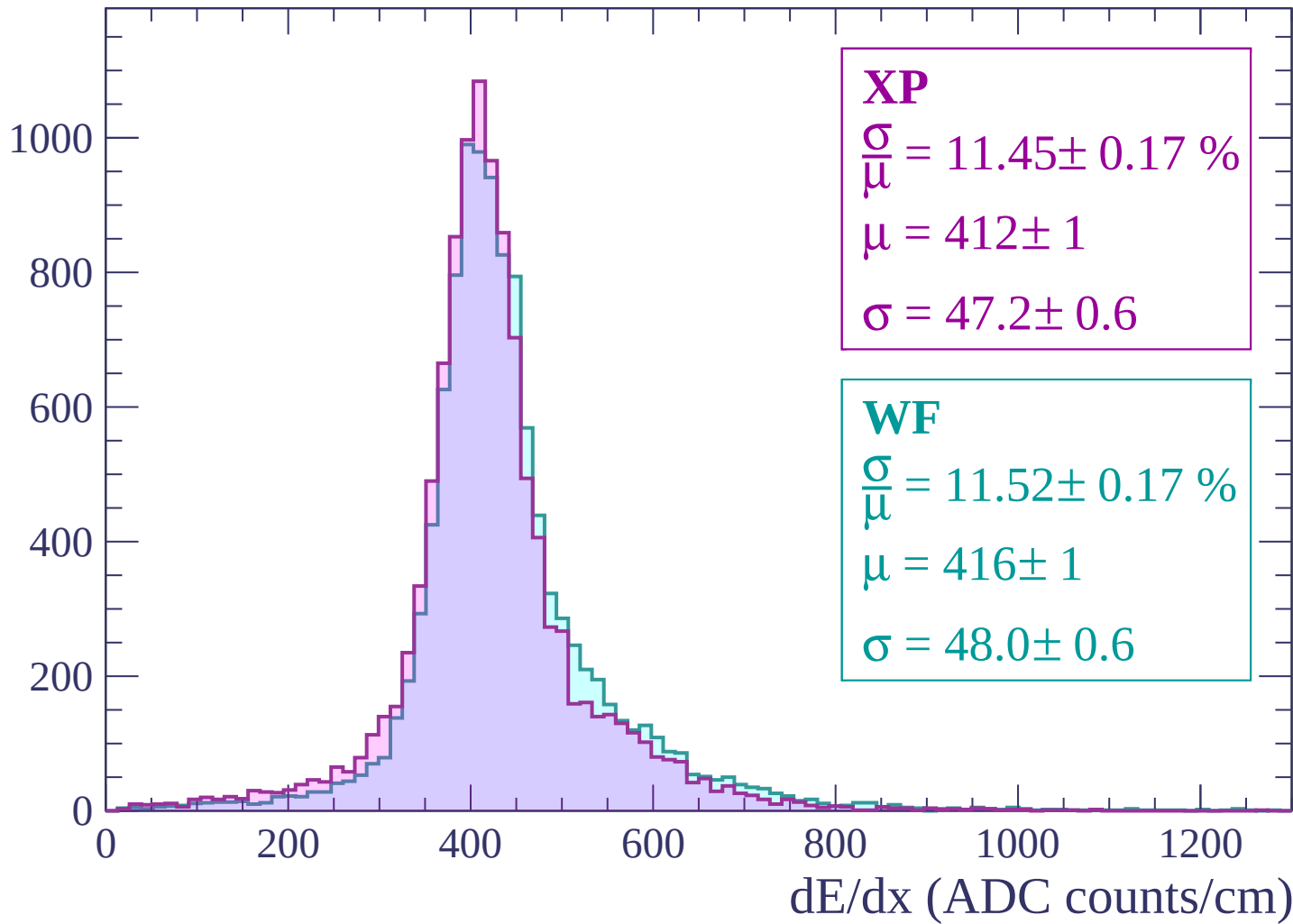
Count



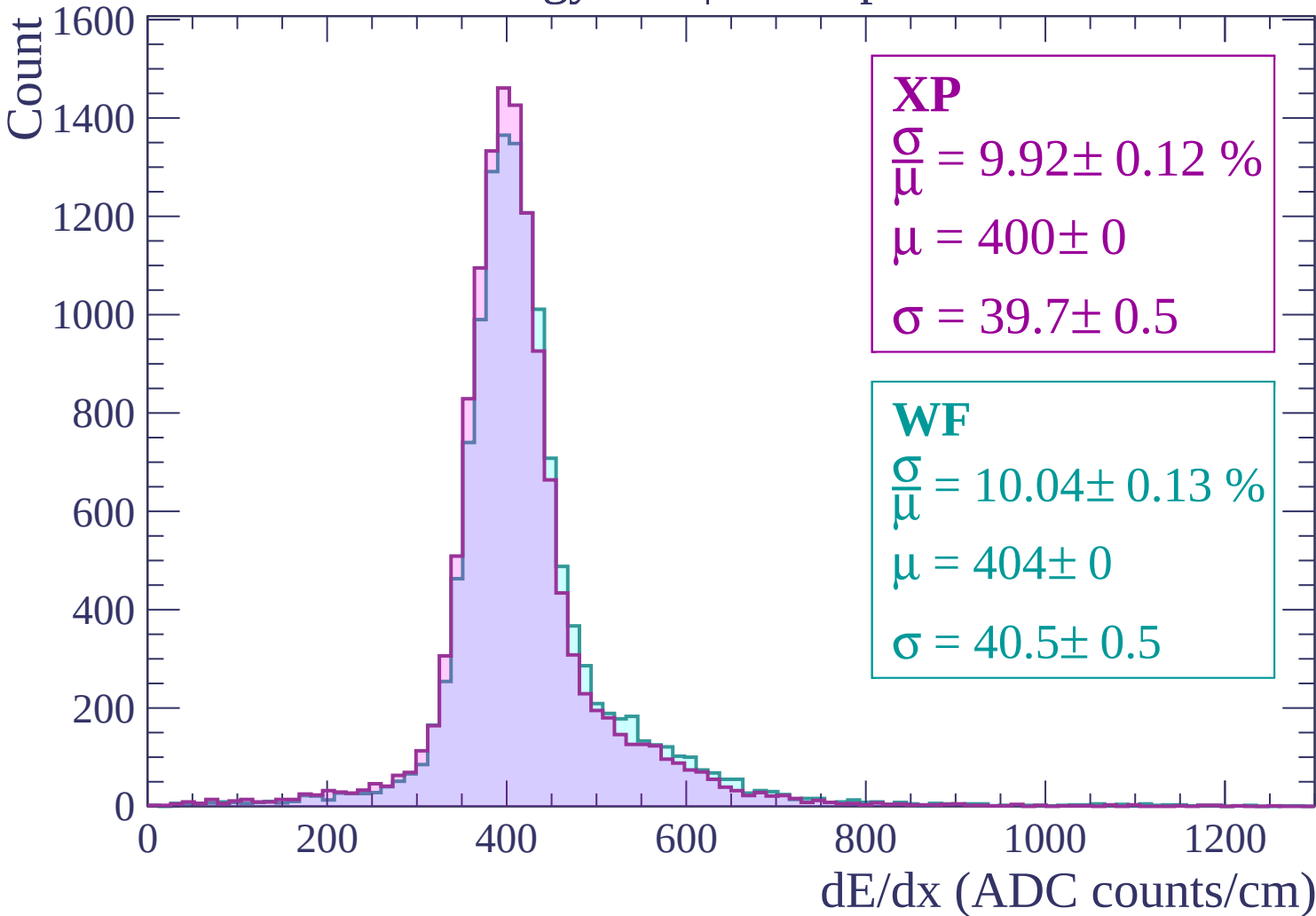
dE/dx (ADC counts/cm)

Energy loss |  $240 < p < 300$

Count



Energy loss |  $300 < p < 360$



Energy loss |  $360 < p < 420$

Count

1600  
1400  
1200  
1000  
800  
600  
400  
200  
0

**XP**

$$\frac{\sigma}{\mu} = 9.67 \pm 0.13 \%$$

$$\mu = 397 \pm 0$$

$$\sigma = 38.4 \pm 0.5$$

**WF**

$$\frac{\sigma}{\mu} = 10.21 \pm 0.13 \%$$

$$\mu = 401 \pm 0$$

$$\sigma = 41.0 \pm 0.5$$

0

200

400

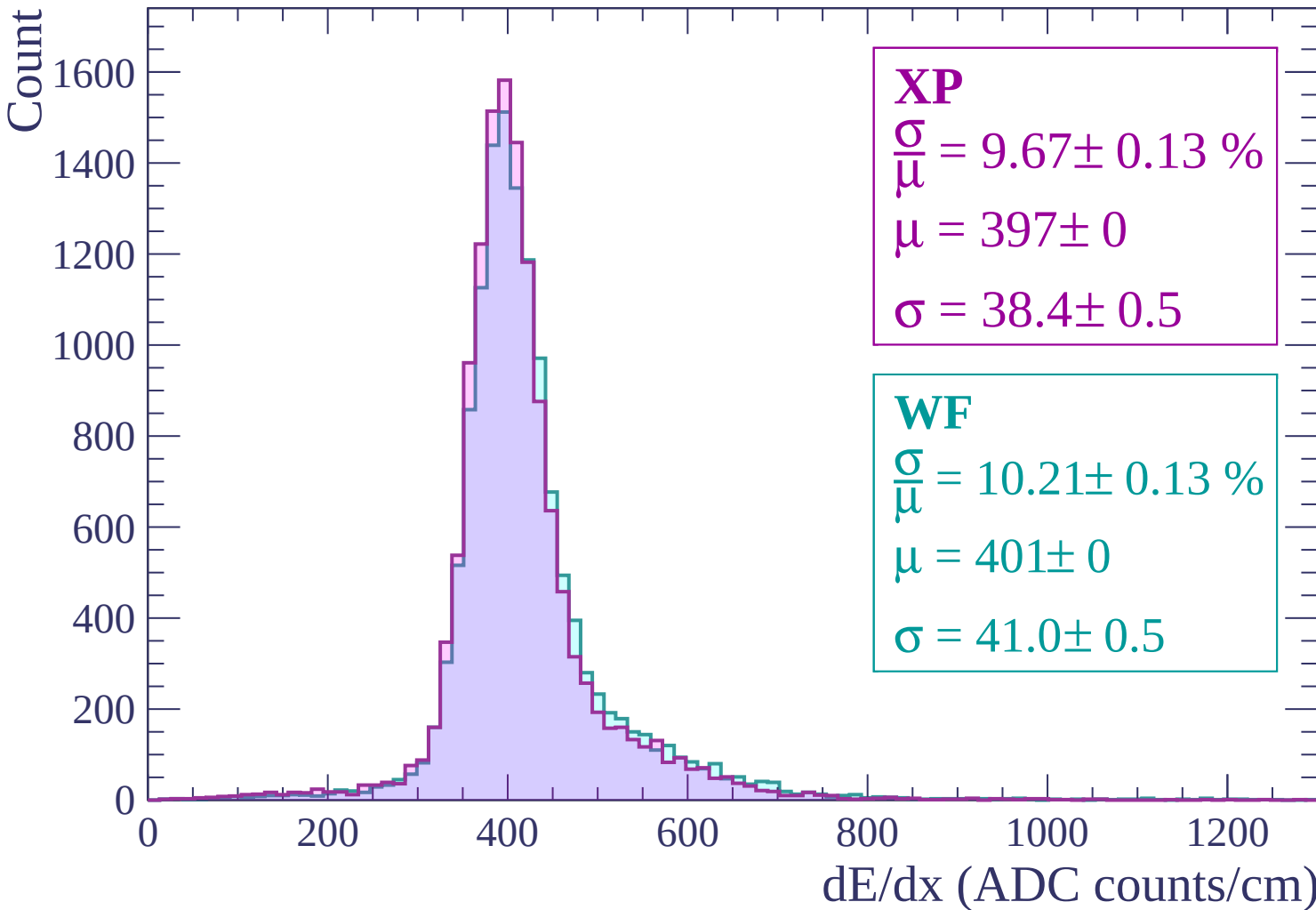
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss |  $420 < p < 480$

Count

1600  
1400  
1200  
1000  
800  
600  
400  
200  
0

**XP**

$$\frac{\sigma}{\mu} = 9.28 \pm 0.12 \%$$

$$\mu = 399 \pm 0$$

$$\sigma = 37.1 \pm 0.4$$

**WF**

$$\frac{\sigma}{\mu} = 9.89 \pm 0.12 \%$$

$$\mu = 403 \pm 0$$

$$\sigma = 39.8 \pm 0.5$$

0

200

400

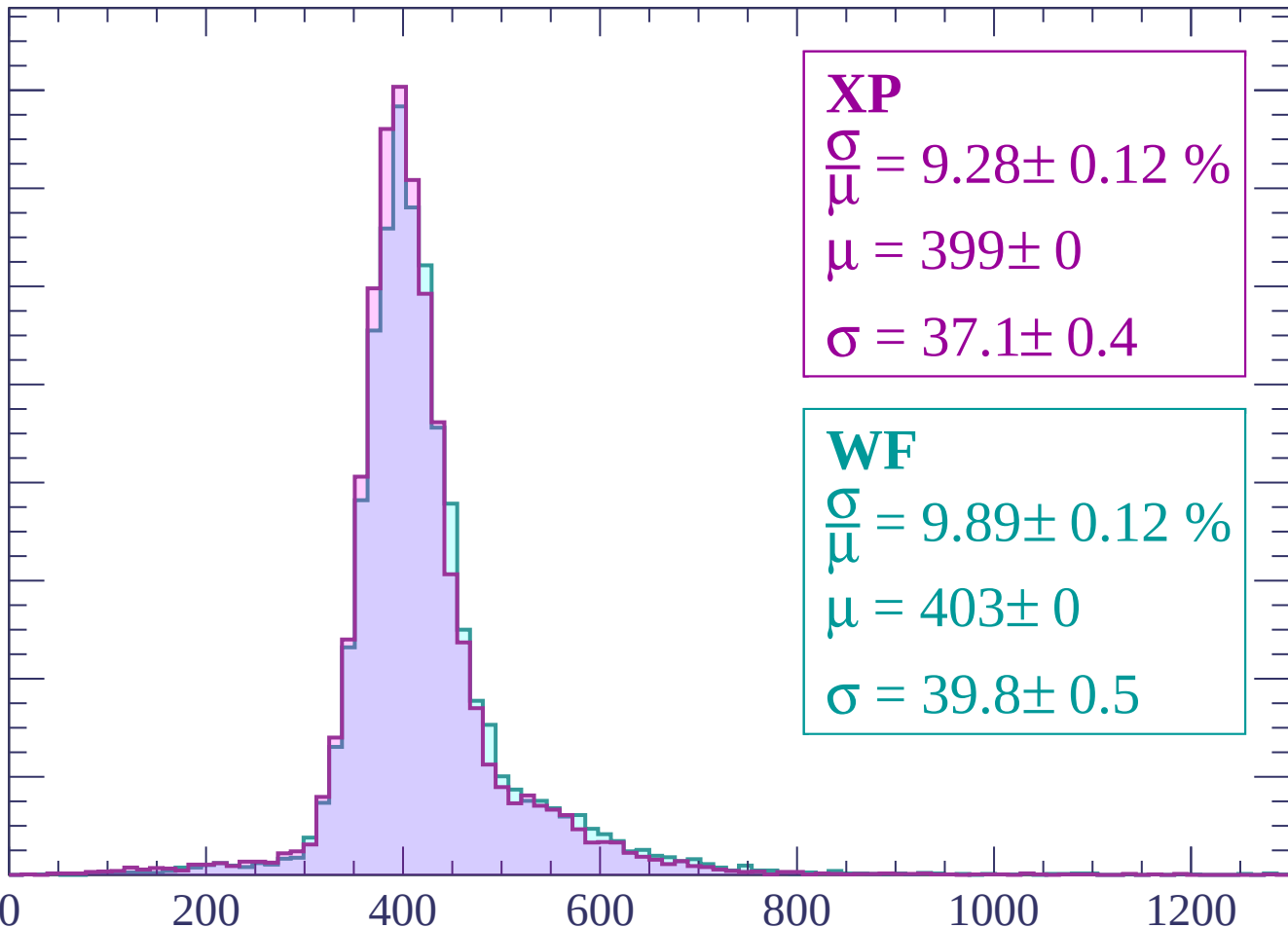
600

800

1000

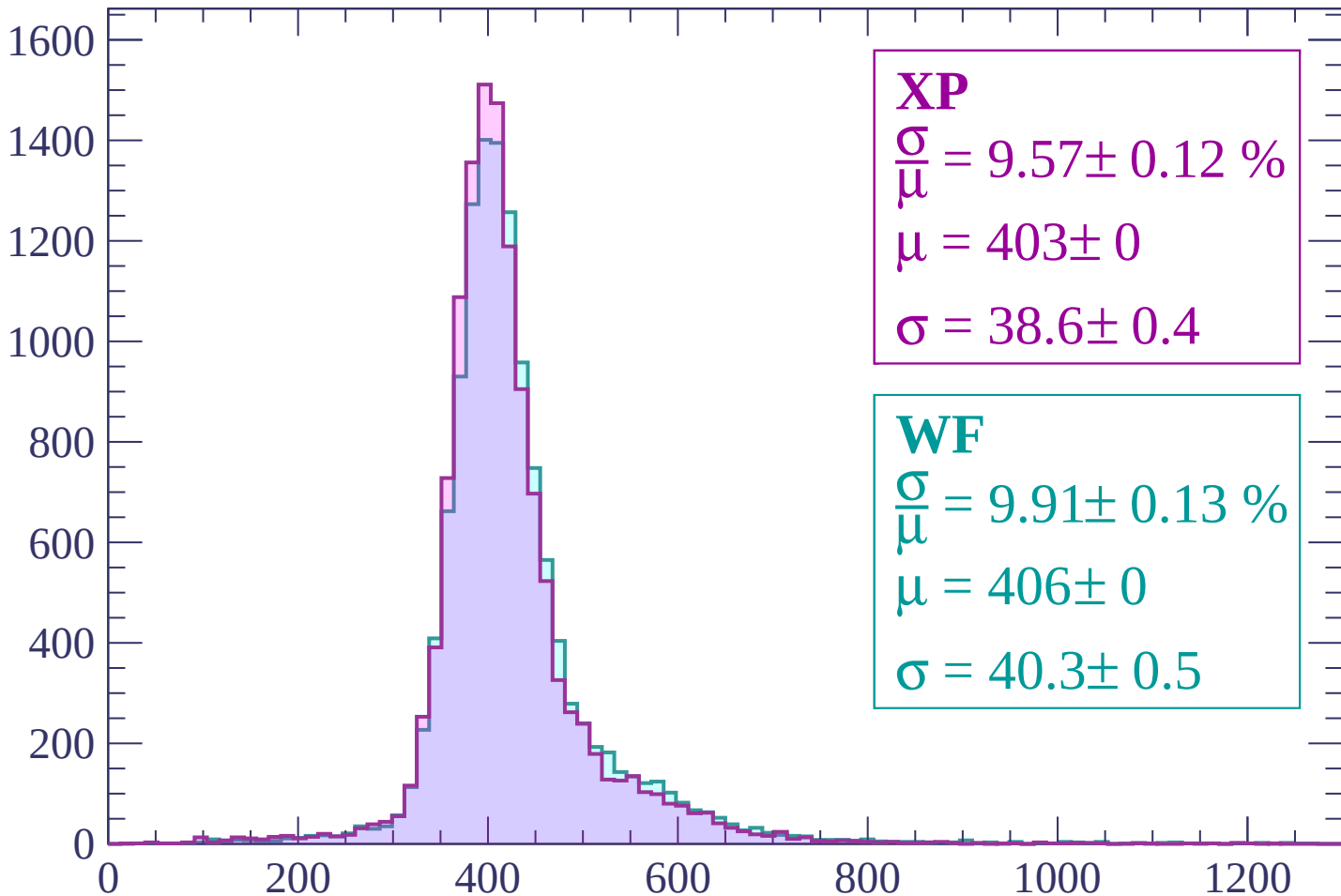
1200

dE/dx (ADC counts/cm)



Energy loss |  $480 < p < 540$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.57 \pm 0.12 \%$$

$$\mu = 403 \pm 0$$

$$\sigma = 38.6 \pm 0.4$$

**WF**

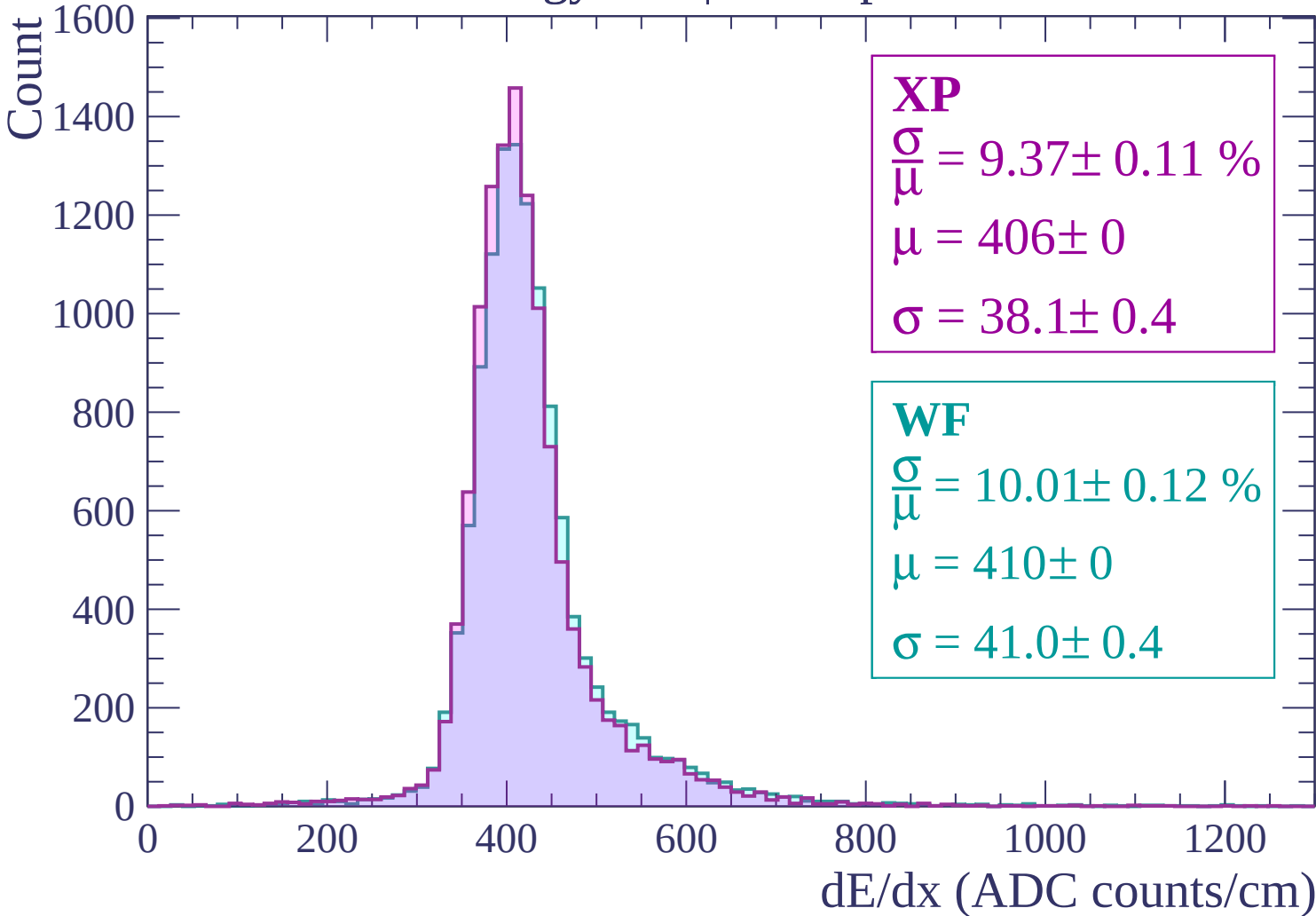
$$\frac{\sigma}{\mu} = 9.91 \pm 0.13 \%$$

$$\mu = 406 \pm 0$$

$$\sigma = 40.3 \pm 0.5$$

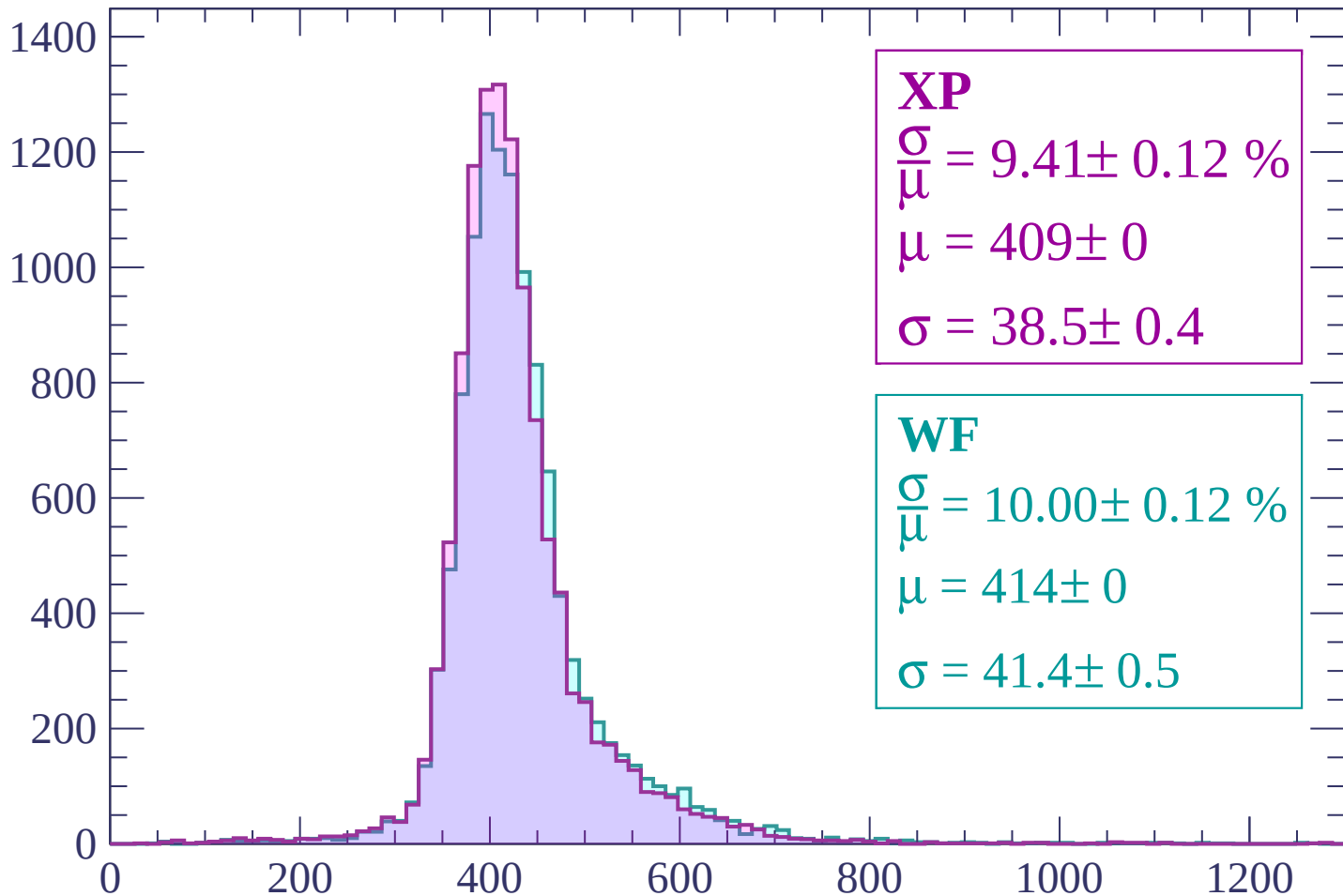
dE/dx (ADC counts/cm)

Energy loss |  $540 < p < 600$



Energy loss |  $600 < p < 660$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.41 \pm 0.12 \%$$

$$\mu = 409 \pm 0$$

$$\sigma = 38.5 \pm 0.4$$

**WF**

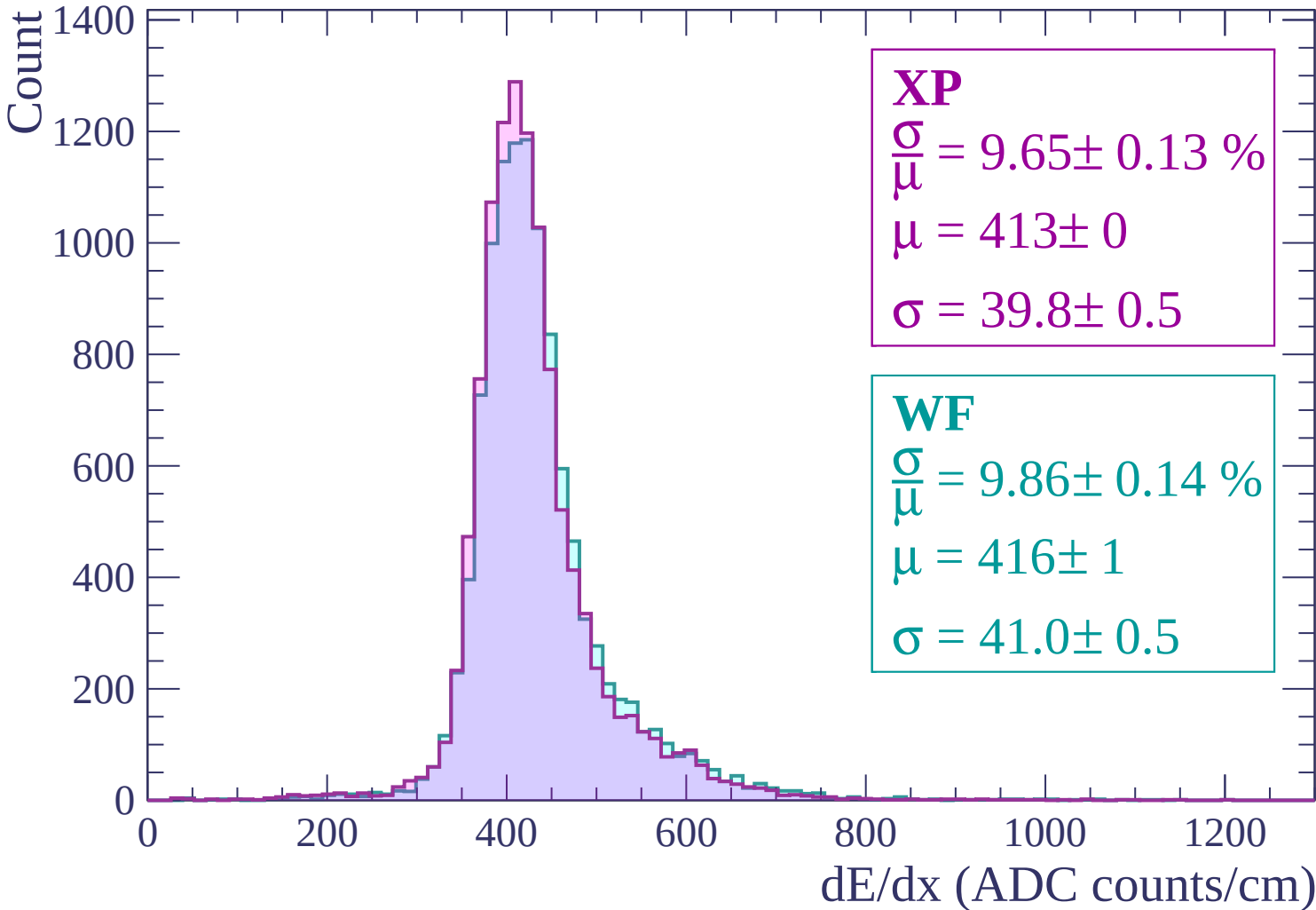
$$\frac{\sigma}{\mu} = 10.00 \pm 0.12 \%$$

$$\mu = 414 \pm 0$$

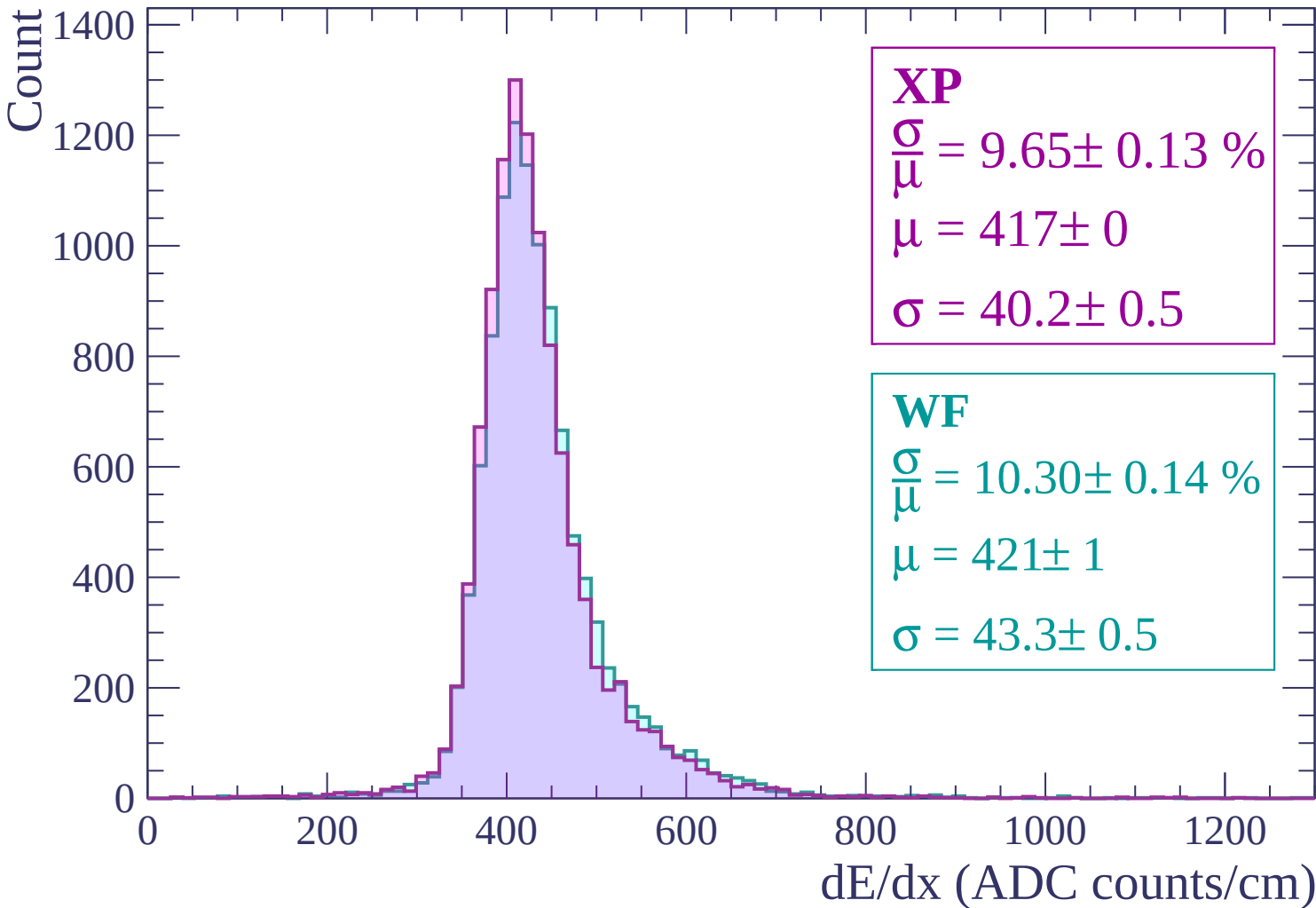
$$\sigma = 41.4 \pm 0.5$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $660 < p < 720$



Energy loss |  $720 < p < 780$



Energy loss |  $780 < p < 840$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

**XP**

$$\frac{\sigma}{\mu} = 9.36 \pm 0.13 \%$$

$$\mu = 420 \pm 1$$

$$\sigma = 39.3 \pm 0.5$$

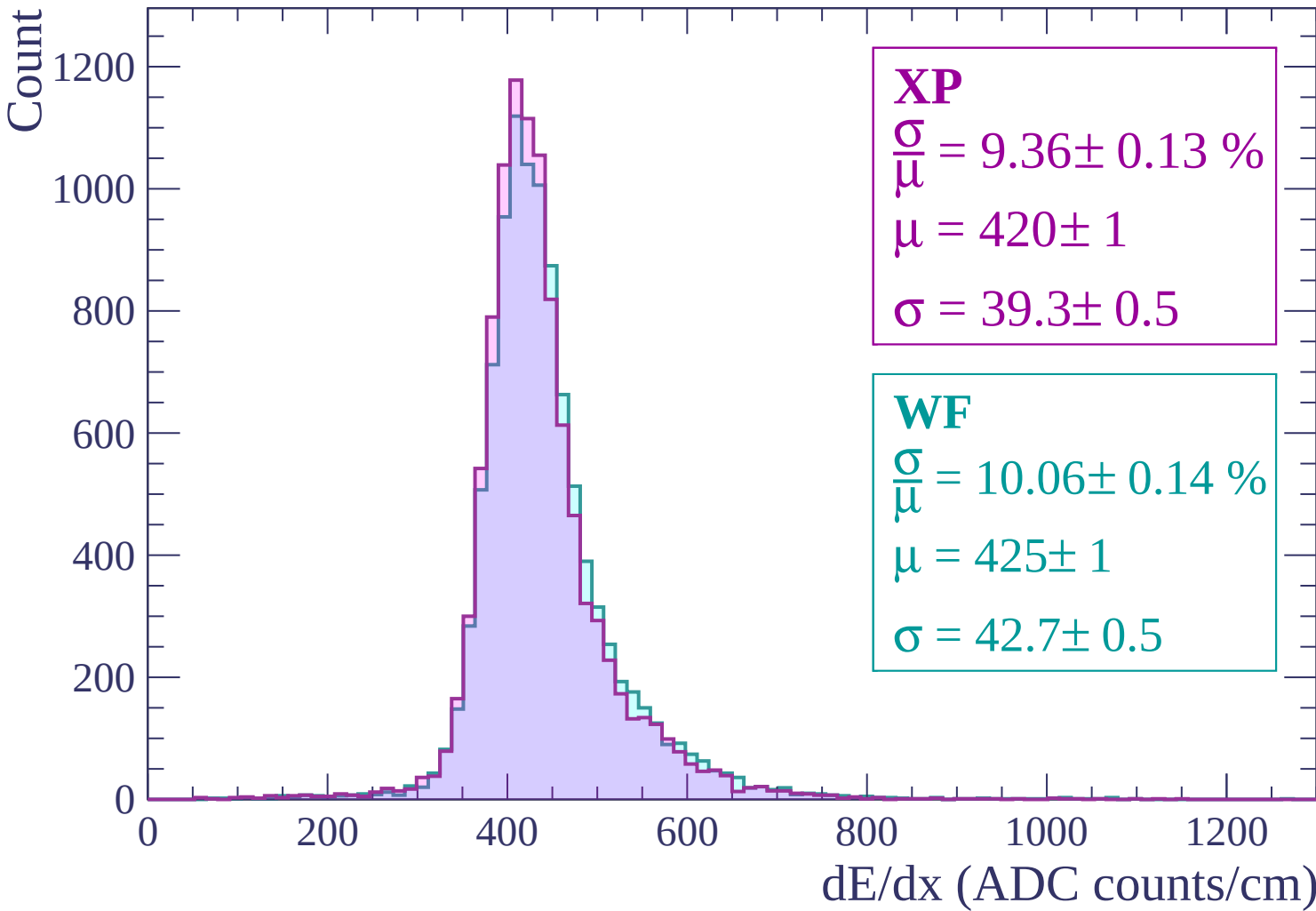
**WF**

$$\frac{\sigma}{\mu} = 10.06 \pm 0.14 \%$$

$$\mu = 425 \pm 1$$

$$\sigma = 42.7 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss |  $840 < p < 900$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

**XP**

$$\frac{\sigma}{\mu} = 9.77 \pm 0.13 \%$$

$$\mu = 425 \pm 1$$

$$\sigma = 41.5 \pm 0.5$$

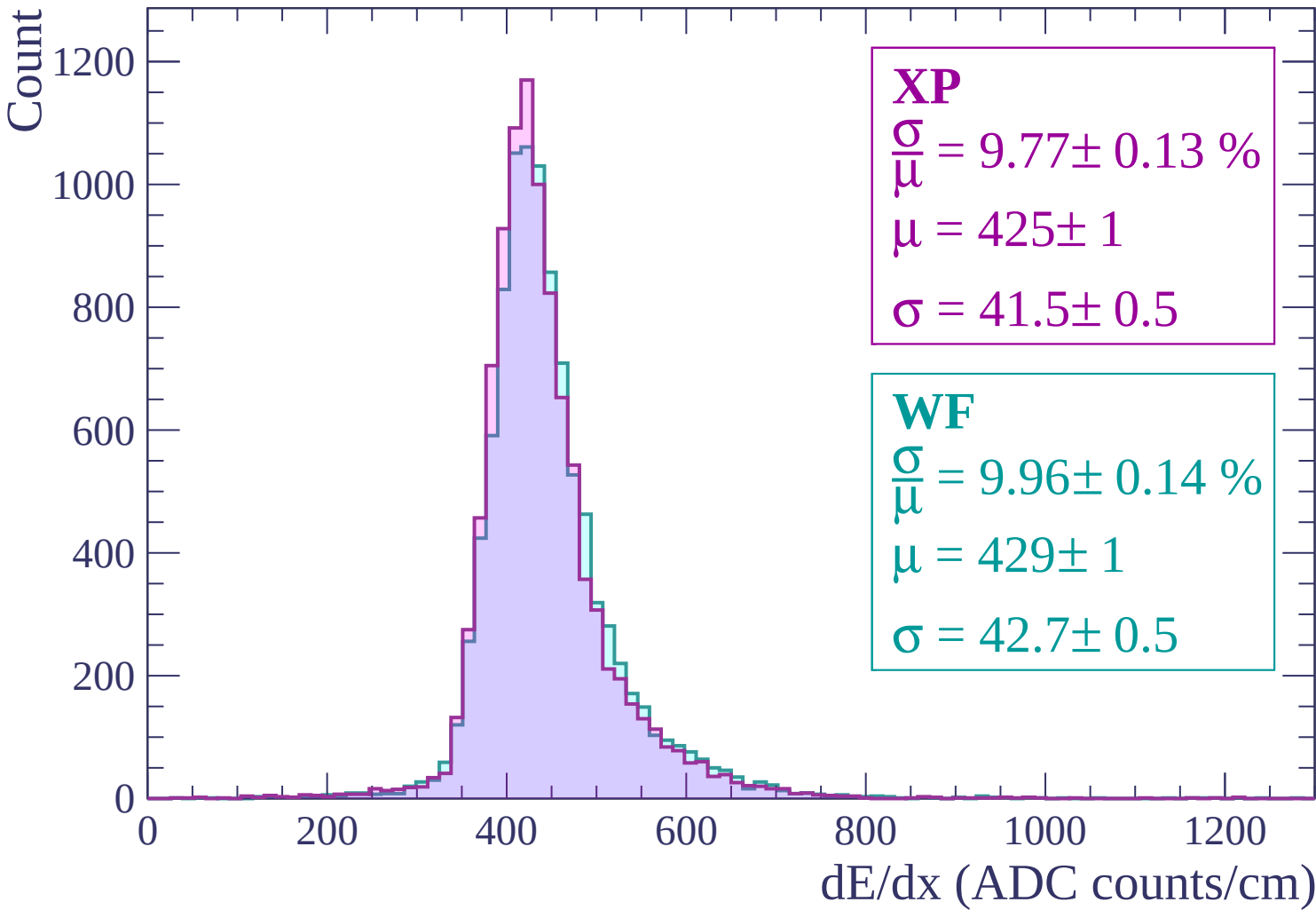
**WF**

$$\frac{\sigma}{\mu} = 9.96 \pm 0.14 \%$$

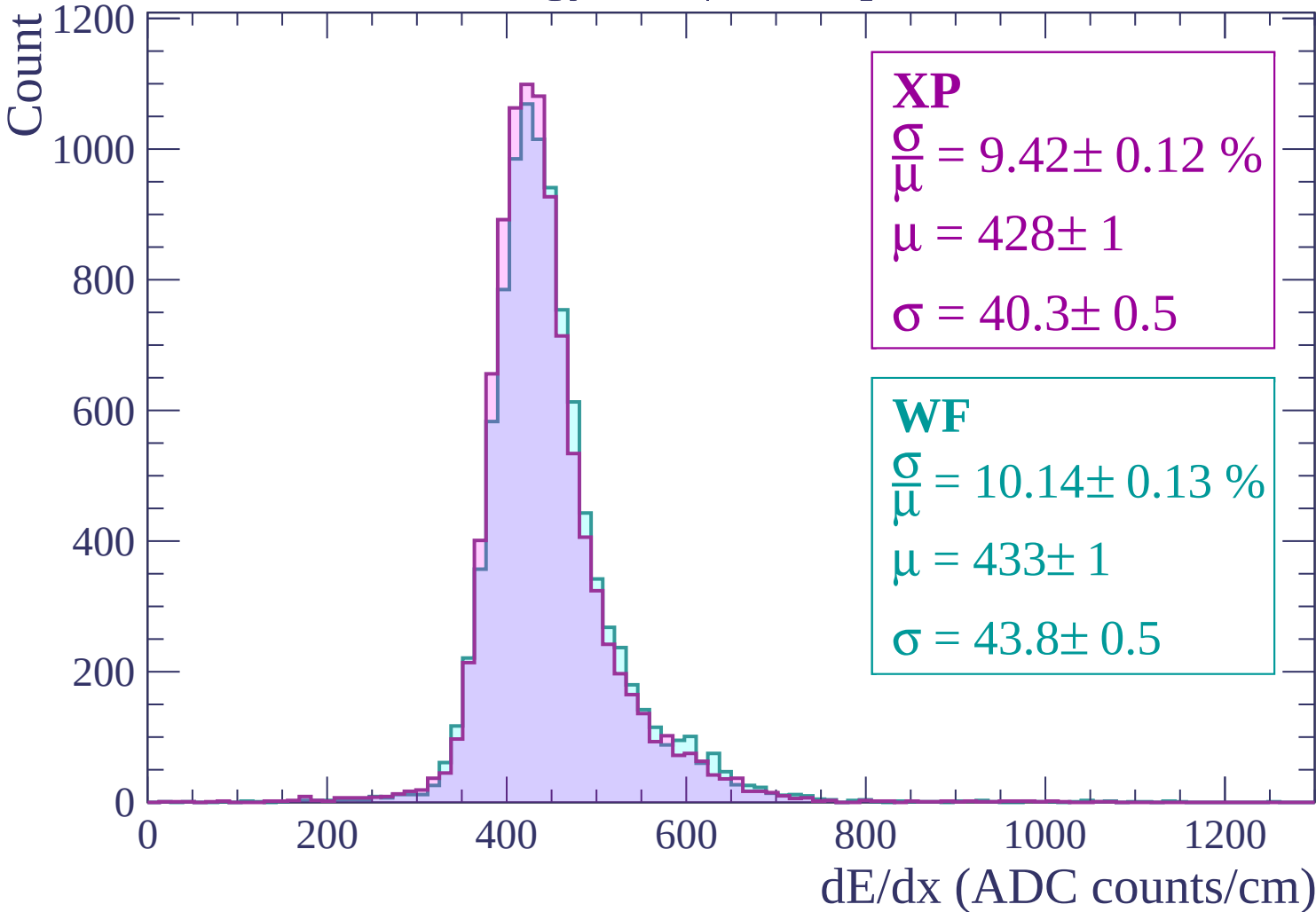
$$\mu = 429 \pm 1$$

$$\sigma = 42.7 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss |  $900 < p < 960$



Energy loss |  $960 < p < 1020$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$$\frac{\sigma}{\mu} = 9.52 \pm 0.15 \%$$

$$\mu = 431 \pm 1$$

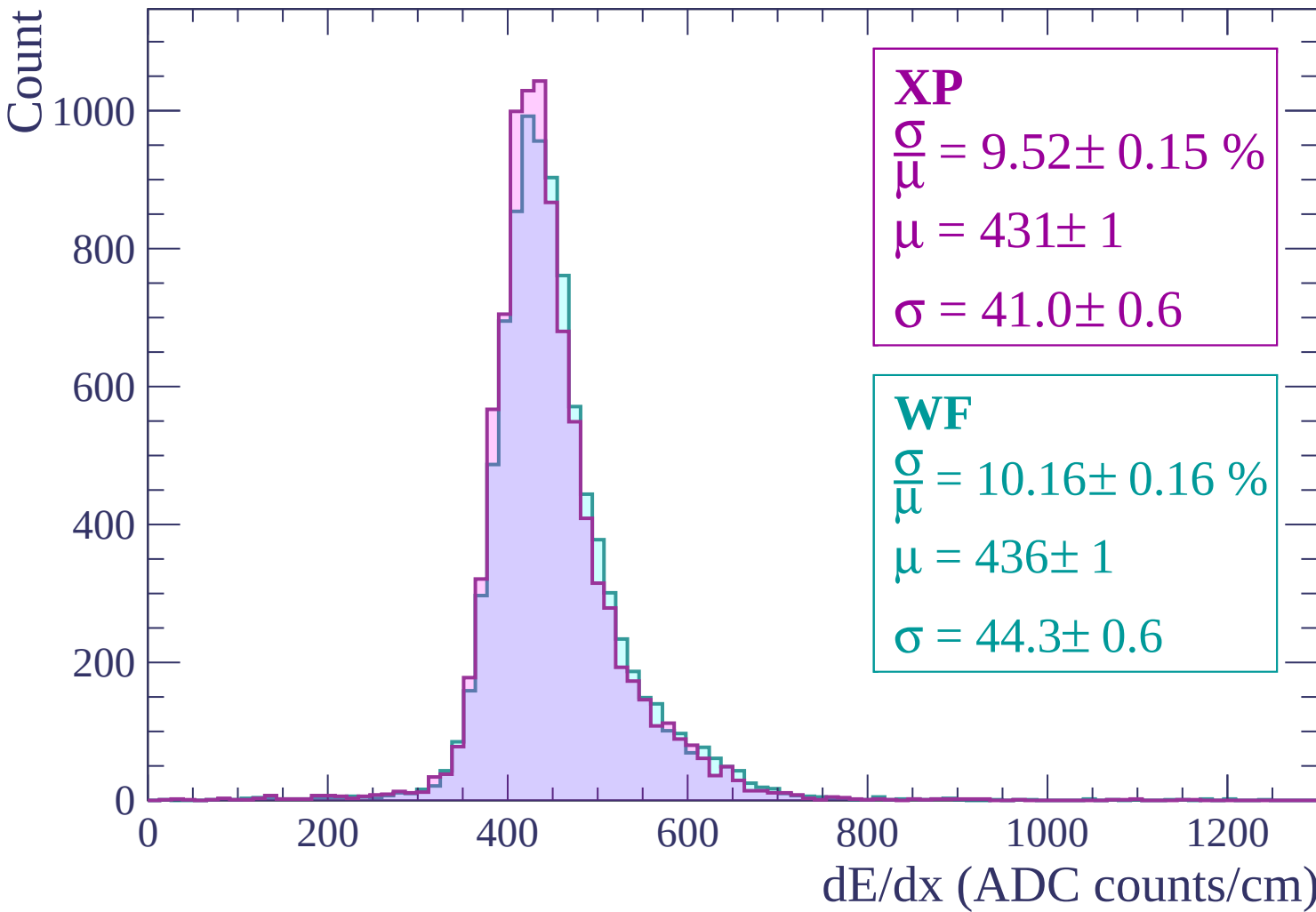
$$\sigma = 41.0 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 10.16 \pm 0.16 \%$$

$$\mu = 436 \pm 1$$

$$\sigma = 44.3 \pm 0.6$$



Energy loss |  $1020 < p < 1080$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

**XP**

$$\frac{\sigma}{\mu} = 9.76 \pm 0.14 \%$$

$$\mu = 436 \pm 1$$

$$\sigma = 42.6 \pm 0.6$$

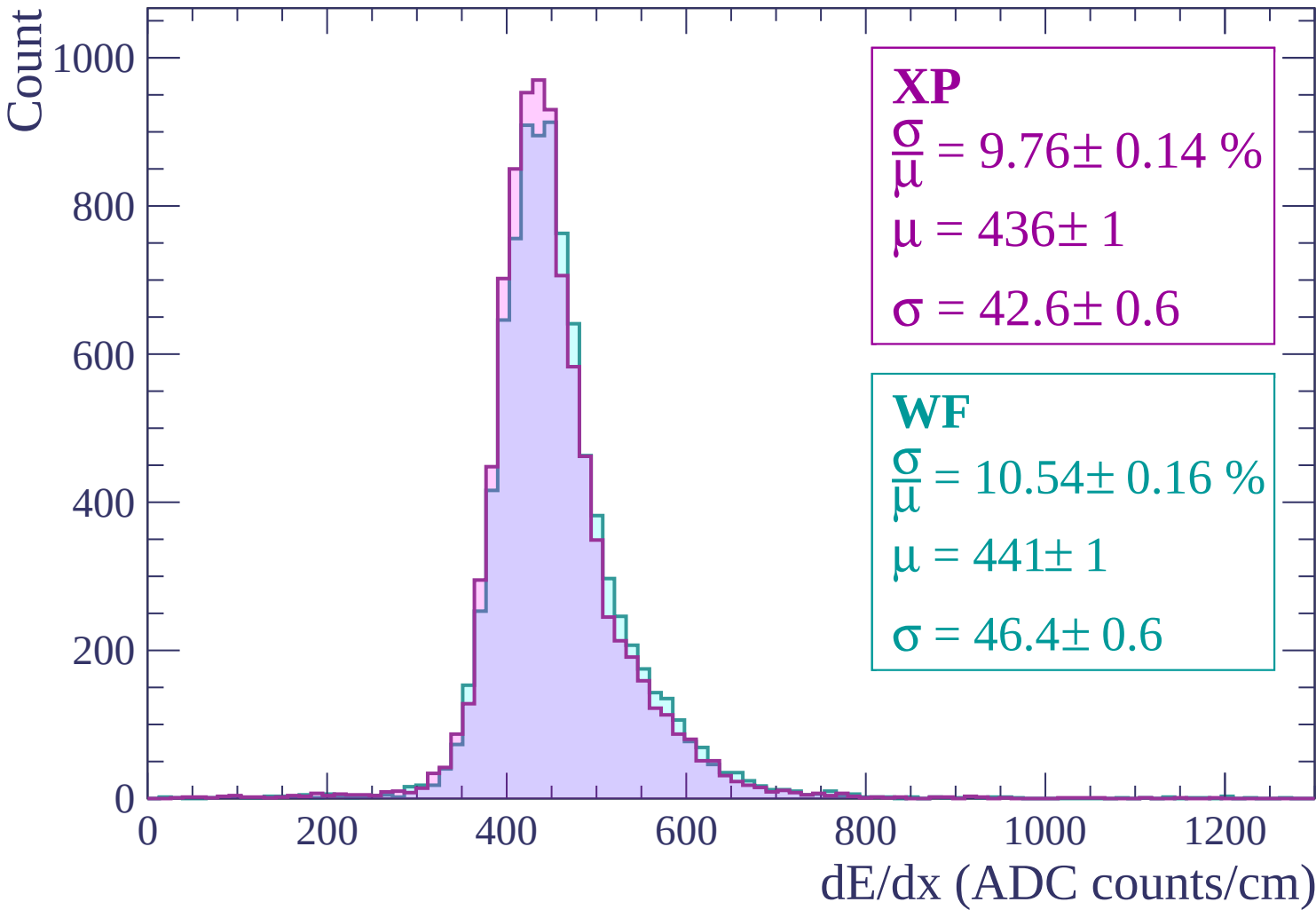
**WF**

$$\frac{\sigma}{\mu} = 10.54 \pm 0.16 \%$$

$$\mu = 441 \pm 1$$

$$\sigma = 46.4 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss |  $1080 < p < 1140$

Count

1000

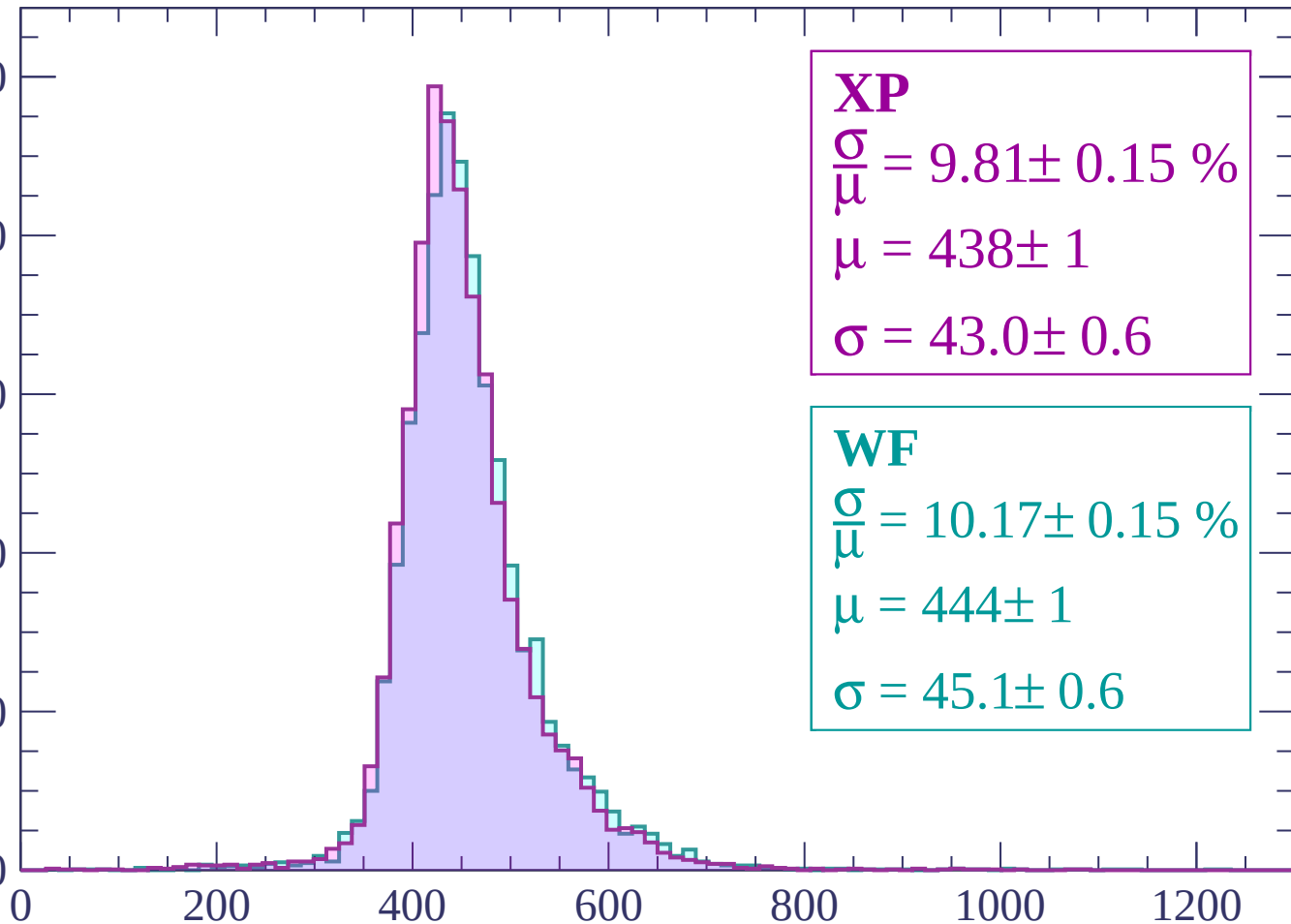
800

600

400

200

0



**XP**

$$\frac{\sigma}{\mu} = 9.81 \pm 0.15 \%$$

$$\mu = 438 \pm 1$$

$$\sigma = 43.0 \pm 0.6$$

**WF**

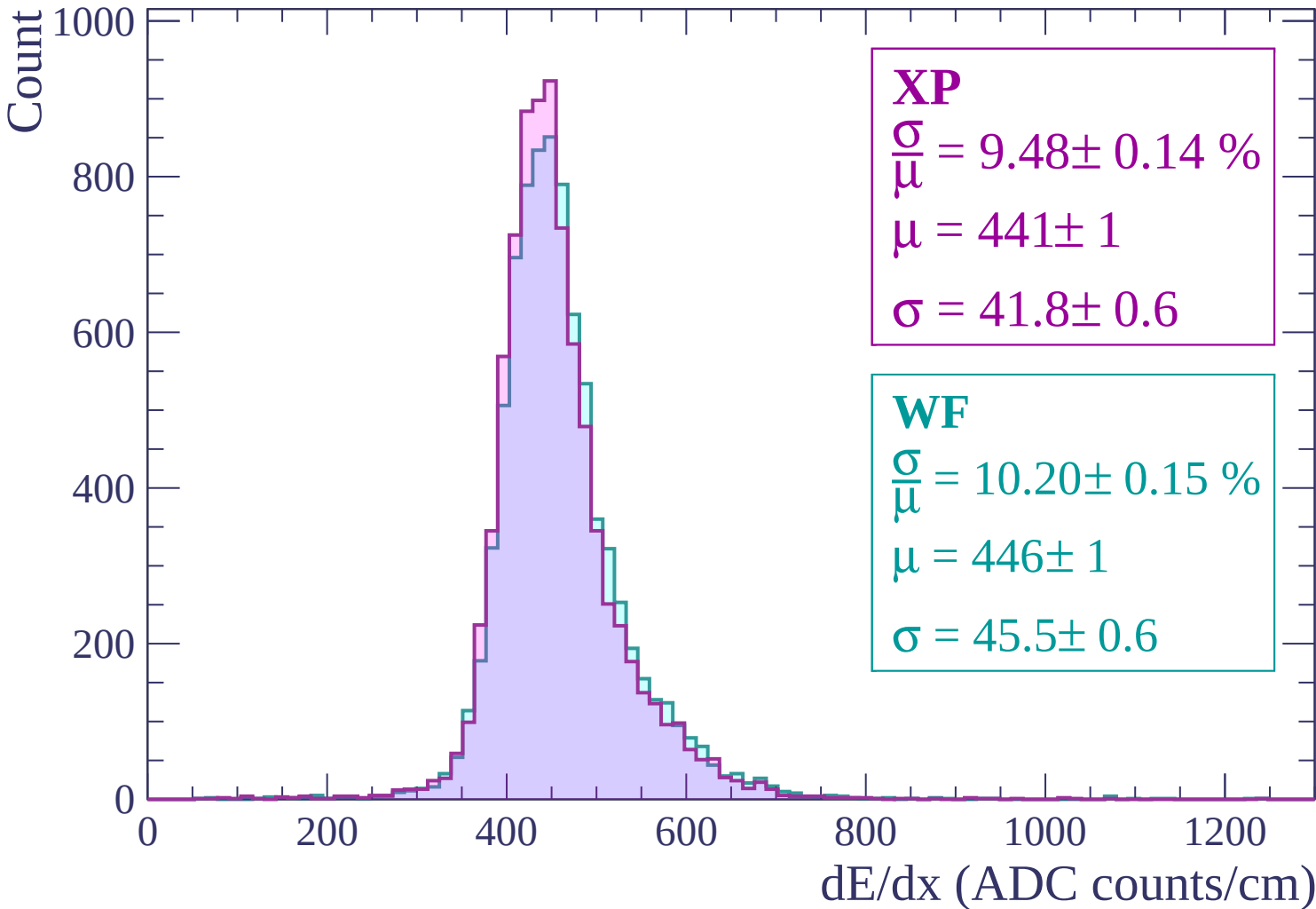
$$\frac{\sigma}{\mu} = 10.17 \pm 0.15 \%$$

$$\mu = 444 \pm 1$$

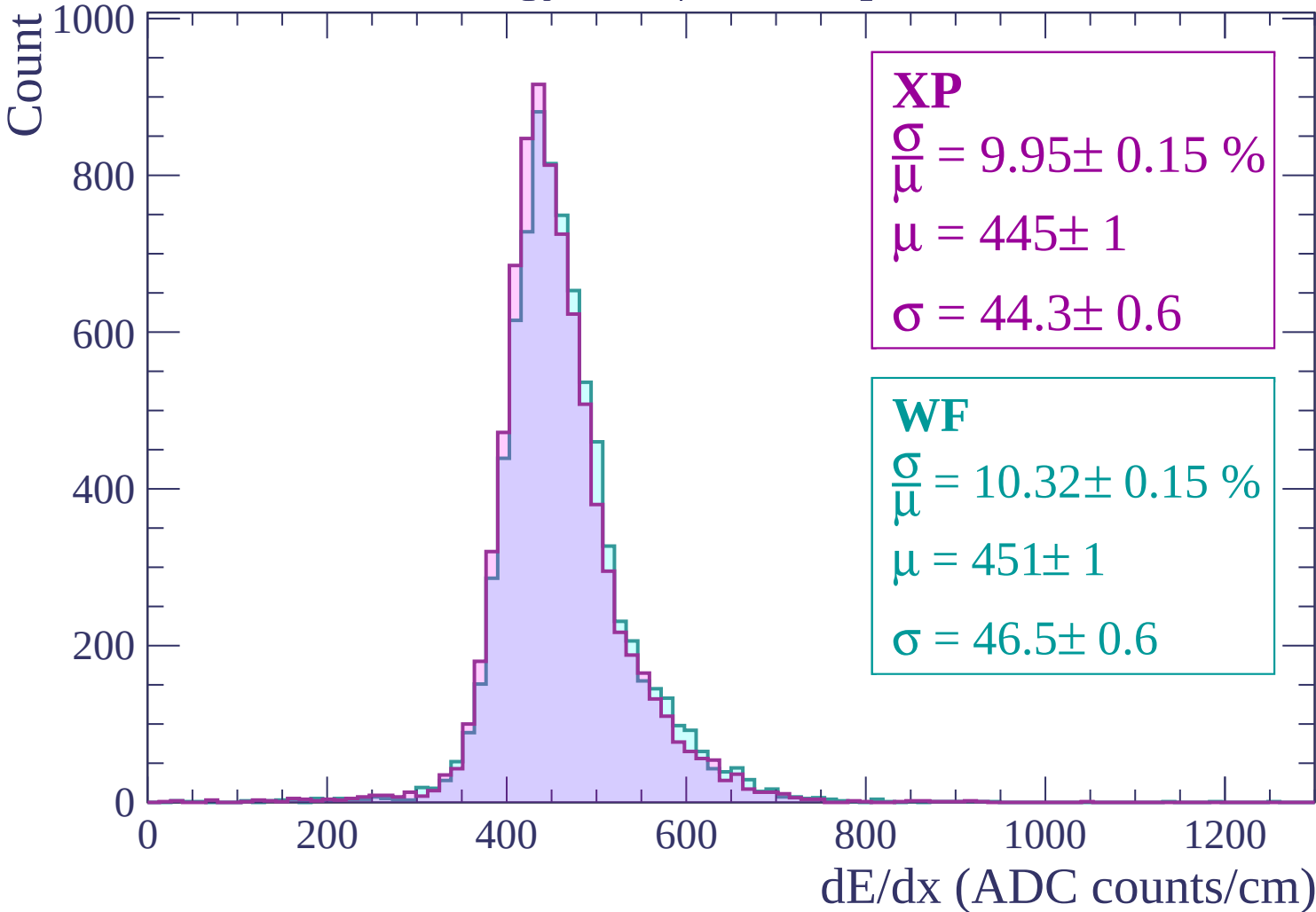
$$\sigma = 45.1 \pm 0.6$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1140 < p < 1200$

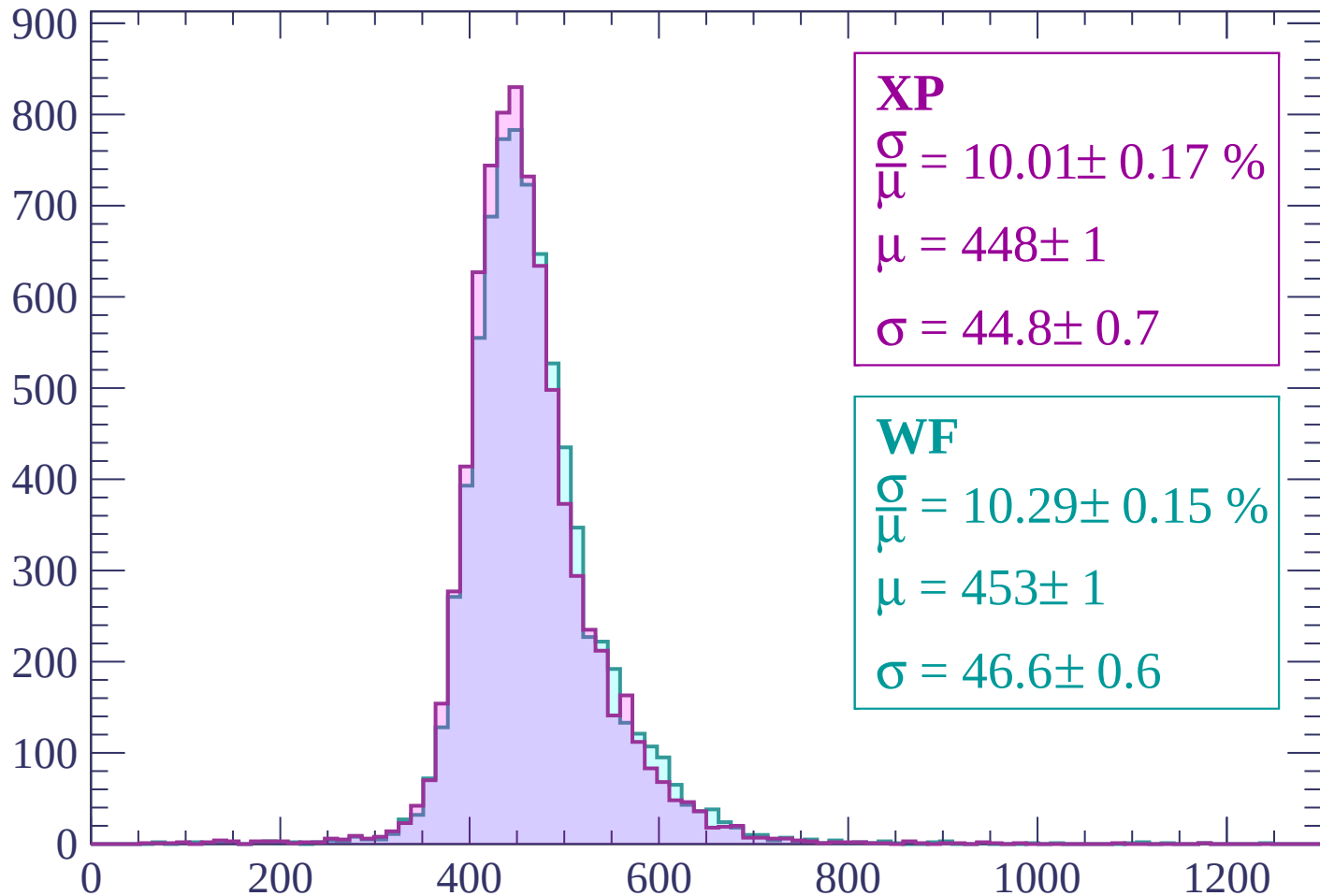


Energy loss |  $1200 < p < 1260$



Energy loss |  $1260 < p < 1320$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.01 \pm 0.17 \%$$

$$\mu = 448 \pm 1$$

$$\sigma = 44.8 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 10.29 \pm 0.15 \%$$

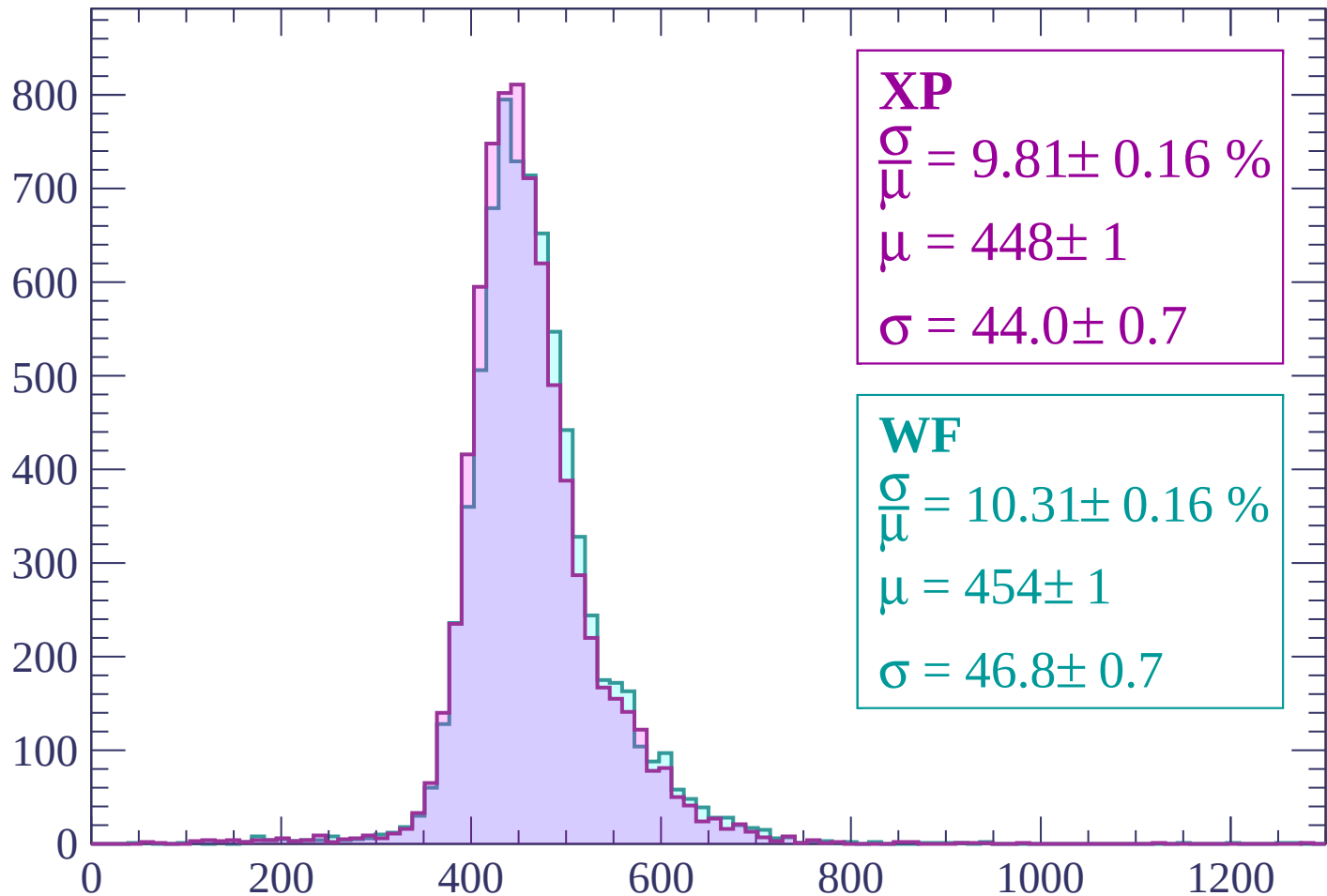
$$\mu = 453 \pm 1$$

$$\sigma = 46.6 \pm 0.6$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1320 < p < 1380$

Count



**XP**

$$\frac{\sigma}{\bar{\mu}} = 9.81 \pm 0.16 \%$$

$$\mu = 448 \pm 1$$

$$\sigma = 44.0 \pm 0.7$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 10.31 \pm 0.16 \%$$

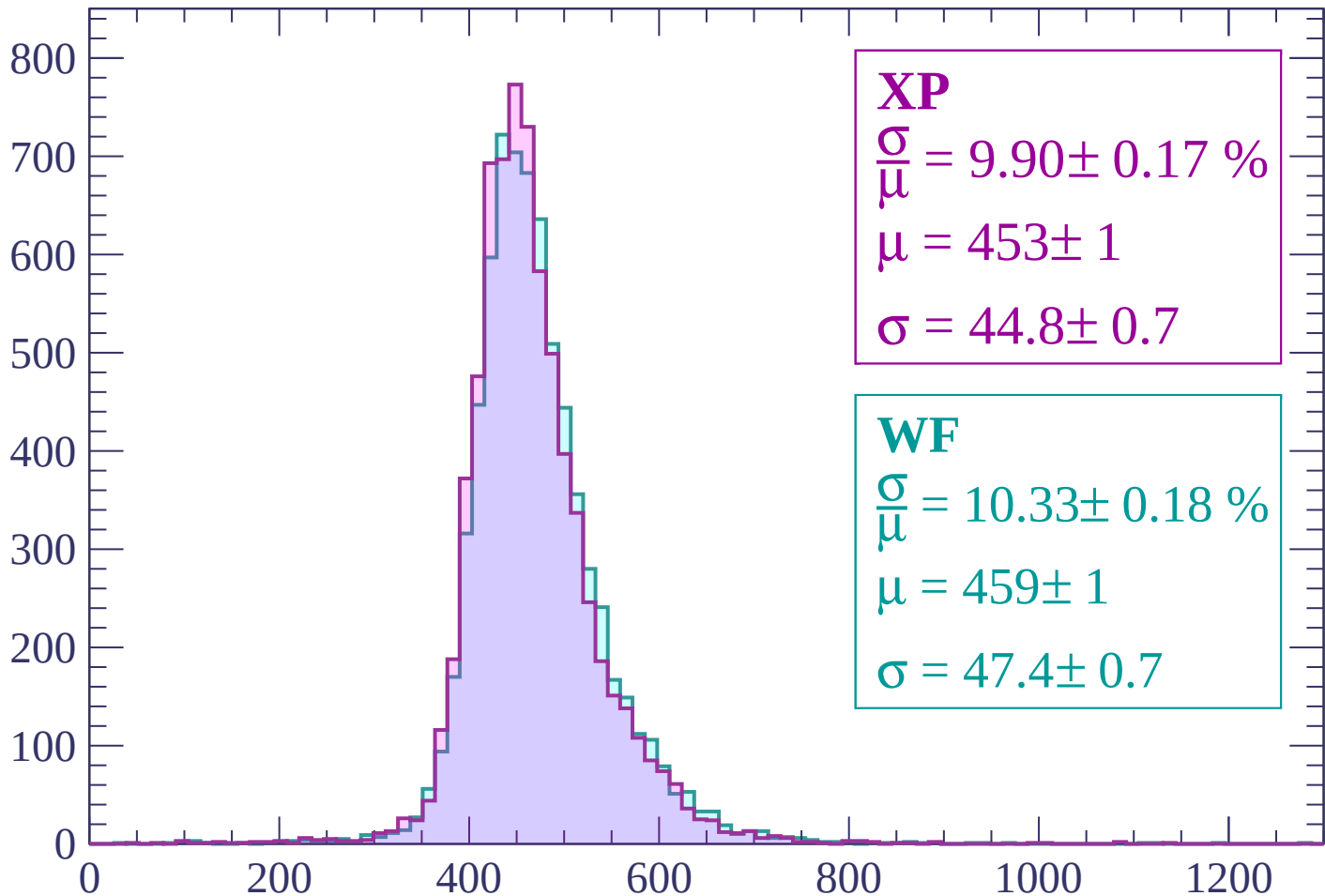
$$\mu = 454 \pm 1$$

$$\sigma = 46.8 \pm 0.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1380 < p < 1440$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.90 \pm 0.17 \%$$

$$\mu = 453 \pm 1$$

$$\sigma = 44.8 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 10.33 \pm 0.18 \%$$

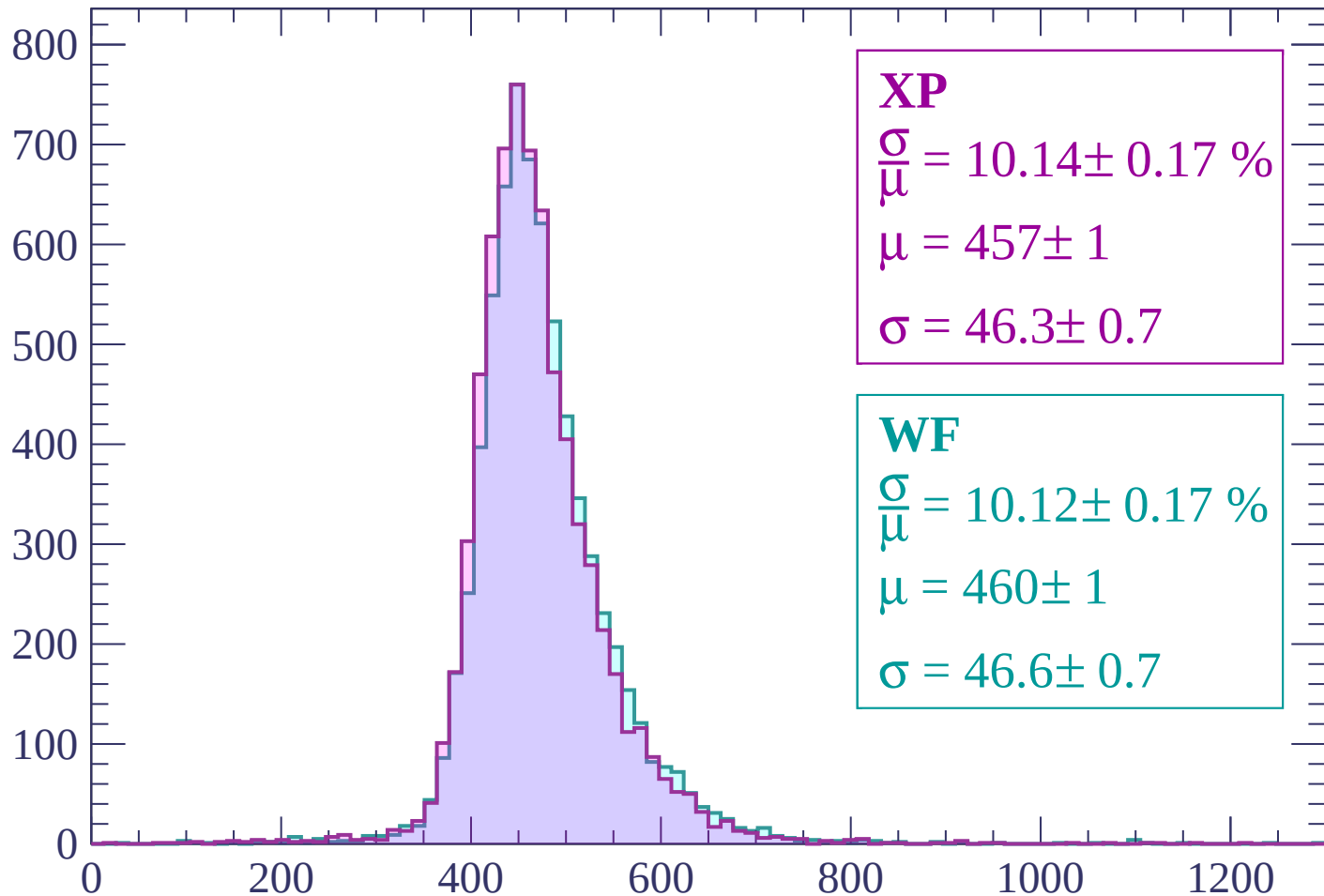
$$\mu = 459 \pm 1$$

$$\sigma = 47.4 \pm 0.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1440 < p < 1500$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.14 \pm 0.17 \%$$

$$\mu = 457 \pm 1$$

$$\sigma = 46.3 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 10.12 \pm 0.17 \%$$

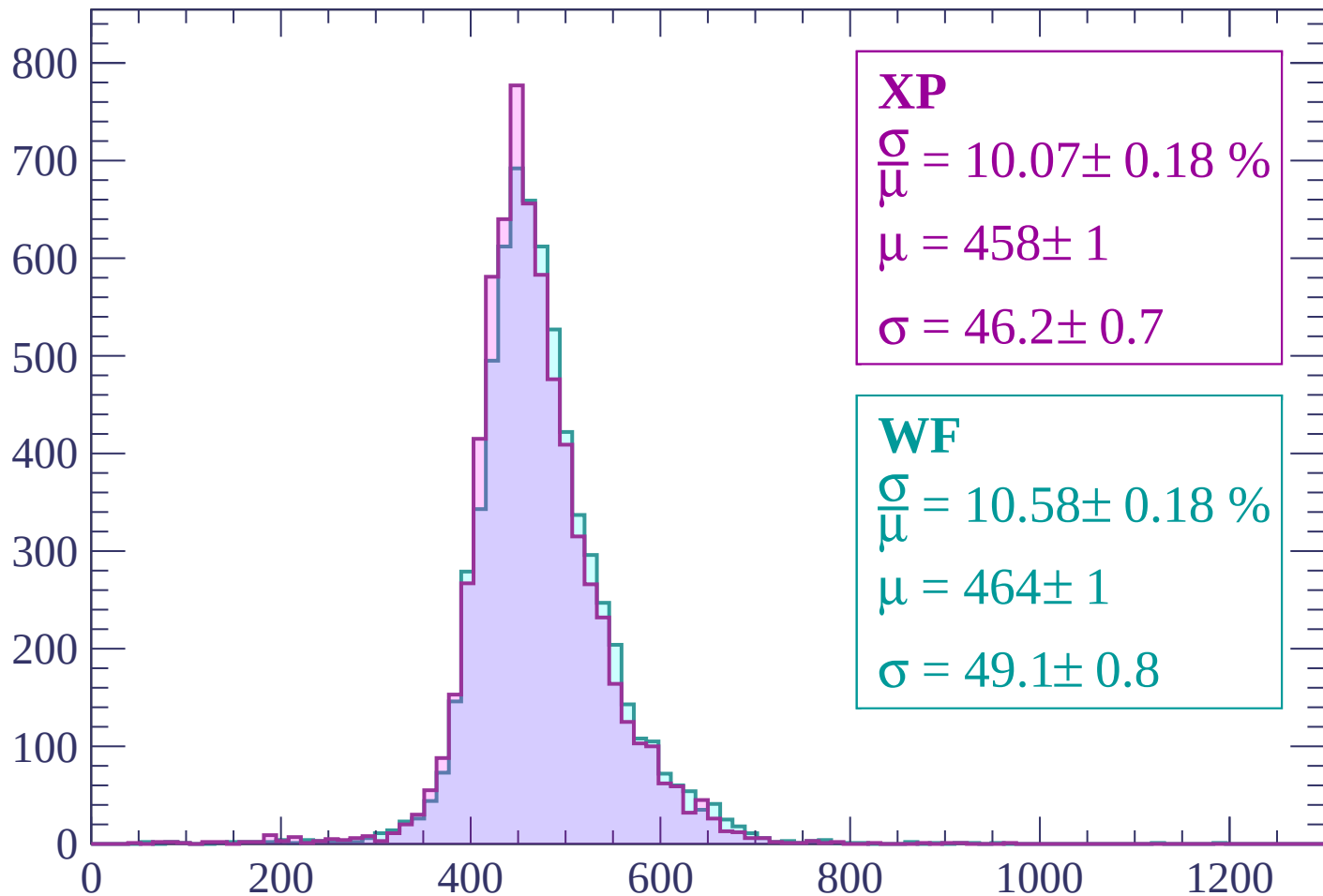
$$\mu = 460 \pm 1$$

$$\sigma = 46.6 \pm 0.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1500 < p < 1560$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.07 \pm 0.18 \%$$

$$\mu = 458 \pm 1$$

$$\sigma = 46.2 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 10.58 \pm 0.18 \%$$

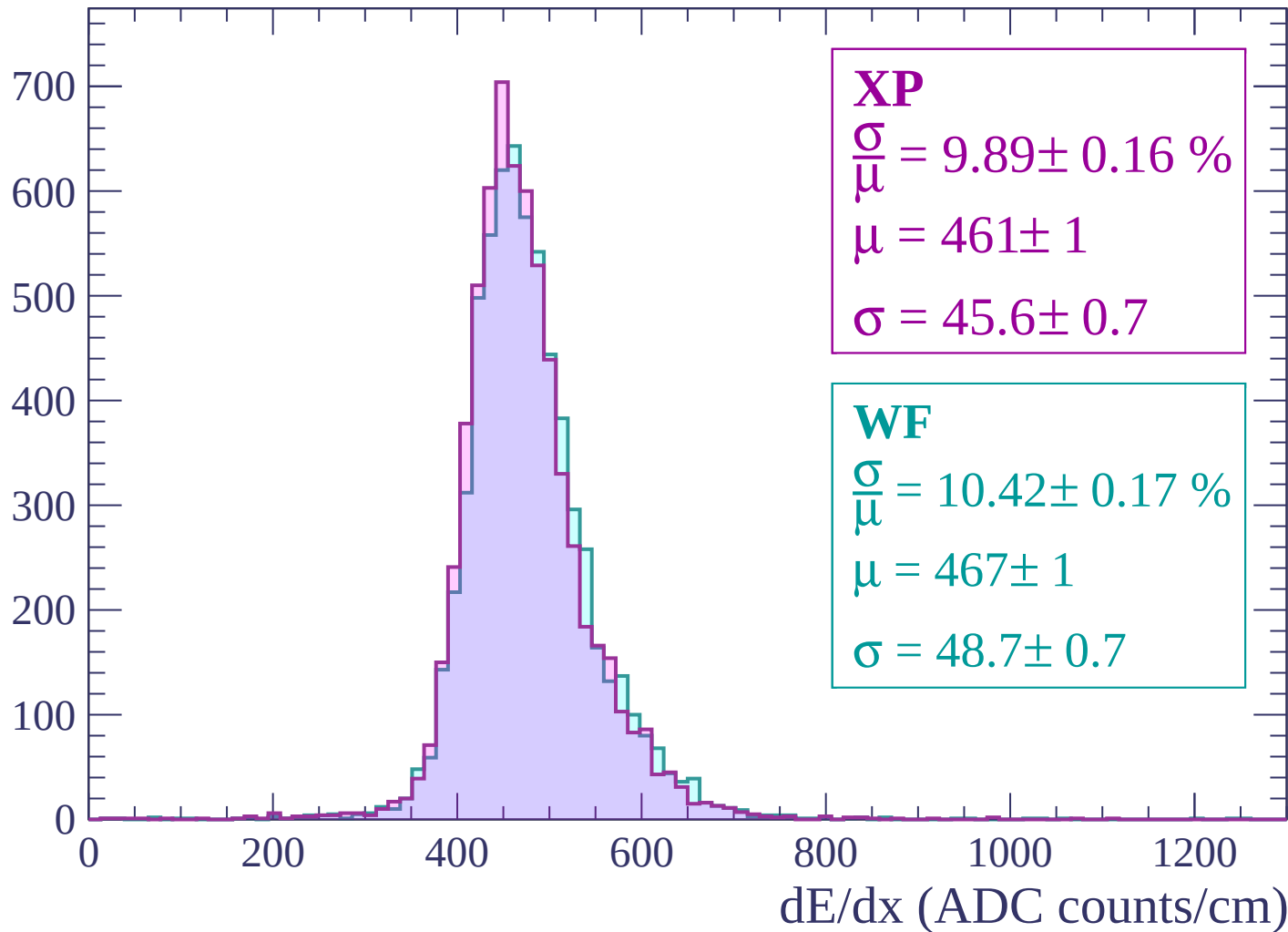
$$\mu = 464 \pm 1$$

$$\sigma = 49.1 \pm 0.8$$

$dE/dx$  (ADC counts/cm)

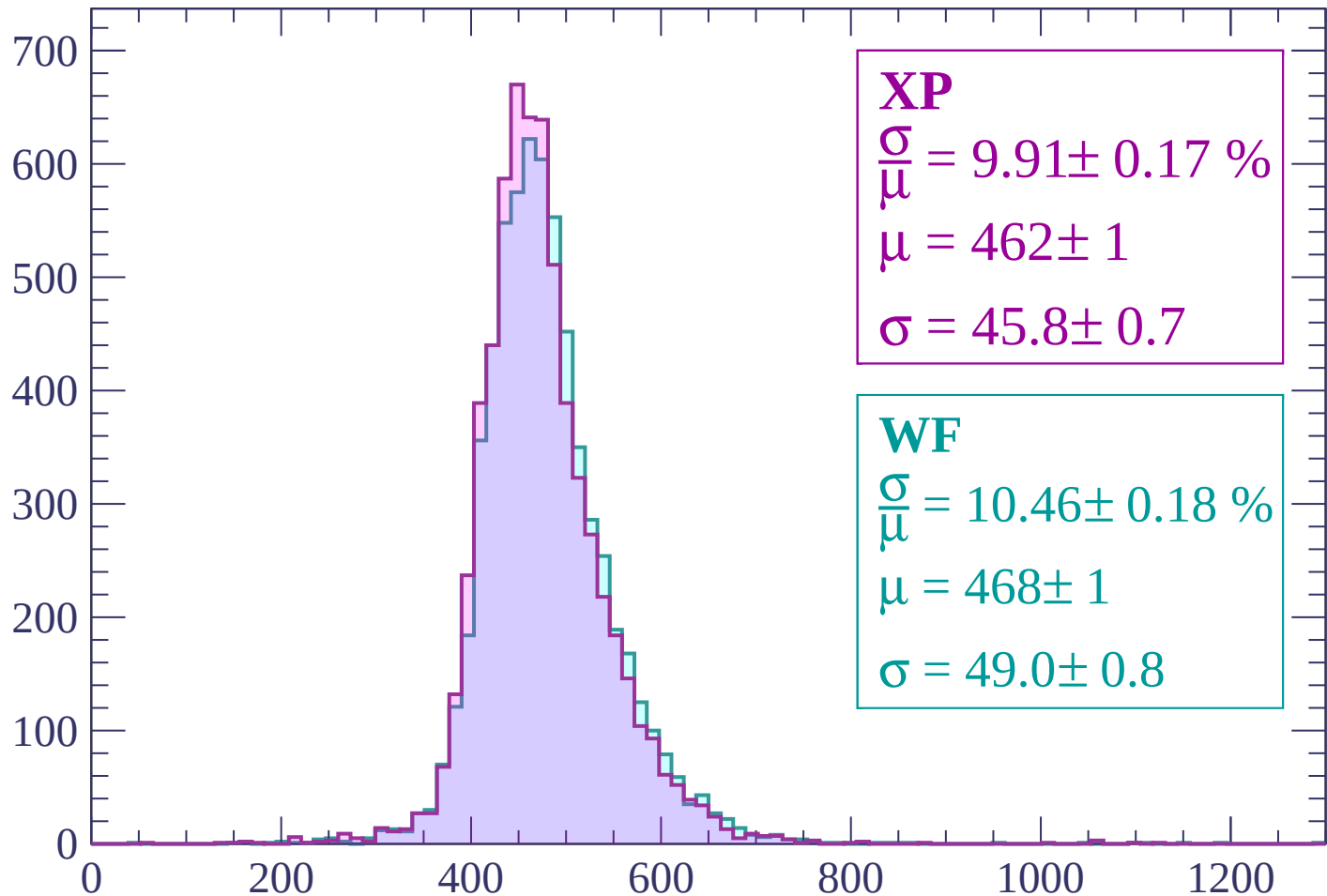
Energy loss |  $1560 < p < 1620$

Count



Energy loss |  $1620 < p < 1680$

Count



**XP**

$$\frac{\sigma}{\bar{\mu}} = 9.91 \pm 0.17 \%$$

$$\mu = 462 \pm 1$$

$$\sigma = 45.8 \pm 0.7$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 10.46 \pm 0.18 \%$$

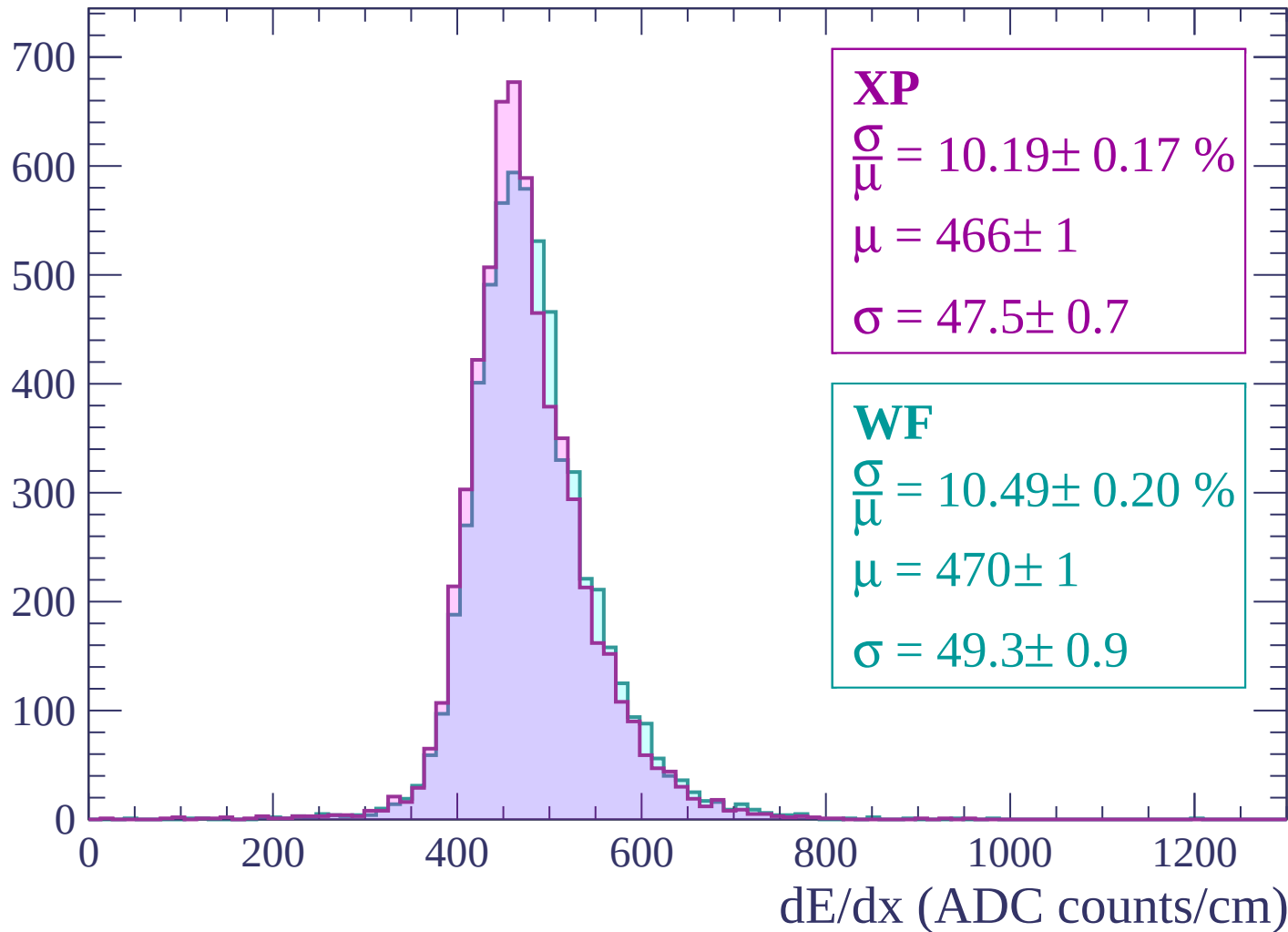
$$\mu = 468 \pm 1$$

$$\sigma = 49.0 \pm 0.8$$

dE/dx (ADC counts/cm)

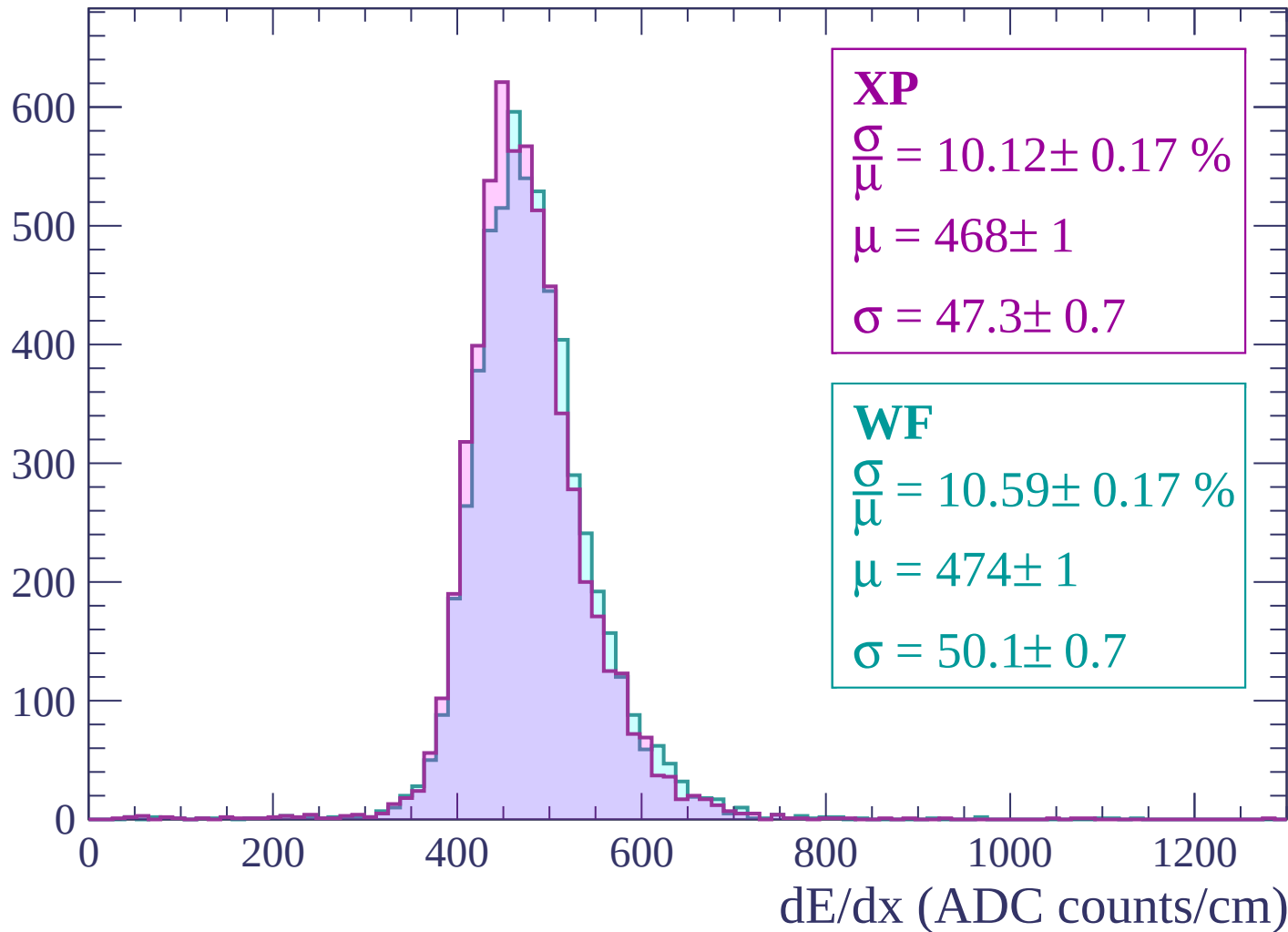
Energy loss |  $1680 < p < 1740$

Count



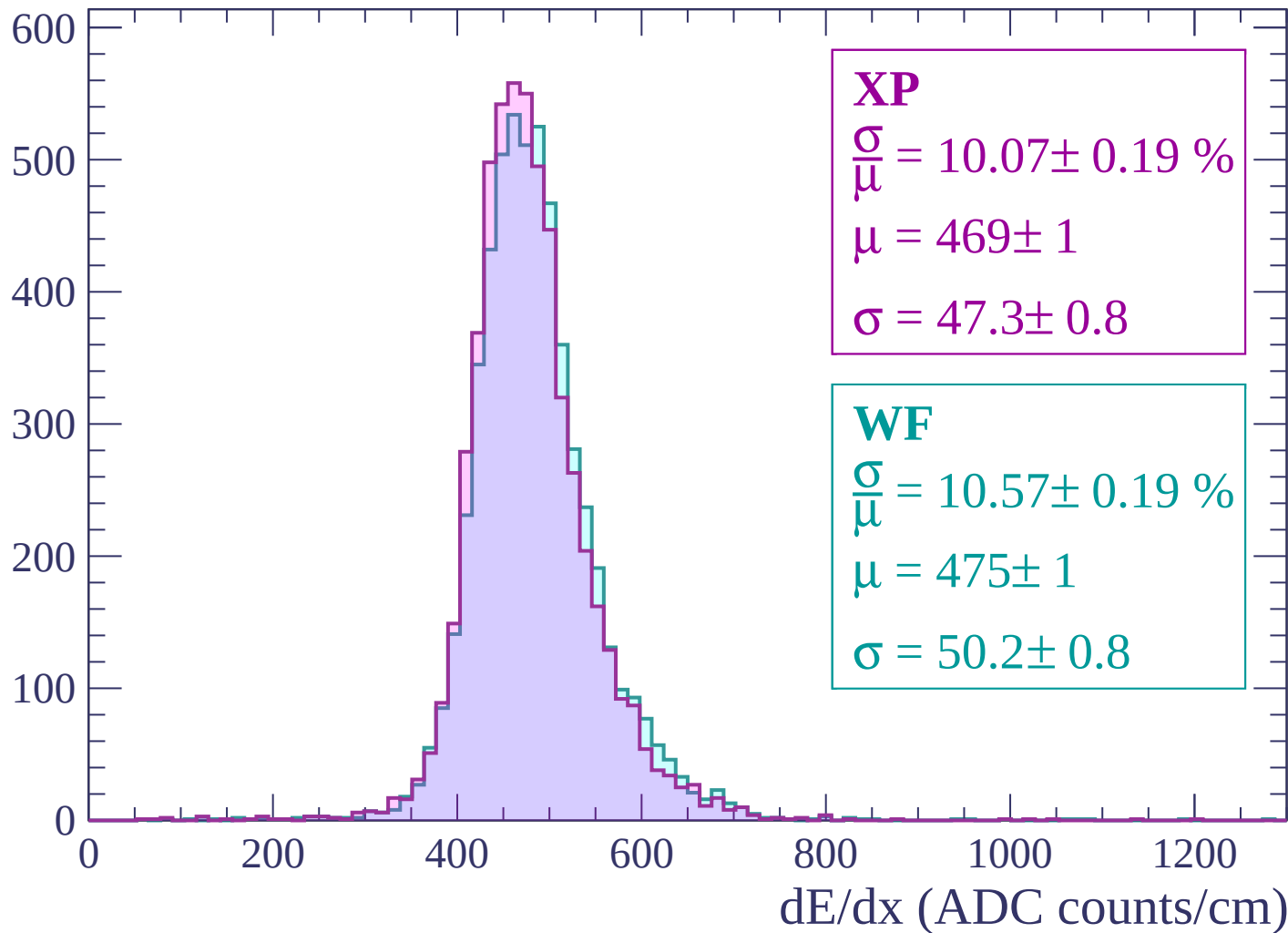
Energy loss |  $1740 < p < 1800$

Count



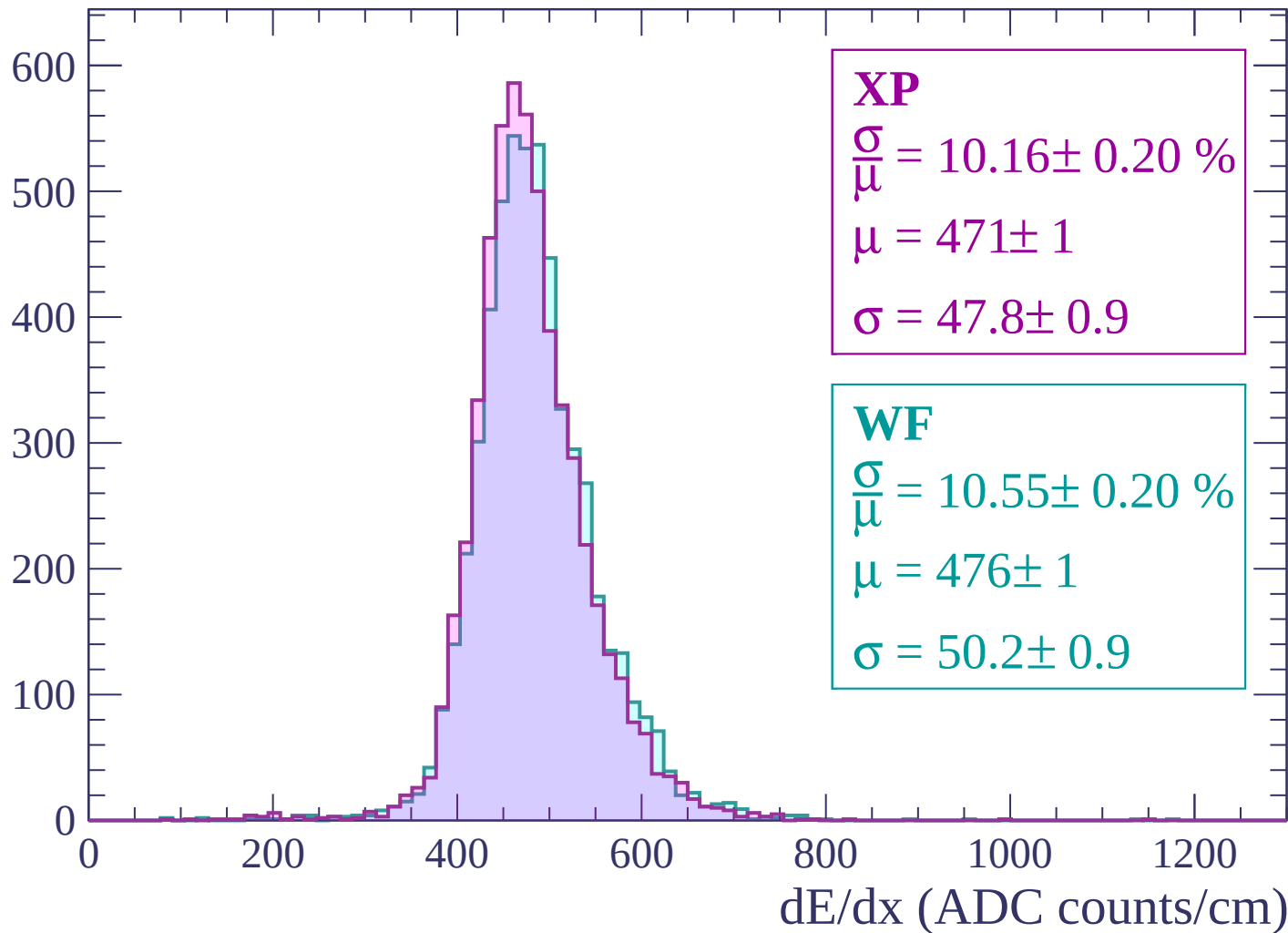
Energy loss |  $1800 < p < 1860$

Count

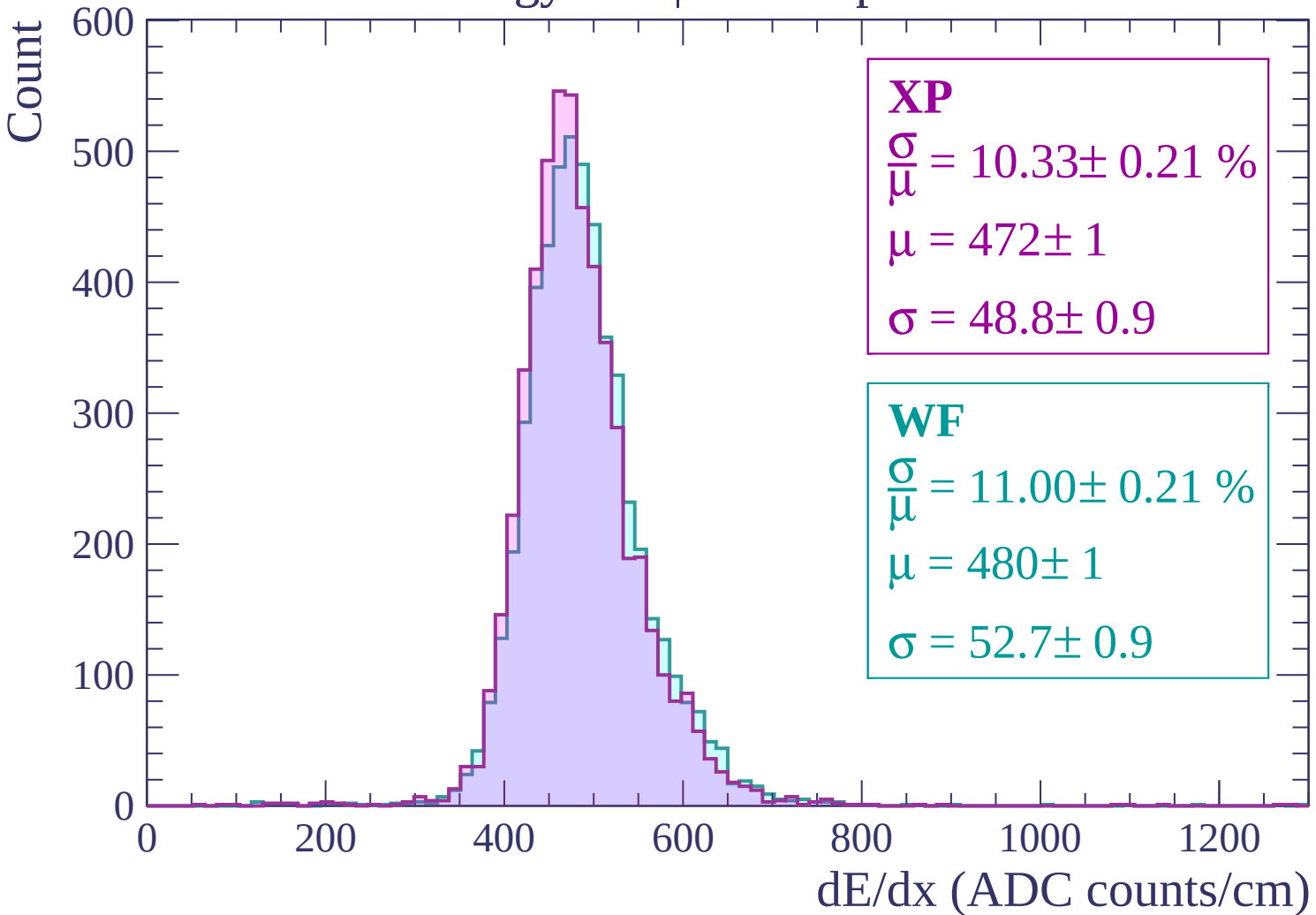


# Energy loss | $1860 < p < 1920$

Count

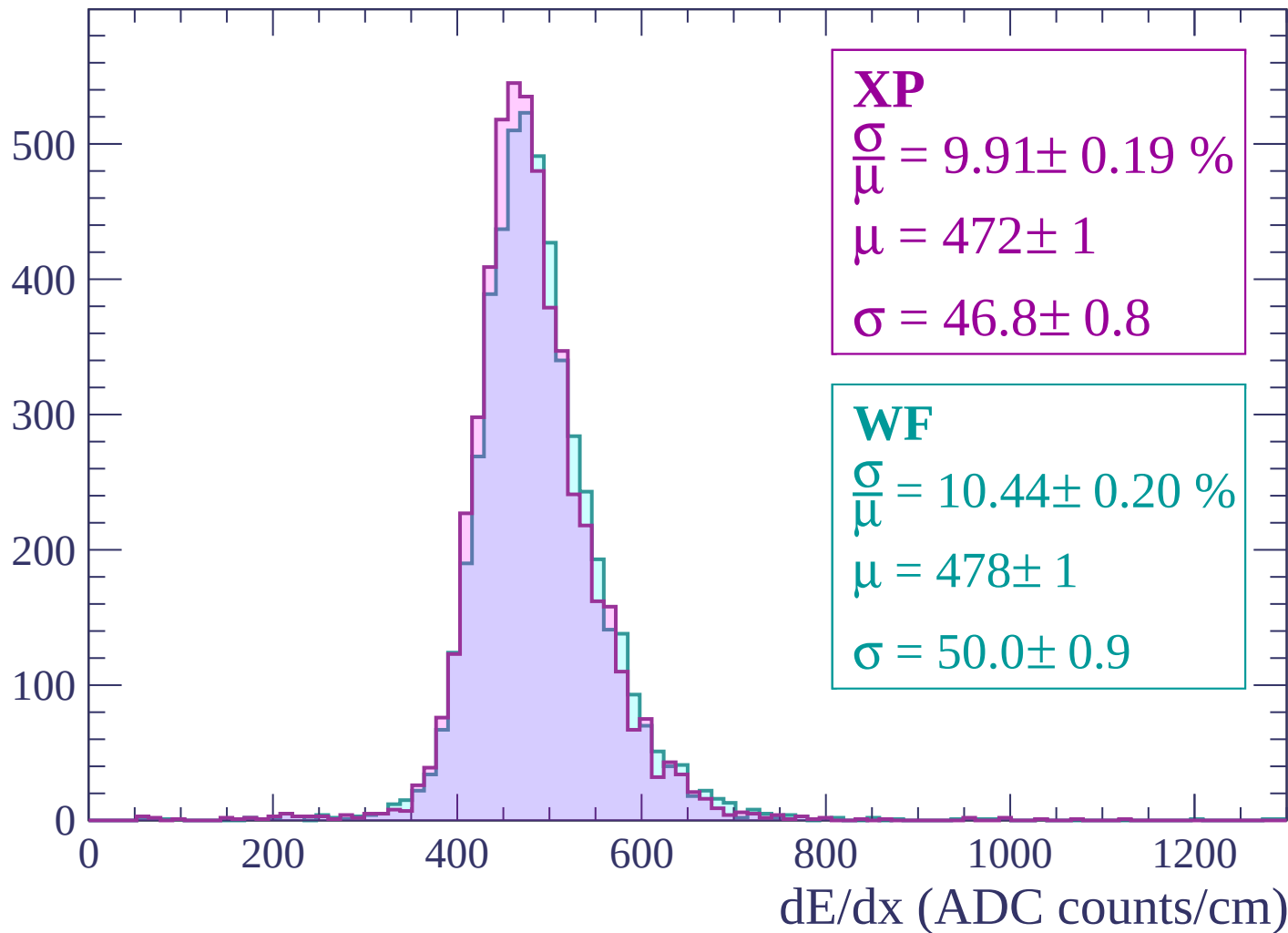


# Energy loss | $1920 < p < 1980$



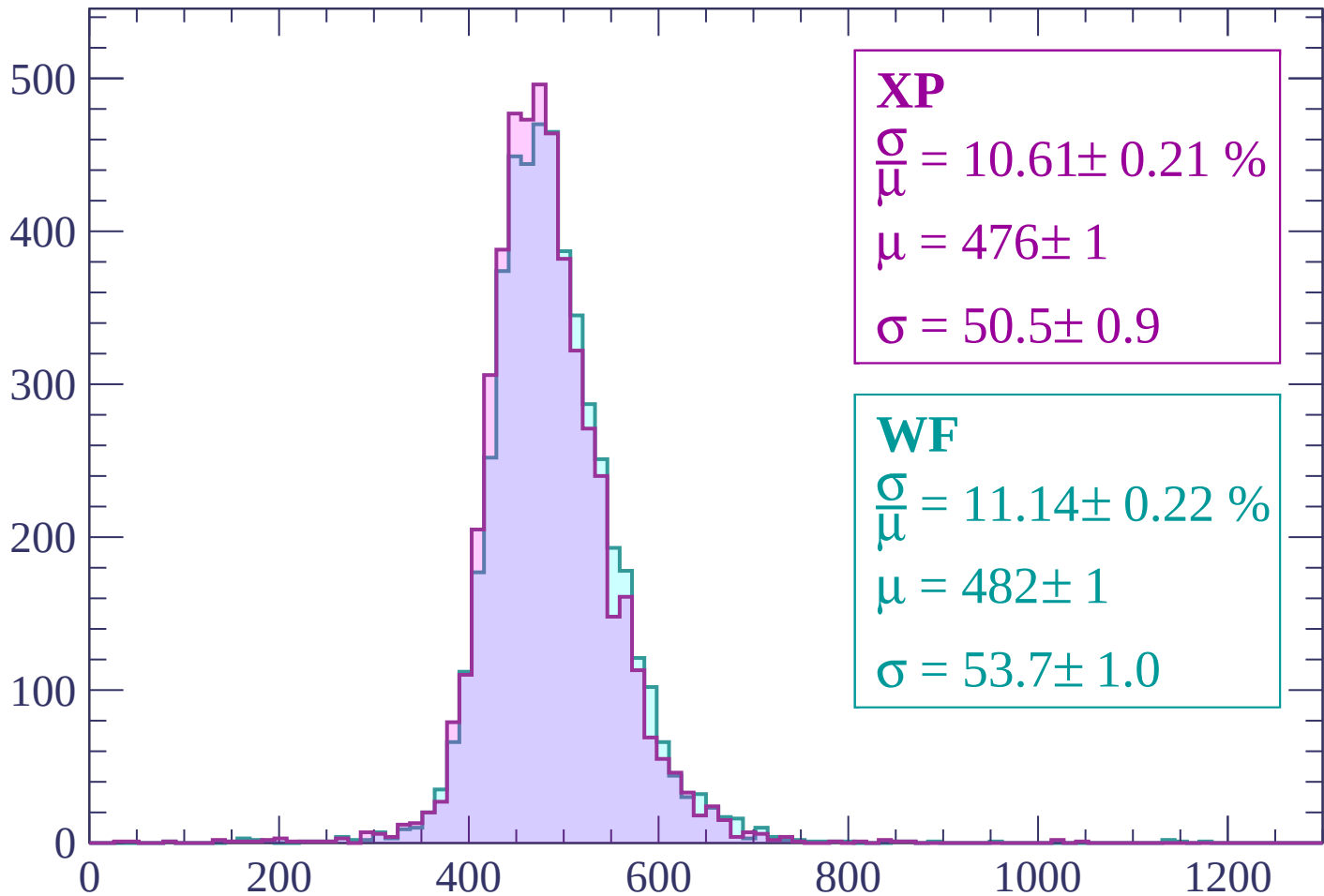
Energy loss |  $1980 < p < 2040$

Count



# Energy loss | $2040 < p < 2100$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.61 \pm 0.21 \%$$

$$\mu = 476 \pm 1$$

$$\sigma = 50.5 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 11.14 \pm 0.22 \%$$

$$\mu = 482 \pm 1$$

$$\sigma = 53.7 \pm 1.0$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $2100 < p < 2160$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

**XP**

$$\frac{\sigma}{\mu} = 10.43 \pm 0.21 \%$$

$$\mu = 480 \pm 1$$

$$\sigma = 50.1 \pm 0.9$$

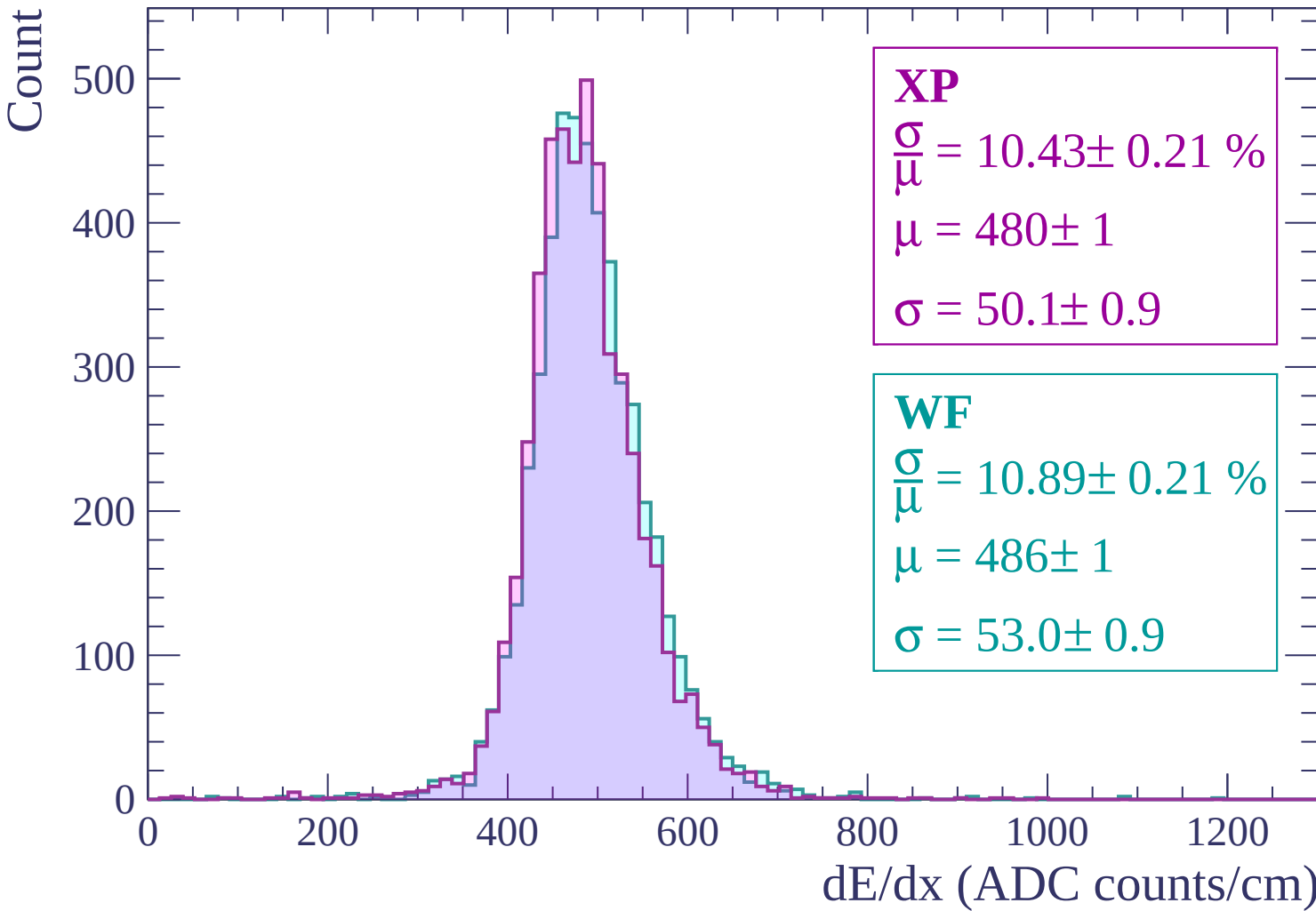
**WF**

$$\frac{\sigma}{\mu} = 10.89 \pm 0.21 \%$$

$$\mu = 486 \pm 1$$

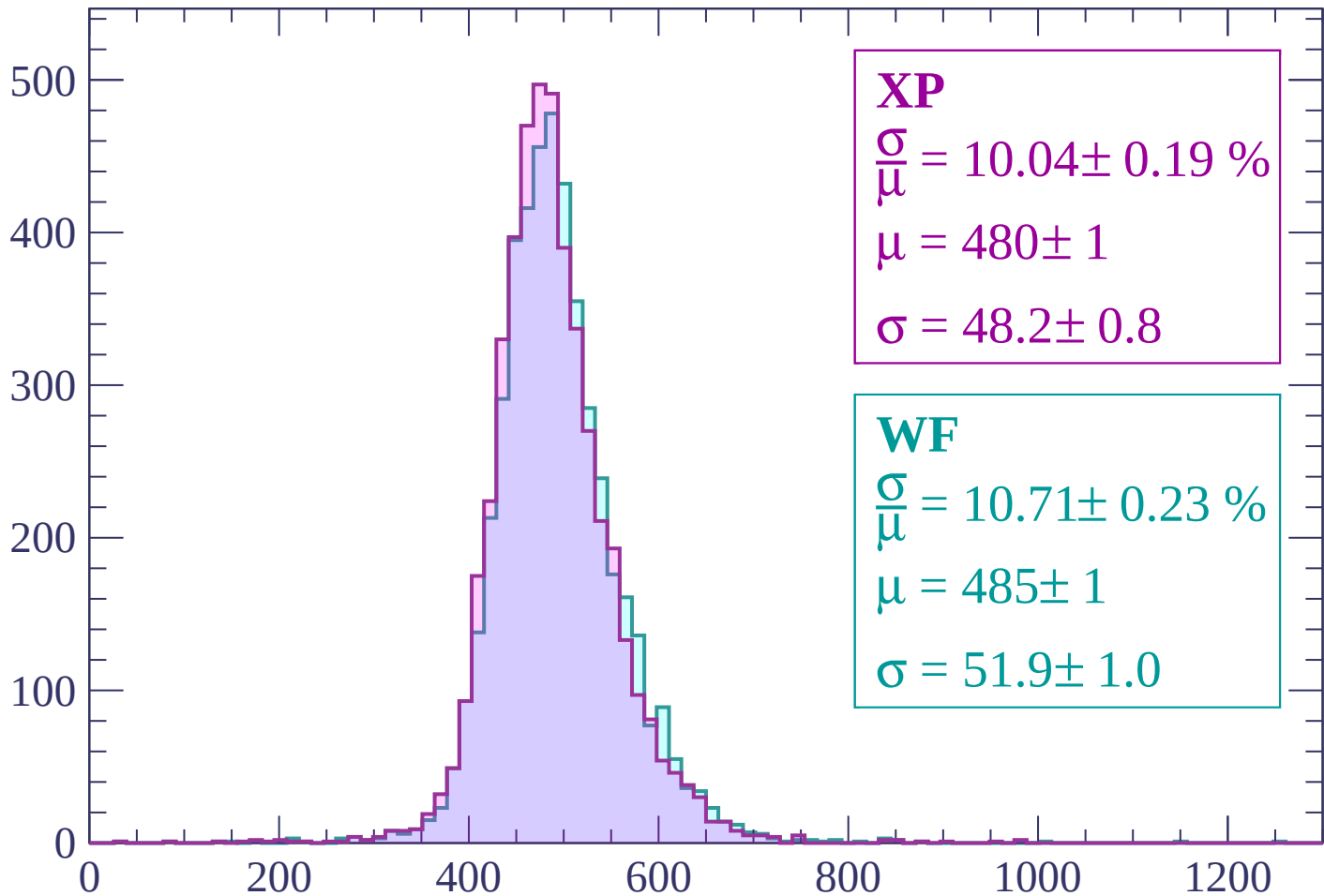
$$\sigma = 53.0 \pm 0.9$$

dE/dx (ADC counts/cm)



# Energy loss | $2160 < p < 2220$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.04 \pm 0.19 \%$$

$$\mu = 480 \pm 1$$

$$\sigma = 48.2 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 10.71 \pm 0.23 \%$$

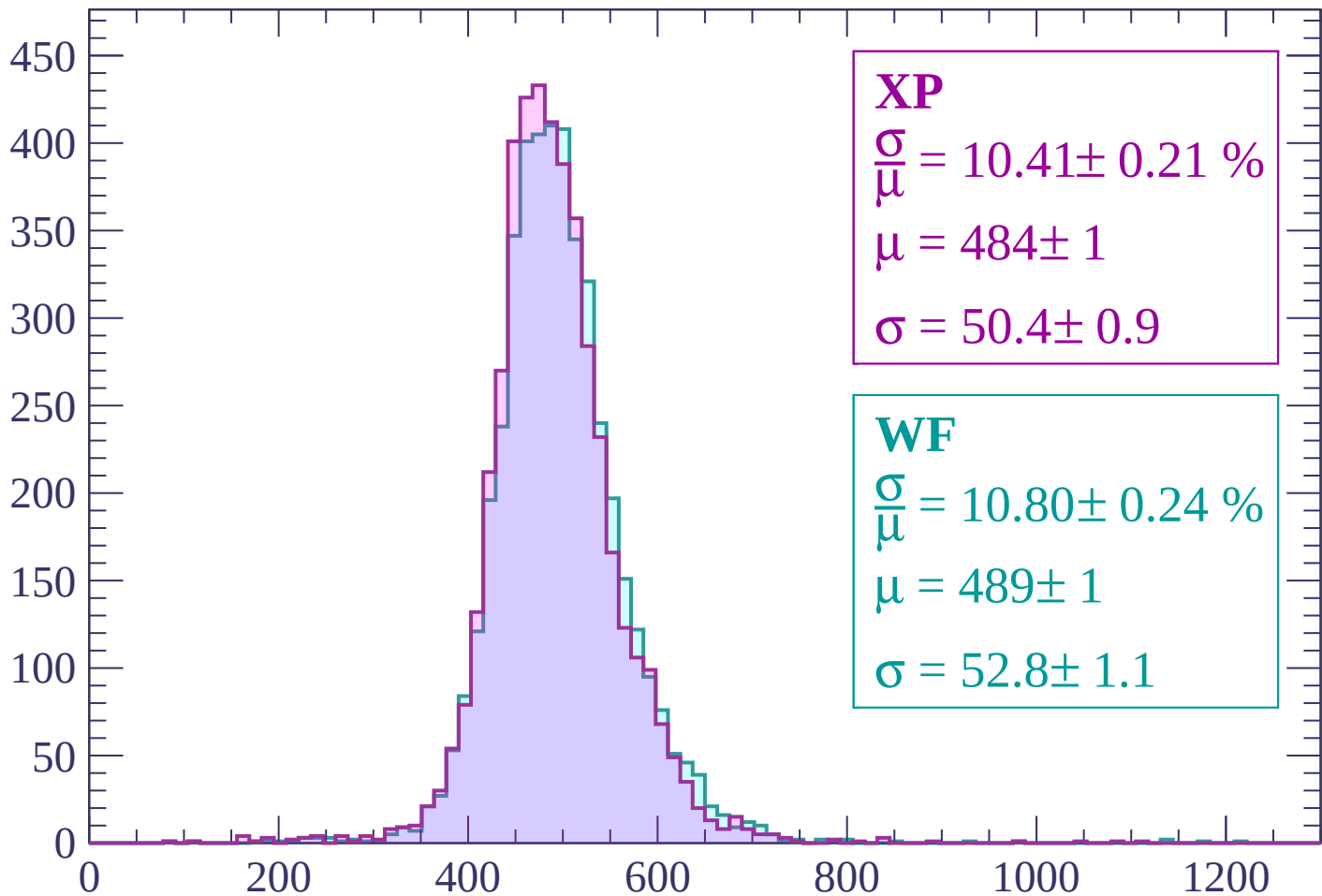
$$\mu = 485 \pm 1$$

$$\sigma = 51.9 \pm 1.0$$

$dE/dx$  (ADC counts/cm)

# Energy loss | $2220 < p < 2280$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.41 \pm 0.21 \%$$

$$\mu = 484 \pm 1$$

$$\sigma = 50.4 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 10.80 \pm 0.24 \%$$

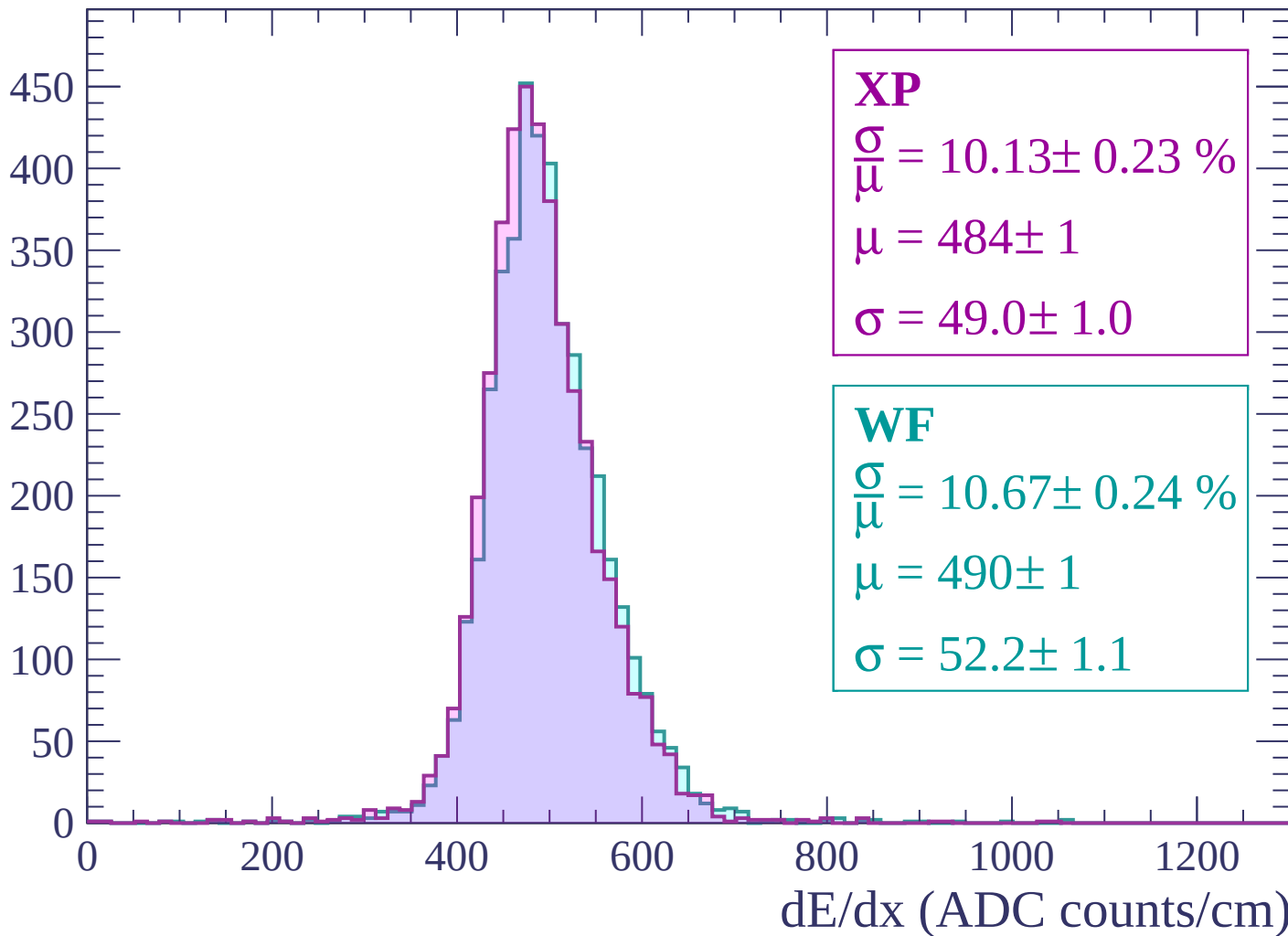
$$\mu = 489 \pm 1$$

$$\sigma = 52.8 \pm 1.1$$

$dE/dx$  (ADC counts/cm)

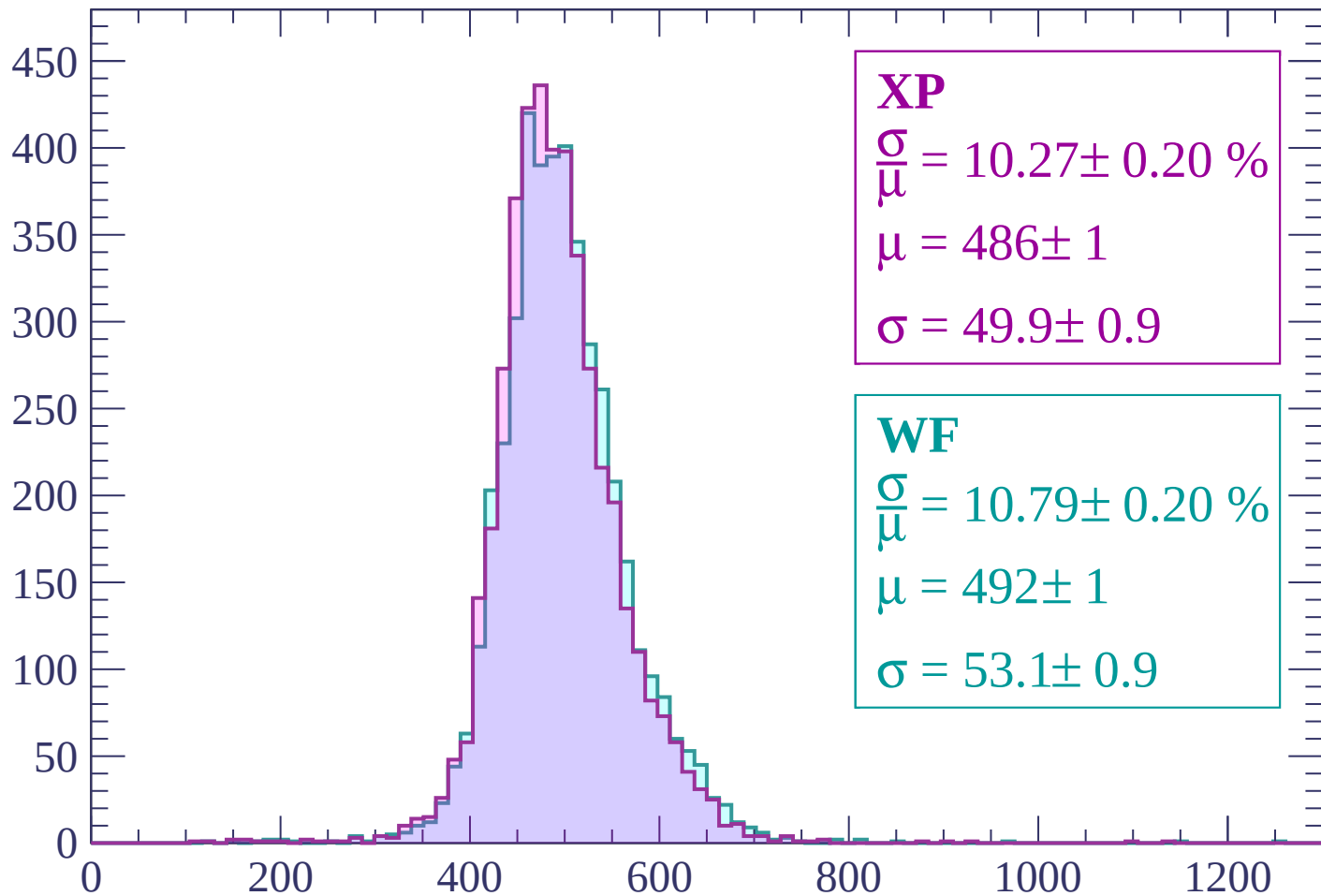
Energy loss |  $2280 < p < 2340$

Count



Energy loss |  $2340 < p < 2400$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.27 \pm 0.20 \%$$

$$\mu = 486 \pm 1$$

$$\sigma = 49.9 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 10.79 \pm 0.20 \%$$

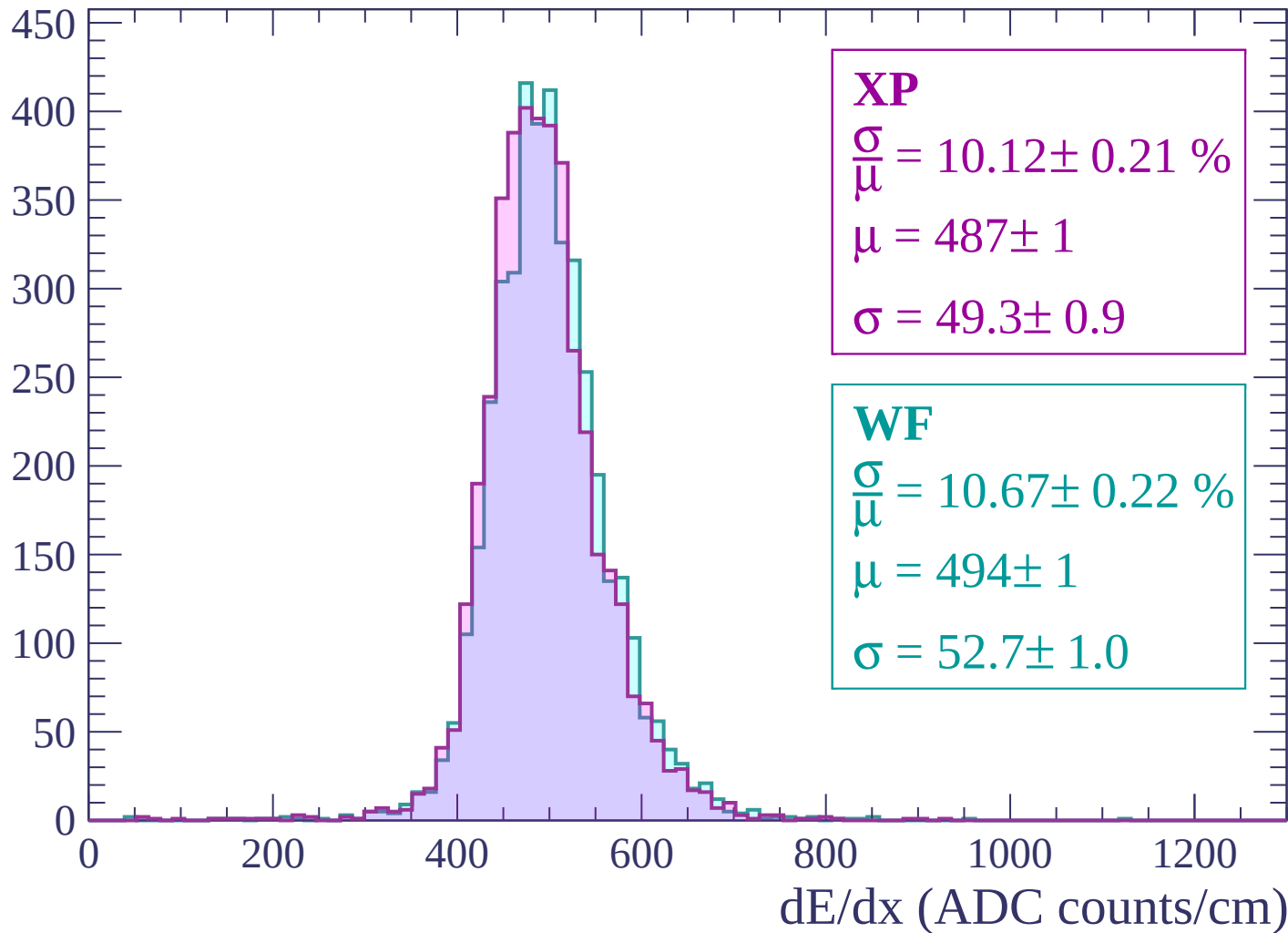
$$\mu = 492 \pm 1$$

$$\sigma = 53.1 \pm 0.9$$

$dE/dx$  (ADC counts/cm)

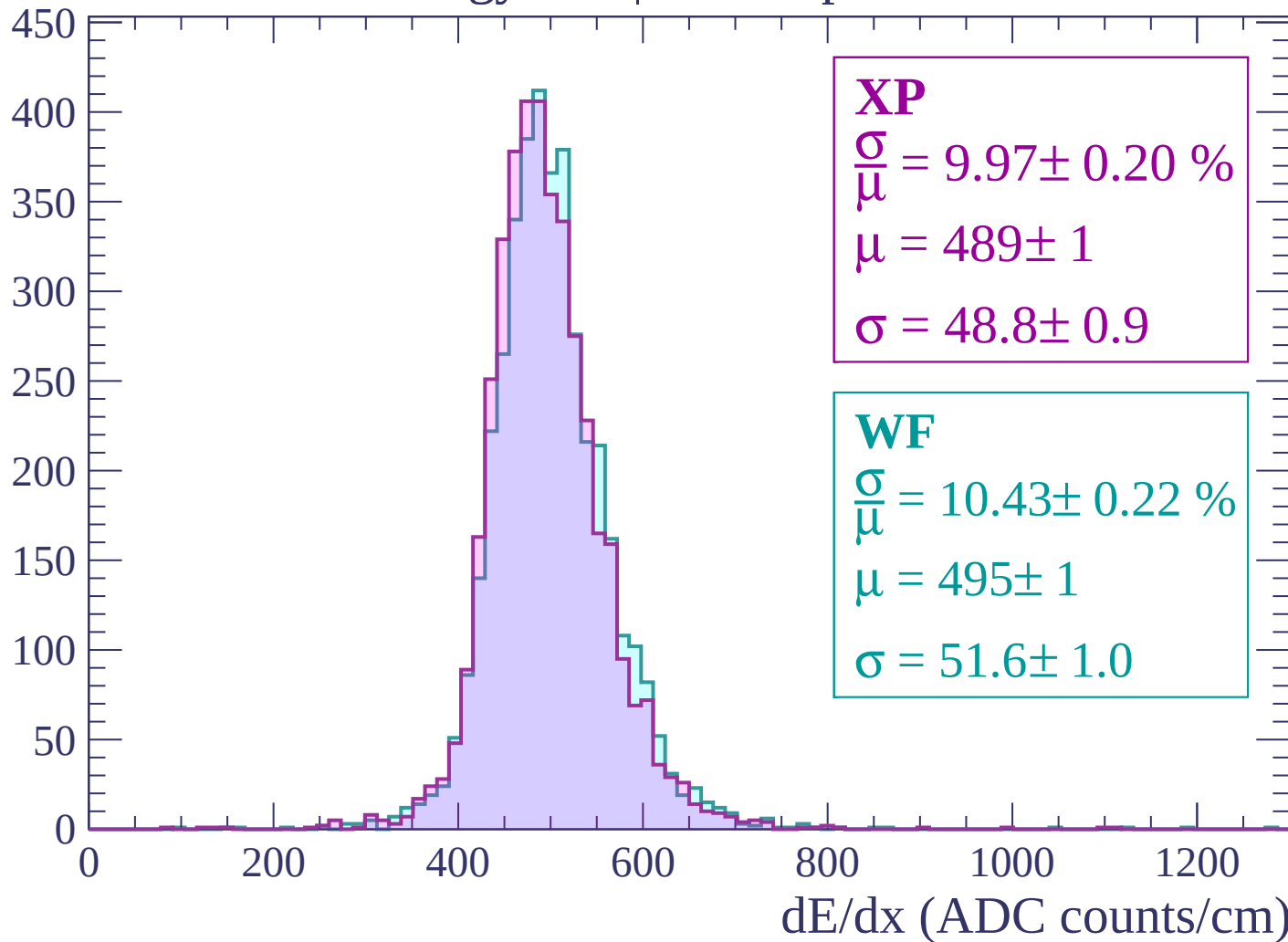
Energy loss |  $2400 < p < 2460$

Count



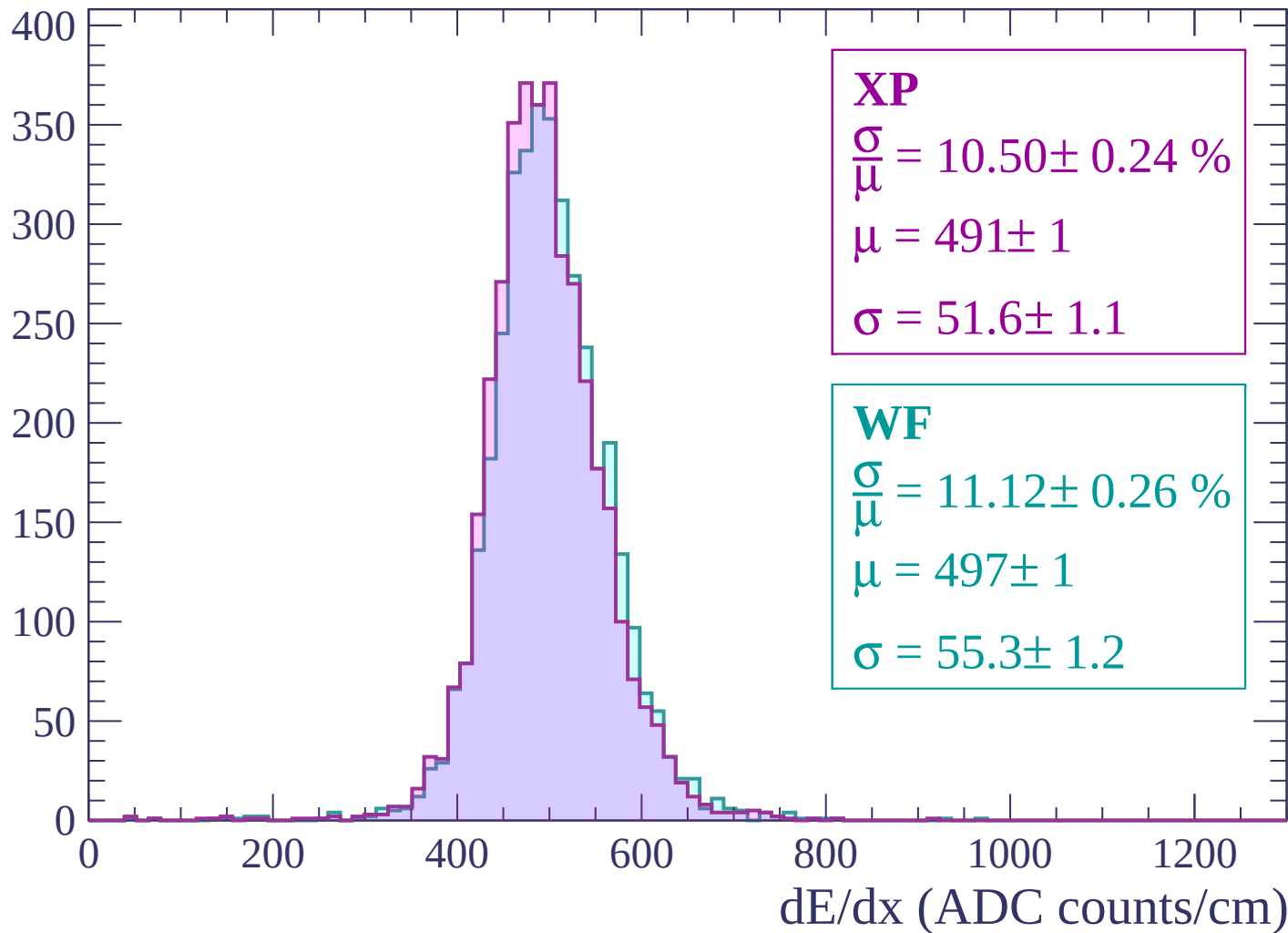
Energy loss |  $2460 < p < 2520$

Count



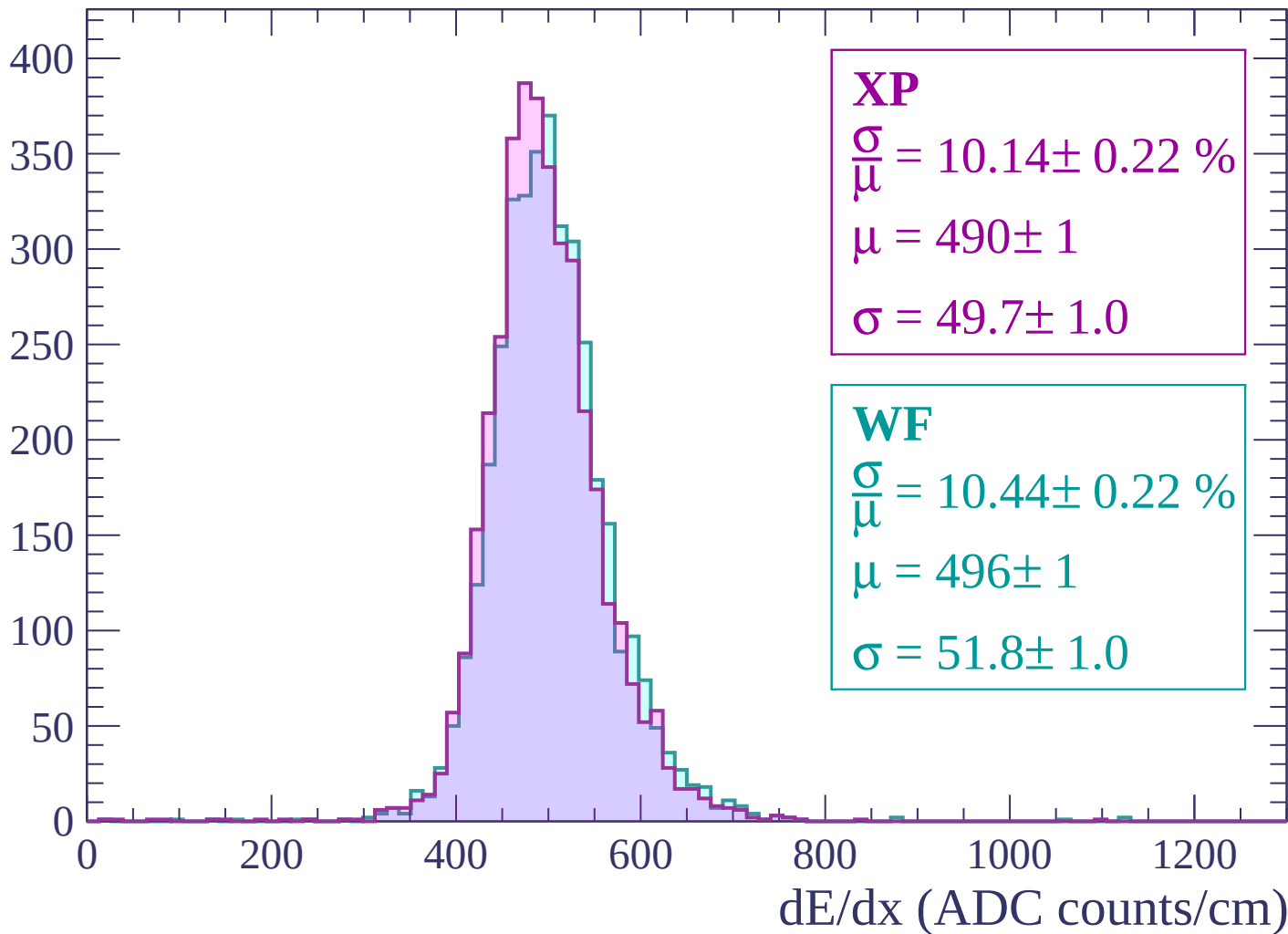
Energy loss |  $2520 < p < 2580$

Count



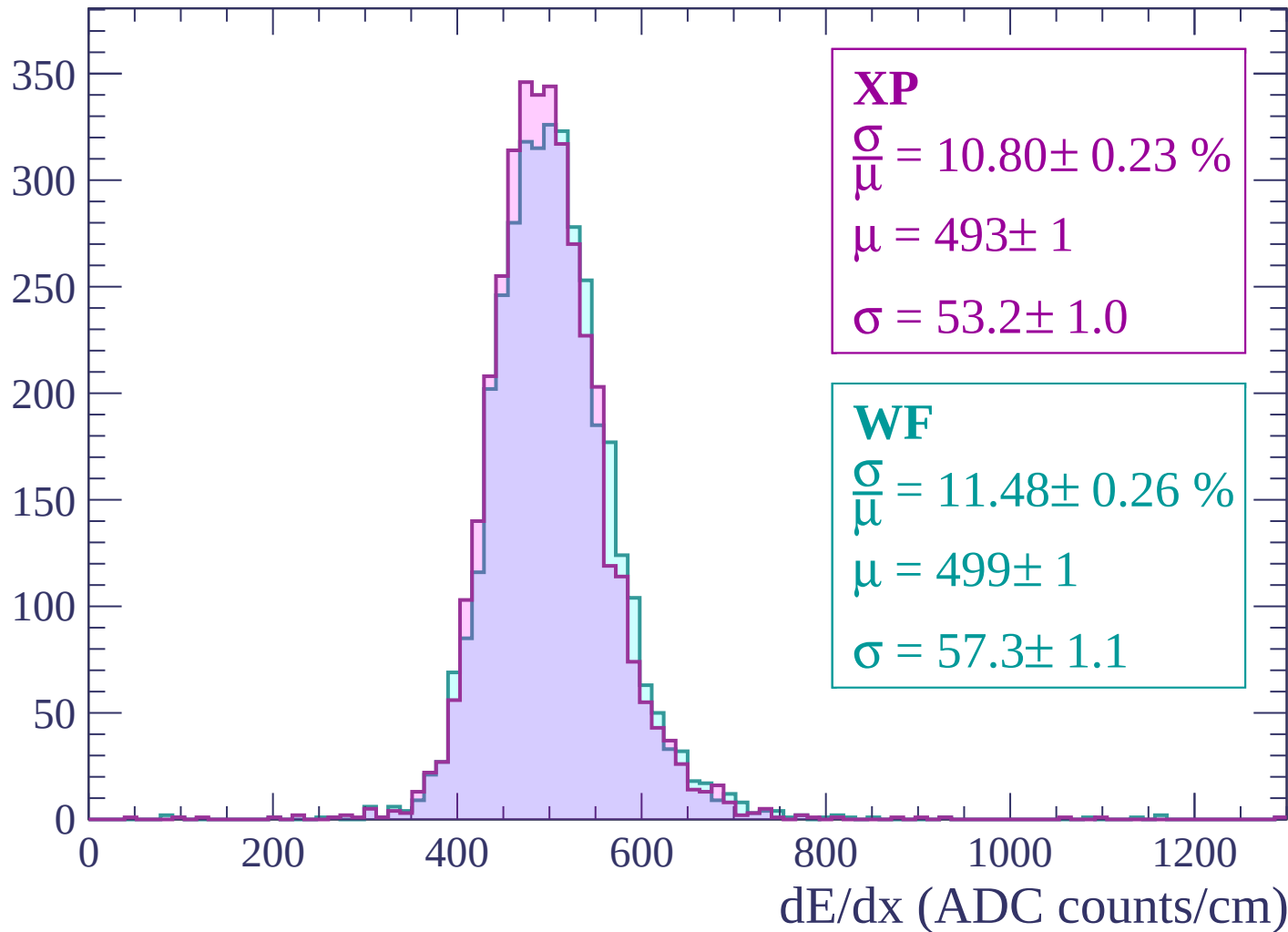
Energy loss |  $2580 < p < 2640$

Count



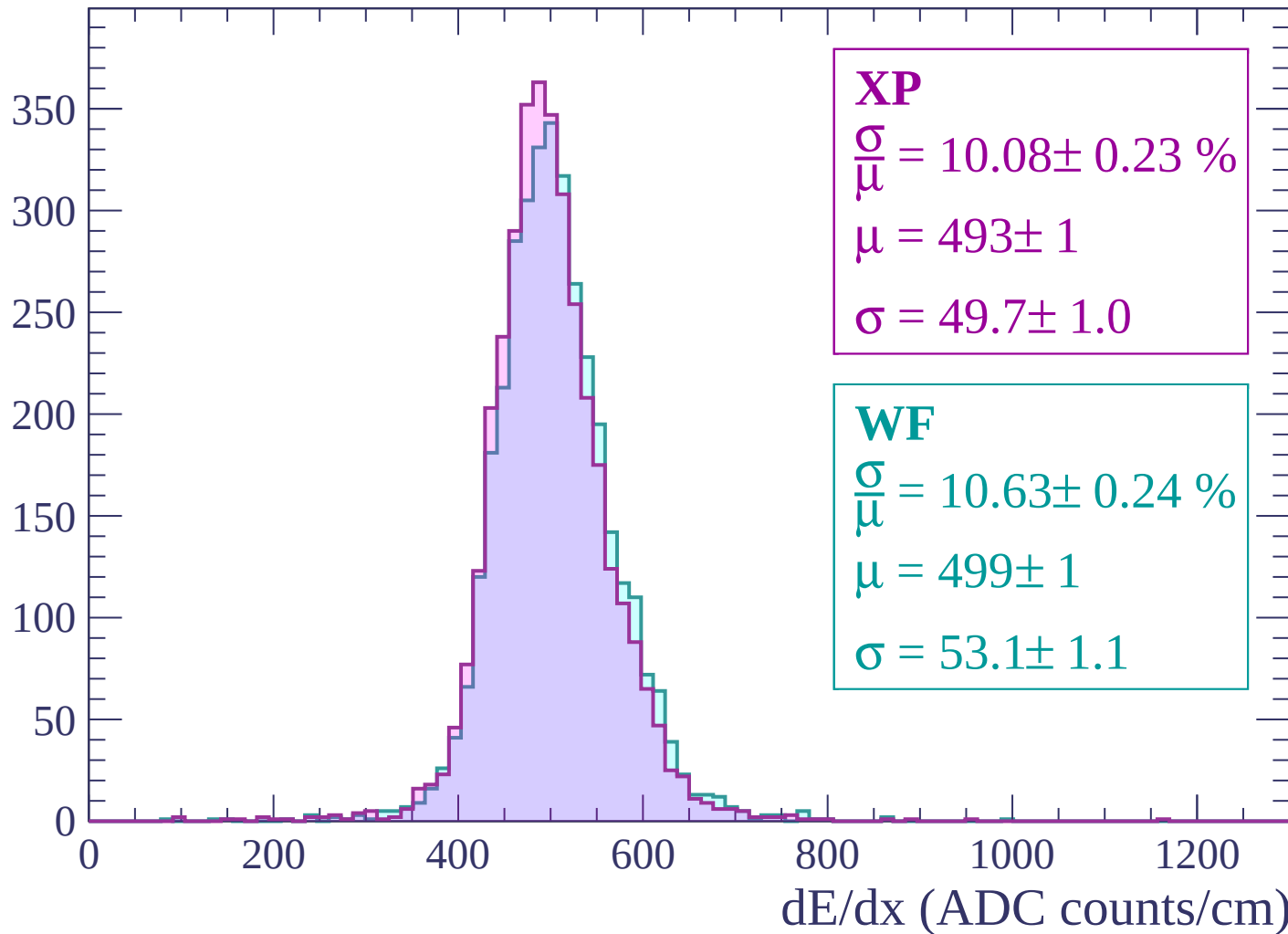
Energy loss |  $2640 < p < 2700$

Count



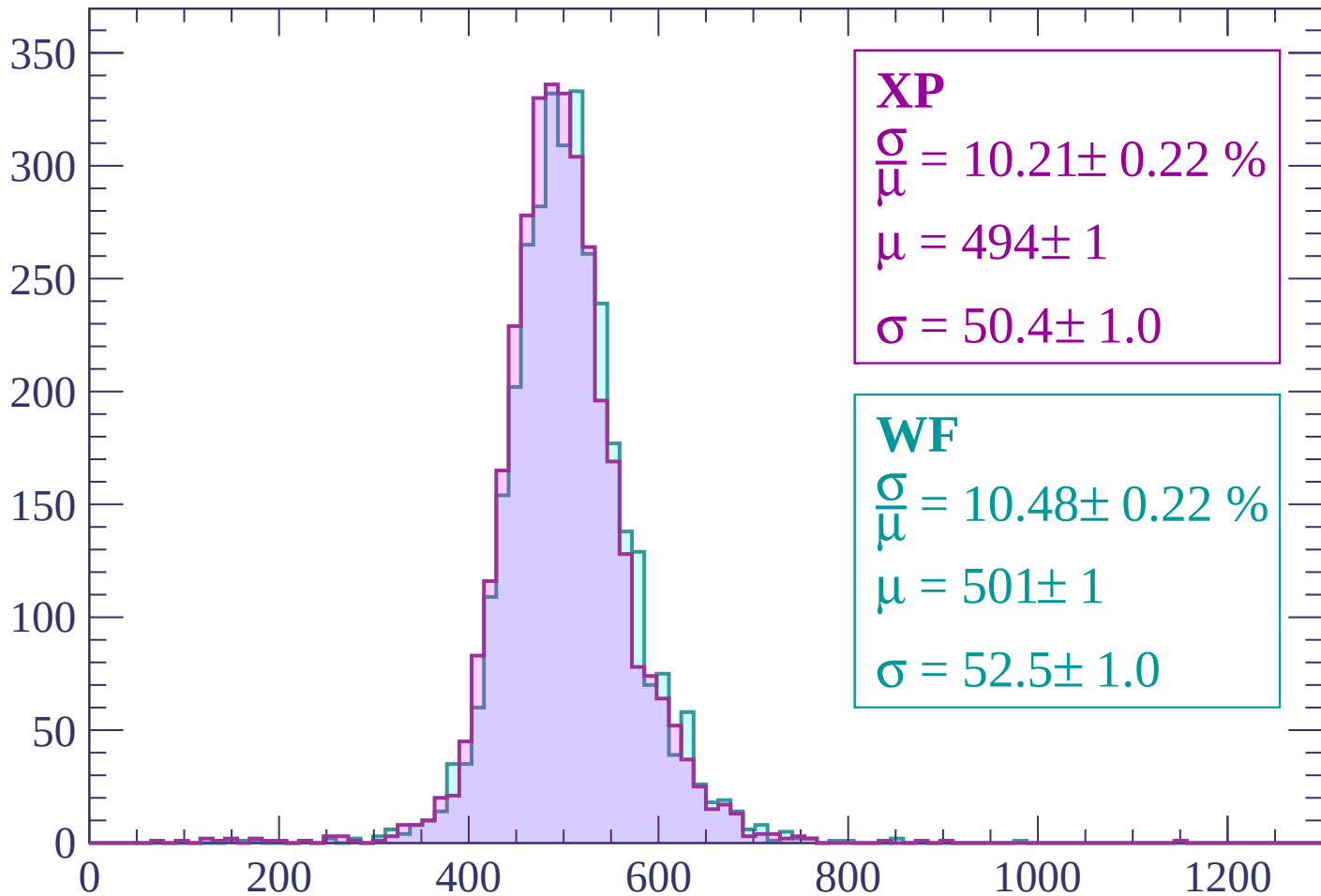
Energy loss |  $2700 < p < 2760$

Count



Energy loss |  $2760 < p < 2820$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.21 \pm 0.22 \%$$

$$\mu = 494 \pm 1$$

$$\sigma = 50.4 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 10.48 \pm 0.22 \%$$

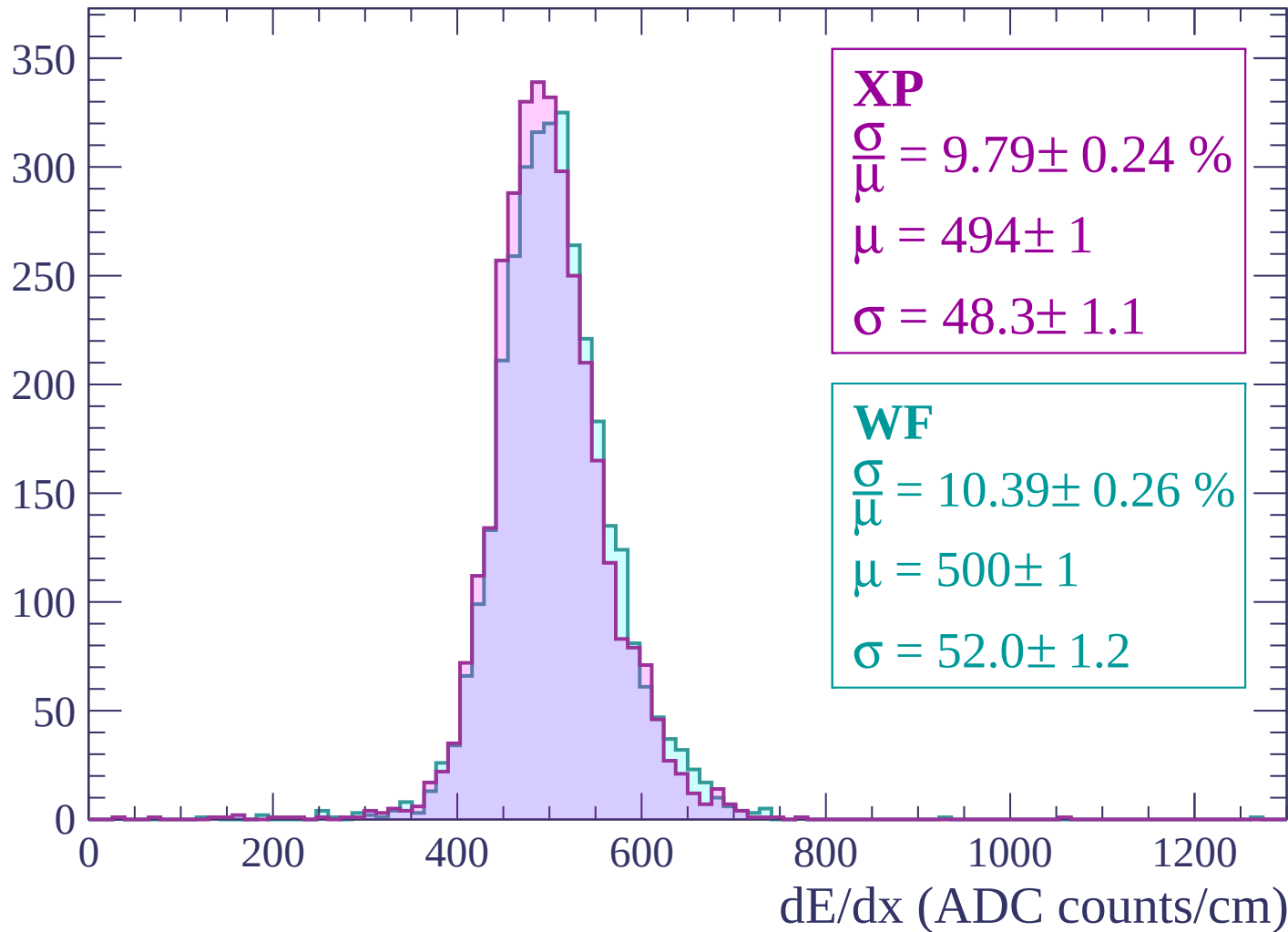
$$\mu = 501 \pm 1$$

$$\sigma = 52.5 \pm 1.0$$

$dE/dx$  (ADC counts/cm)

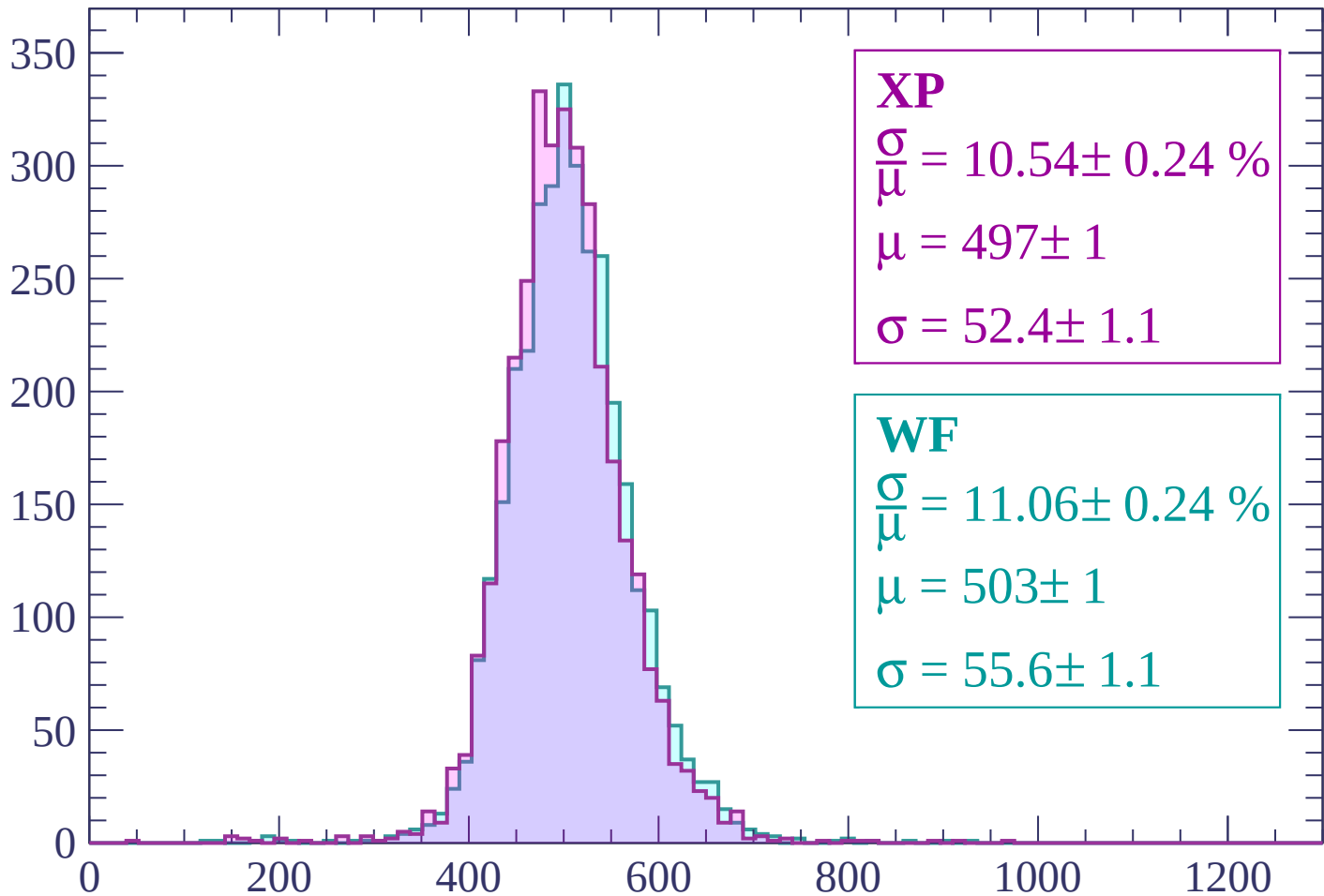
Energy loss |  $2820 < p < 2880$

Count



Energy loss |  $2880 < p < 2940$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.54 \pm 0.24 \%$$

$$\mu = 497 \pm 1$$

$$\sigma = 52.4 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 11.06 \pm 0.24 \%$$

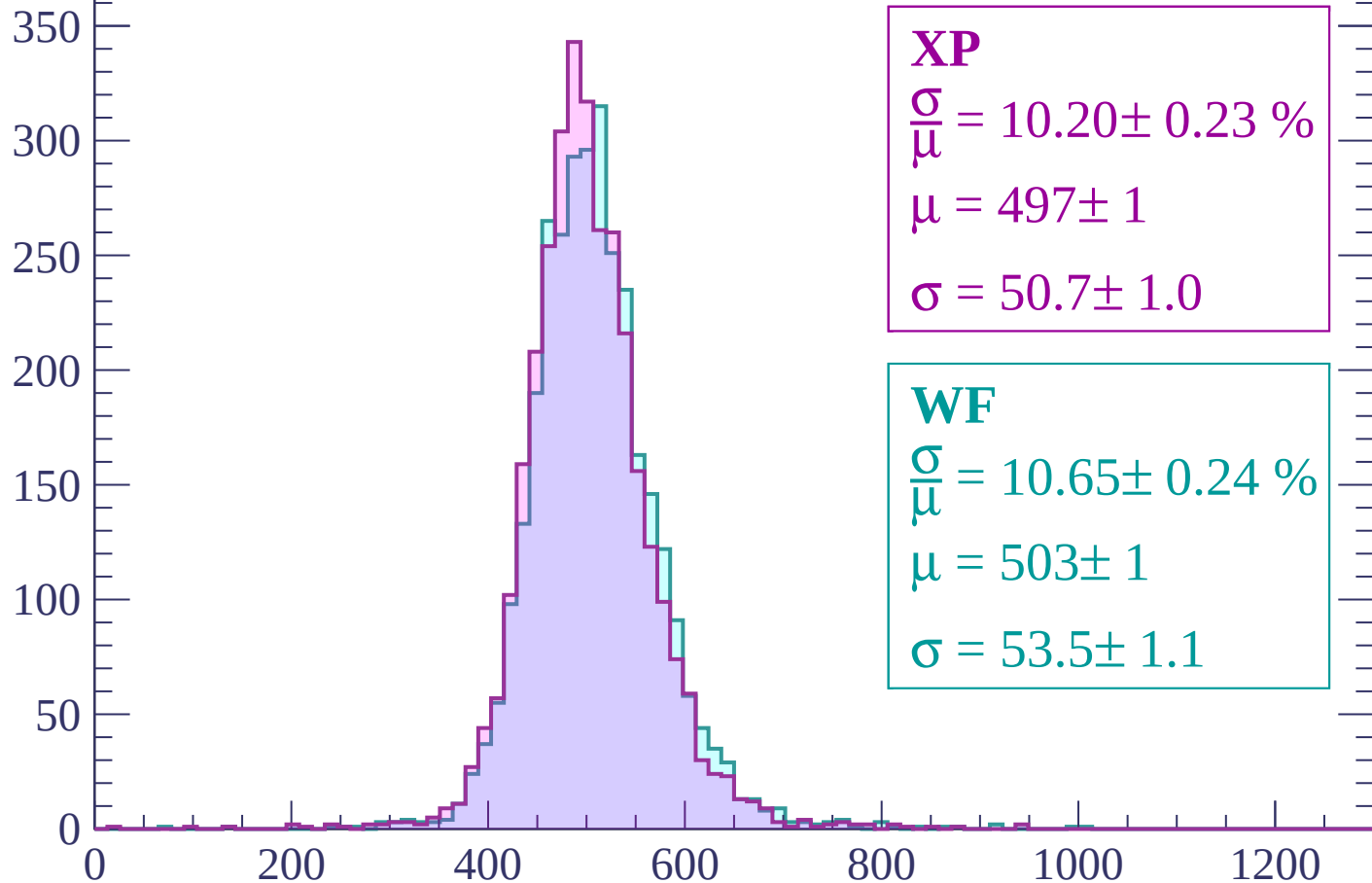
$$\mu = 503 \pm 1$$

$$\sigma = 55.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss |  $2940 < p < 3000$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.20 \pm 0.23 \%$$

$$\mu = 497 \pm 1$$

$$\sigma = 50.7 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 10.65 \pm 0.24 \%$$

$$\mu = 503 \pm 1$$

$$\sigma = 53.5 \pm 1.1$$

$dE/dx$  (ADC counts/cm)

# Energy loss | $3000 < p < 3060$

Count

